Event tree และ Guard tree

สกัดจากแบบจำลอง Event-B 198 ตัว (model.txt ที่ได้จาก ChronoSeq-B โดยไม่แก้ไข) · มีเงื่อนไขจากแผนภาพ 146 ตัว · ไม่มีเงื่อนไข 52 ตัว

1. หลักการ

guard ในแบบจำลองแบ่งได้ 2 กลุ่ม

กลุ่ม	ตัวอย่าง	ใช้ทำ
guard ควบคุมลำดับข้อความ (สูตรแปลงสร้างให้ทุกข้อความ)	`m ∉ sentMessages`, `p ∈ receivedMessages`, `currentMessage = ∅`	Event tree
guard เงื่อนไขจากแผนภาพ (loop / alt / opt)	`cycleNo < 8`, `mode = GOLD`, `cycleNo mod 3 = 0`	Guard tree

การตั้งรหัส guard
- **ตัวอักษร** = จุดตัดสินใจ: guard ที่นิพจน์ด้านซ้ายเหมือนกันใช้ตัวอักษรเดียวกัน (`mode = GOLD` กับ `mode = SILVER` เป็นตัว b ทั้งคู่)
- **ตัวเลข** = ทางเลือกของจุดตัดสินใจนั้น เรียงตามลำดับที่พบในไฟล์ (b1, b2, b3)
- ตัวอักษรเรียงตามลำดับที่พบในไฟล์ (a, b, c ...)

การตั้งรหัส event (ใช้หลักเดียวกับ guard · send และ receive เป็นคนละ event)
- **ตัวอักษร** = ลำดับขั้น: event ที่ต่อกันเป็นเส้นตรงได้ตัวอักษรใหม่ทีละขั้น (a1 → b1 → c1)
- **ตัวเลข** = ทางเลือกตรงจุดแยกกิ่ง: event ขั้นเดียวกันของแต่ละกิ่งใช้ตัวอักษรเดียวกัน (กิ่ง GOLD / SILVER / BRONZE → m1 / m2 / m3 แล้วขั้นถัดไป n1 / n2 / n3) · ท้าย loop แยกเป็น ทำต่อ (o1) กับวนรอบใหม่ (o2)
- event ที่หลายกิ่งมารวมกันได้ตัวอักษรใหม่ (q1) · ในวงเล็บ (guard ...) คือรหัส guard ของ event นั้น

การสร้าง tree
- **Event tree**: send m → receive m และ receive p → send m เมื่อ send m มี guard `p ∈ receivedMessages` · alt แตกเป็นหลายกิ่ง · หลัง alt ที่มี guard แบบ `p ∈ receivedMessages ∨ q ∈ receivedMessages` ทุกกิ่งต่อด้วย event เดียวกัน · event นับรอบ loop (checkloop) อยู่ท้าย loop และมีลูกศรวงกลับไปจุดเริ่ม loop
- ในเอกสารนี้ event tree แสดงแบบย่อ: event ที่ต่อกันเป็นเส้นตรงเขียนในบรรทัดเดียว (a1 → b1 → c1) และแตกบรรทัดเฉพาะจุดที่แยกกิ่ง · รูปเต็มอยู่ใน `models/<id>/tree.txt`
- **Guard tree**: เดินตามแต่ละเส้นทางของ event tree แล้วเก็บรหัส guard ตามลำดับที่พบครั้งแรก จึงได้ทั้งเงื่อนไขซ้อนกัน (loop ที่มี alt ข้างใน) และเงื่อนไขที่มาต่อกัน (opt แล้วตามด้วย alt)

2. ตัวอย่าง

ตัวอย่างหลักการ: guard แรกคือ `age > 50` และจุดถัดไปมี 2 ทางเลือก `salary ≥ 500` กับ `salary < 500`

a1	age > 50
b1	salary ≥ 500
b2	salary < 500

```
(root)
└── a1  age > 50
```

```
    ├── b1  salary ≥ 500
    └── b2  salary < 500

เส้นทาง:  a1 -> b1
          a1 -> b2
```

---

model_141 PharmacyDelivery — opt แล้วตามด้วย alt 3 ทาง (เงื่อนไขมาต่อกัน)

Guard tree

a1	needAck = 1
b1	mode = GOLD
b2	mode = SILVER
b3	mode = BRONZE

```
(root)
└── a1  needAck = 1
    ├── b1  mode = GOLD
    ├── b2  mode = SILVER
    └── b3  mode = BRONZE
```

เส้นทาง

```
a1 -> b1
a1 -> b2
a1 -> b3
```

Event tree

a1 sendsyncSignature_1
b1 receivesyncSignature_1
c1 sendvalidateSignature_2
d1 receivevalidateSignature_2
e1 sendfetchRoute_3_opt1 (guard a1)
f1 receivefetchRoute_3_opt1 (guard a1)
g1 sendqueryRoute_4_opt1 (guard a1)
h1 receivequeryRoute_4_opt1 (guard a1)
i1 sendconfirmRoute_5
j1 receiveconfirmRoute_5
k1 sendissuePackage_6
l1 receiveissuePackage_6
m1 sendcancelSignature_7 (guard b1)
m2 sendqueryPackage_9 (guard b2)
m3 sendscheduleSignature_11 (guard b3)
n1 receivecancelSignature_7 (guard b1)
n2 receivequeryPackage_9 (guard b2)
n3 receivescheduleSignature_11 (guard b3)
o1 sendconfirmPackage_8 (guard b1)
o2 sendackSignature_10 (guard b2)
p1 receiveconfirmPackage_8 (guard b1)
p2 receiveackSignature_10 (guard b2)
q1 sendapproveRoute_12
r1 receiveapproveRoute_12

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1 → o1 → p1 → q1 → r1
    ├── m2 → n2 → o2 → p2 → q1 → r1
    └── m3 → n3 → q1 → r1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 -> n2 -> o2 -> p2 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m3 -> n3 -> q1 -> r1
```

model_109 HospitalLabRequest — loop ที่มี alt อยู่ข้างใน และมี loop ซ้อนในกิ่งแรก (เงื่อนไขซ้อนกัน)

Guard tree

a1	cycleNo < 8
b1	cycleNo mod 3 = 0
c1	attemptNo < 10
b2	cycleNo mod 3 ≠ 0

```
(root)
└── a1  cycleNo < 8
    ├── b1  cycleNo mod 3 = 0
    │   └── c1  attemptNo < 10
    └── b2  cycleNo mod 3 ≠ 0
```

เส้นทาง

```
a1 -> b1 -> c1
a1 -> b2
```

Event tree

a1 sendcheckOrder_1
b1 receivecheckOrder_1
c1 sendrequestResult_2
d1 receiverequestResult_2
e1 sendconfirmOrder_3 (guard a1)
f1 receiveconfirmOrder_3 (guard a1)
g1 sendscheduleOrder_4 (guard a1 b1)
g2 sendreportSample_9 (guard a1 b2)
h1 receivescheduleOrder_4 (guard a1 b1)
h2 receivereportSample_9 (guard a1 b2)
i1 sendackSample_5 (guard a1 b1 c1)
j1 receiveackSample_5 (guard a1 b1 c1)
k1 sendapproveSample_6 (guard a1 b1 c1)
l1 receiveapproveSample_6 (guard a1 b1 c1)
m1 sendissueResult_7 (guard a1 b1 c1)
n1 receiveissueResult_7 (guard a1 b1 c1)
o1 sendreportResult_8 (guard a1 b1)
o2 checkloop_ackSample_5_loop2 (guard a1 b1 c1)
p1 receivereportResult_8 (guard a1 b1)
q1 sendrequestSample_10 (guard a1)
r1 receiverequestSample_10 (guard a1)
s1 sendreserveSample_11
s2 checkloop_confirmOrder_3_loop1 (guard a1)
t1 receivereserveSample_11

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1
    │   ├── o1 → p1 → q1 → r1
    │   │   ├── s1 → t1
    │   │   └── s2 ↺ e1
    │   └── o2 ↺ i1
    └── g2 → h2 → q1 → r1
        ├── s1 → t1
        └── s2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o2 ↺ i1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> q1 -> r1 -> s2 ↺ e1
```

3. สรุปทั้งชุด

จำนวนเส้นทางใน guard tree	0	1	2	3	4	6
จำนวนแบบจำลอง	52	71	67	5	2	1

0 = ไม่มีเงื่อนไขจากแผนภาพ · ไฟล์ของแต่ละแบบจำลอง: `models/<id>/tree.txt` (รูป tree เต็ม) และ `tree.json` · รวมทุกแบบจำลอง: `data/trees.jsonl`, `trees/guard_codes.csv`, `trees/event_codes.csv` · **model_001**: model.txt มีเฉพาะชื่อ event จึงอ่าน event และ guard จากไฟล์ proof obligation ของ Rodin (.bpo) แทน · มี 1 event ที่ไฟล์ Rodin ไม่เก็บชื่อไว้ (แสดงเป็น unnamed_event_N, guard `retry < 3`) · แบบจำลองนี้แปลงด้วยสูตรรุ่นแรก ส่วน loop และ par จึงไม่มี guard อ้างข้อความก่อนหน้า event tree จึงแยกเป็นหลายกิ่งจาก root

4. รหัสและเส้นทางของแต่ละแบบจำลอง

ซ้าย: รหัส guard, guard tree และเส้นทาง · ขวา: รหัส event (ในวงเล็บคือ guard ของ event นั้น), event tree แบบย่อ และเส้นทาง · ↺ x = จบรอบ loop แล้วกลับไปที่ event x

การเงินและการชำระเงิน · Finance & Payment (14)

model_001 Payment & Notification · pass

Guard

a1	Login = 1
b1	amount_1 ≥ 1000
b2	amount_1 < 1000
c1	retry < 3
c2	retry = 3
d1	par_1 = 2

```
(root)
├── a1  Login = 1
│   ├── b1  amount_1 ≥ 1000
│   │   └── c1  retry < 3
│   └── b2  amount_1 < 1000
│       └── c1  retry < 3
└── c2  retry = 3
    └── d1  par_1 = 2
```

เส้นทาง

```
a1 -> b1 -> c1
a1 -> b2 -> c1
c2 -> d1
```

Event

a1 sendsubmitPayment_1_opt1 (guard a1)
a2 sendvalidatePayment_7_loop3 (guard c2)
b1 receivesubmitPayment_1_opt1 (guard a1)
b2 receivevalidatePayment_7_loop3 (guard c2)
c1 sendpaymentDetails_2_opt1 (guard a1)
c2 sendvalidationResult_8_loop3 (guard c2)
d1 receivePaymentDetails_2_opt1 (guard a1)
d2 recivevalidationResult_8_loop3 (guard c2)
e1 sendprocessLargePayment_3_alt2 (guard b1)
e2 sendProcessStandarddPayment_5_alt2 (guard b2)
f1 receiveprocessLargePayment_3_alt2 (guard b1)
f2 receiveProcessStandarddPayment_5_alt2 (guard b2)
g1 sendapprovalRequired_4_alt2 (guard b1)
g2 sendpaymentConfirmation_6_alt2 (guard b2)
h1 reciveapprovalRequired_4_alt2 (guard b1)
h2 recivepaymentConfirmation_6_alt2 (guard b2)
i1 unnamed_event_N (guard c1)
j1 sendsendSMSNotification_9_par4
k1 sendsendEmailNotification_10_par4
k2 receivedsendSMSNotification_9_par4 (guard d1)
l1 receivedsendEmailNotification_10_par4 (guard d1)
m1 sendpaymentStatus_11
n1 receivepaymentStatus_11

```
(root) INITIALISATION
├── a1 → b1 → c1 → d1
│   ├── e1 → f1 → g1 → h1 → i1
│   └── e2 → f2 → g2 → h2 → i1
└── a2 → b2 → c2 → d2 → j1
    ├── k1 → l1 → m1 → n1
    └── k2
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g2 -> h2 -> i1
a2 -> b2 -> c2 -> d2 -> j1 -> k1 -> l1 -> m1 -> n1
a2 -> b2 -> c2 -> d2 -> j1 -> k2
```

model_003 ATMWithdrawal · pass

Guard

a1	pinAttempt < 3
b1	pinValid = 1
b2	pinValid = 0

```
(root)
└── a1  pinAttempt < 3
    ├── b1  pinValid = 1
    └── b2  pinValid = 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

a1 sendinsertCard_1
b1 receiveinsertCard_1
c1 sendrequestPIN_2
d1 receiverequestPIN_2
e1 sendenterPIN_3
f1 receiveenterPIN_3
g1 sendinvalidPIN_4 (guard a1)
h1 receiveinvalidPIN_4 (guard a1)
i1 sendreenterPIN_5 (guard a1)
j1 receivereenterPIN_5 (guard a1)
k1 sendselectAmount_6 (guard b1)
k2 sendretainCard_8 (guard b2)
k3 checkloop_invalidPIN_4_loop1 (guard a1)
l1 receiveselectAmount_6 (guard b1)
l2 receiveretainCard_8 (guard b2)
m1 senddispenseCash_7 (guard b1)
n1 receivedispenseCash_7 (guard b1)

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1 → m1 → n1
    ├── k2 → l2
    └── k3 ↺ g1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 -
> l2
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k3 ↺
g1
```

model_004 BankTransfer · fail

Guard

a1	riskFlag = 0
a2	riskFlag = 1

```
(root)
├── a1  riskFlag = 0
└── a2  riskFlag = 1
```

เส้นทาง

```
a1
a2
```

Event

a1 sendinitiateTransfer_1
b1 receiveinitiateTransfer_1
c1 sendcheckFraud_2
d1 receivecheckFraud_2
e1 sendriskScore_3
f1 receiveriskScore_3
g1 sendpostTransfer_4 (guard a1)
g2 sendholdTransfer_6 (guard a2)
h1 receivepostTransfer_4 (guard a1)
h2 receiveholdTransfer_6 (guard a2)
i1 sendpostingDone_5 (guard a1)
j1 receivepostingDone_5 (guard a1)
k1 sendtransferStatus_7
l1 receivetransferStatus_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    └── g2 → h2 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1
```

model_025 InsuranceClaim · pass

Guard

a1	hasDocs = 1
b1	claimAmount ≤ 5000
b2	claimAmount > 5000

```
(root)
└── a1  hasDocs = 1
    ├── b1  claimAmount ≤ 5000
    └── b2  claimAmount > 5000
```

เส้นทาง

a1 -> b1
a1 -> b2

Event

a1 sendsubmitClaim_1
b1 receivesubmitClaim_1
c1 senduploadDocs_2_opt1 (guard a1)
d1 receiveuploadDocs_2_opt1 (guard a1)
e1 sendapprovalNotice_3 (guard b1)
e2 sendassignAssessor_4 (guard b2)
f1 receiveapprovalNotice_3 (guard b1)
f2 receiveassignAssessor_4 (guard b2)
g1 sendpayout_8
h1 receivepayout_8
i1 sendtransferDone_9
j1 receivetransferDone_9
k1 sendnotifyClaimant_10
l1 receivenotifyClaimant_10
m1 sendsiteVisit_5 (guard b2)
n1 receivesiteVisit_5 (guard b2)
o1 sendvisitDone_6 (guard b2)
p1 receivevisitDone_6 (guard b2)
q1 sendassessmentReport_7 (guard b2)
r1 receiveassessmentReport_7 (guard b2)

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1
    └── e2 → f2 → m1 → n1 → o1 → p1 → q1 → r1 →
          g1 → h1 → i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_056 BankBranchQueue · pass

Guard

a1	passNo < 11

```
(root)
└── a1  passNo < 11
```

เส้นทาง

a1

Event

a1 sendqueryCounter_1
b1 receivequeryCounter_1
c1 sendassignService_2
d1 receiveassignService_2
e1 sendscheduleService_3 (guard a1)
f1 receivescheduleService_3 (guard a1)
g1 sendsyncTicket_4 (guard a1)
h1 receivesyncTicket_4 (guard a1)
i1 sendfetchTicket_5 (guard a1)
j1 receivefetchTicket_5 (guard a1)
k1 sendackCounter_6 (guard a1)
l1 receiveackCounter_6 (guard a1)
m1 sendnotifyService_7 (guard a1)
n1 receivenotifyService_7 (guard a1)
o1 sendqueryTicket_8
o2 checkloop_scheduleService_3_loop 1 (guard a1)
p1 receivequeryTicket_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1
    └── o2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o2 ↺ e1

model_066 CanteenPayment · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendupdateBalance_1
b1 receiveupdateBalance_1
c1 sendlogTopup_2
d1 receivelogTopup_2
e1 sendrequestTopup_3
f1 receiverequestTopup_3
g1 sendassignBalance_4
h1 receiveassignBalance_4
i1 sendcheckTopup_5
j1 receivecheckTopup_5
k1 sendreleaseMeal_6
l1 receivereleaseMeal_6
m1 sendreportMeal_7
n1 receivereportMeal_7
o1 sendupdateTopup_8
p1 receiveupdateTopup_8
q1 sendscheduleMeal_9
r1 receivescheduleMeal_9
s1 sendsendBalance_10
t1 receivesendBalance_10
u1 sendscheduleBalance_11
v1 receivescheduleBalance_11
w1 sendvalidateMeal_12
x1 receivevalidateMeal_12
y1 sendconfirmTopup_13
z1 receiveconfirmTopup_13

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1 → u1 → v1 → w1 → x1 →
      y1 → z1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1 -> y1 -> z1

model_094 FleetFuelCard · fail

Guard

a1	temp < 50
a2	temp ≥ 50

```
(root)
├── a1  temp < 50
└── a2  temp ≥ 50
```

เส้นทาง

a1
a2

Event

a1 sendreleaseLimit_1
b1 receivereleaseLimit_1
c1 sendreportCard_2
d1 receivereportCard_2
e1 sendsendFuel_3 (guard a1)
e2 sendcancelFuel_8 (guard a2)
f1 receivesendFuel_3 (guard a1)
f2 receivecancelFuel_8 (guard a2)
g1 sendissueLimit_4 (guard a1)
h1 receiveissueLimit_4 (guard a1)
i1 sendissueCard_5 (guard a1)
j1 receiveissueCard_5 (guard a1)
k1 sendapproveCard_6 (guard a1)
l1 receiveapproveCard_6 (guard a1)
m1 sendissueFuel_7 (guard a1)
n1 receiveissueFuel_7 (guard a1)
o1 sendvalidateCard_9
p1 receivevalidateCard_9

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1 →
    │     m1 → n1 → o1 → p1
    └── e2 → f2 → o1 → p1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> o1 -> p1

model_100 GasStationPrepay · pass

Guard

a1	express = 1
b1	attemptNo < 4

```
(root)
└── a1  express = 1
    └── b1  attemptNo < 4
```

เส้นทาง

```
a1 -> b1
```

Event

a1 sendnotifyPrepay_1
b1 receivenotifyPrepay_1
c1 sendvalidatePrepay_2
d1 receivevalidatePrepay_2
e1 sendlogFuel_3_opt1 (guard a1)
f1 receivelogFuel_3_opt1 (guard a1)
g1 sendreserveReceipt_4
h1 receivereserveReceipt_4
i1 sendassignFuel_5
j1 receiveassignFuel_5
k1 sendreleaseReceipt_6 (guard b1)
l1 receivereleaseReceipt_6 (guard b1)
m1 sendfetchFuel_7 (guard b1)
n1 receivefetchFuel_7 (guard b1)
o1 sendrequestFuel_8
o2 checkloop_releaseReceipt_6_loop1 (guard b1)
p1 receiverequestFuel_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1
    └── o2 ↺ k1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o2 ↺ k1
```

model_113 InsuranceTelematics · pass

Guard

a1	express = 1
b1	roundNo < 6
c1	urgent = 1
c2	urgent = 0

```
(root)
└── a1  express = 1
    └── b1  roundNo < 6
        ├── c1  urgent = 1
        └── c2  urgent = 0
```

เส้นทาง

```
a1 -> b1 -> c1
a1 -> b1 -> c2
```

Event

a1 sendconfirmRating_1
b1 receiveconfirmRating_1
c1 sendfetchPremium_2
d1 receivefetchPremium_2
e1 sendsyncRating_3_opt1 (guard a1)
f1 receivesyncRating_3_opt1 (guard a1)
g1 sendnotifyRating_4_opt1 (guard a1 b1)
h1 receivenotifyRating_4_opt1 (guard a1 b1)
i1 sendconfirmTrip_5_opt1 (guard a1)
i2 checkloop_notifyRating_4_loop1 (guard a1 b1)
j1 receiveconfirmTrip_5_opt1 (guard a1)
k1 sendconfirmPremium_6
l1 receiveconfirmPremium_6
m1 sendackPremium_7
n1 receiveackPremium_7
o1 sendassignTrip_8 (guard c1)
o2 sendschedulePremium_9 (guard c2)
p1 receiveassignTrip_8 (guard c1)
p2 receiveschedulePremium_9 (guard c2)
q1 sendqueryRating_10
r1 receivequeryRating_10

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1 → m1 → n1
    │   ├── o1 → p1 → q1 → r1
    │   └── o2 → p2 → q1 → r1
    └── i2 ↺ g1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o2 -> p2 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ g1
```

model_124 MarketplaceEscrow · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendvalidateEscrow_1
b1 receivevalidateEscrow_1
c1 sendapproveEscrow_2
d1 receiveapproveEscrow_2
e1 sendlogEscrow_3
f1 receivelogEscrow_3
g1 sendrequestEscrow_4
h1 receiverequestEscrow_4
i1 sendfetchEscrow_5
j1 receivefetchEscrow_5
k1 sendnotifyEscrow_6
l1 receivenotifyEscrow_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_128 MobileTopUp · fail

Guard

a1	t_obs ≤ 13
a2	t_obs > 13

```
(root)
├── a1  t_obs ≤ 13
└── a2  t_obs > 13
```

เส้นทาง

a1
a2

Event

a1 sendnotifyCredit_1
b1 receivenotifyCredit_1
c1 sendconfirmBundle_2
d1 receiveconfirmBundle_2
e1 sendreportBundle_3 (guard a1)
e2 sendcheckBundle_5 (guard a2)
f1 receivereportBundle_3 (guard a1)
f2 receivecheckBundle_5 (guard a2)
g1 sendackCredit_4 (guard a1)
g2 sendassignCredit_6 (guard a2)
h1 receiveackCredit_4 (guard a1)
h2 receiveassignCredit_6 (guard a2)
i1 sendreleaseTopup_7
j1 receivereleaseTopup_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    └── e2 → f2 → g2 → h2 → i1 → j1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g2 -> h2 -> i1 -> j1

model_131 OnlineBankingOTP · fail

Guard

a1	retryNo < 12
b1	retryNo mod 4 = 0
b2	retryNo mod 4 ≠ 0

```
(root)
└── a1  retryNo < 12
    ├── b1  retryNo mod 4 = 0
    └── b2  retryNo mod 4 ≠ 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

a1 sendconfirmTransfer_1
b1 receiveconfirmTransfer_1
c1 sendsendOtp_2
d1 receivesendOtp_2
e1 sendsendTransfer_3 (guard a1)
f1 receivesendTransfer_3 (guard a1)
g1 sendvalidateAccount_4 (guard a1 b1)
g2 sendsyncOtp_6 (guard a1 b2)
h1 receivevalidateAccount_4 (guard a1 b1)
h2 receivesyncOtp_6 (guard a1 b2)
i1 sendupdateOtp_5 (guard a1 b1)
i2 sendapproveOtp_7 (guard a1 b2)
j1 receiveupdateOtp_5 (guard a1 b1)
j2 receiveapproveOtp_7 (guard a1 b2)
k1 sendcancelTransfer_8 (guard a1)
l1 receivecancelTransfer_8 (guard a1)
m1 sendapproveAccount_9 (guard a1)
n1 receiveapproveAccount_9 (guard a1)
o1 sendsendAccount_10 (guard a1)
p1 receivesendAccount_10 (guard a1)
q1 sendissueOtp_11
q2 checkloop_sendTransfer_3_loop1 (guard a1)
r1 receiveissueOtp_11

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1 →
    │     o1 → p1
    │   ├── q1 → r1
    │   └── q2 ↺ e1
    └── g2 → h2 → i2 → j2 → k1 → l1 → m1 → n1 →
          o1 → p1
        ├── q1 → r1
        └── q2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1 -> o1 -> p1 -> q2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> i2 -> j2 -> k1 -
> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> i2 -> j2 -> k1 -
> l1 -> m1 -> n1 -> o1 -> p1 -> q2 ↺ e1
```

model_132 OnlineDonation · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendsendReceipt_1
b1 receivesendReceipt_1
c1 sendqueryGift_2
d1 receivequeryGift_2
e1 sendapproveReceipt_3
f1 receiveapproveReceipt_3
g1 sendnotifyReceipt_4
h1 receivenotifyReceipt_4
i1 sendapproveGift_5
j1 receiveapproveGift_5
k1 sendrequestGift_6
l1 receiverequestGift_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1
```

model_189 TradingOrderGateway · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendcheckFill_1
b1 receivecheckFill_1
c1 sendassignOrder_2
d1 receiveassignOrder_2
e1 sendscheduleFill_3
f1 receivescheduleFill_3
g1 sendscheduleOrder_4
h1 receivescheduleOrder_4
i1 sendnotifyOrder_5
j1 receivenotifyOrder_5
k1 sendackMargin_6
l1 receiveackMargin_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

สุขภาพและการแพทย์ · Healthcare & Medical (26)

model_006 BloodBankRequest · pass

Guard

a1	stockUnits ≥ 2
a2	stockUnits < 2

```
(root)
├── a1  stockUnits ≥ 2
└── a2  stockUnits < 2
```

เส้นทาง

a1
a2

Event

a1 sendrequestBlood_1
b1 receiverequestBlood_1
c1 sendcrossMatch_2 (guard a1)
c2 sendbackorder_5 (guard a2)
d1 receivecrossMatch_2 (guard a1)
d2 receivebackorder_5 (guard a2)
e1 sendprepareCourier_3 (guard a1)
f1 receiveprepareCourier_3 (guard a1)
g1 senddeliverBlood_4 (guard a1)
h1 receivedeliverBlood_4 (guard a1)
i1 sendconfirmReceipt_6
j1 receiveconfirmReceipt_6

```
(root) INITIALISATION
└── a1 → b1
    ├── c1 → d1 → e1 → f1 → g1 → h1 → i1 → j1
    └── c2 → d2 → i1 → j1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c2 -> d2 -> i1 -> j1

model_020 FitnessTracker · pass

Guard

a1	syncRound < 10

```
(root)
└── a1  syncRound < 10
```

เส้นทาง

a1

Event

a1 sendpairDevice_1
b1 receivepairDevice_1
c1 sendpairOK_2
d1 receivepairOK_2
e1 sendsyncSteps_3 (guard a1)
f1 receivesyncSteps_3 (guard a1)
g1 sendsyncAck_4 (guard a1)
h1 receivesyncAck_4 (guard a1)
i1 sendshowDashboard_5
i2 checkloop_syncSteps_3_loop1 (guard a1)
j1 receiveshowDashboard_5

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1
    └── i2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ e1

model_023 HospitalTriage · pass

Guard

a1	severity = CRITICAL
a2	severity = NORMAL

```
(root)
├── a1  severity = CRITICAL
└── a2  severity = NORMAL
```

เส้นทาง
a1
a2

Event

a1 sendregisterPatient_1
b1 receiveregisterPatient_1
c1 sendmeasureVitals_2
d1 receivemeasureVitals_2
e1 sendcallDoctor_3 (guard a1)
e2 sendscheduleVisit_6 (guard a2)
f1 receivecallDoctor_3 (guard a1)
f2 receivescheduleVisit_6 (guard a2)
g1 sendorderLabTest_4 (guard a1)
h1 receiveorderLabTest_4 (guard a1)
i1 sendprescribeDrug_5 (guard a1)
j1 receiveprescribeDrug_5 (guard a1)
k1 senddischargeNote_7
l1 receivedischargeNote_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1
    └── e2 → f2 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> k1 -> l1

model_026 LabSampleTesting · fail

Guard

a1	retestNo < 2

```
(root)
└── a1  retestNo < 2
```

เส้นทาง
a1

Event

a1 sendcollectSample_1
b1 receivecollectSample_1
c1 sendtransportSample_2
d1 receivetransportSample_2
e1 sendsampleReceived_3
f1 receivesampleReceived_3
g1 sendrequestRetest_4 (guard a1)
h1 receiverequestRetest_4 (guard a1)
i1 sendretestResult_5 (guard a1)
j1 receiveretestResult_5 (guard a1)
k1 sendfinalReport_6
k2 checkloop_requestRetest_4_loop1 (guard a1)
l1 receivefinalReport_6
m1 sendnotifyClinic_7
n1 receivenotifyClinic_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1 → m1 → n1
    └── k2 ↺ g1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ g1

model_029 MedicationDispenser · fail

Guard

a1	taken = 1
a2	taken = 0

```
(root)
├── a1  taken = 1
└── a2  taken = 0
```

เส้นทาง
a1
a2

Event

a1 sendscheduleDose_1
b1 receivescheduleDose_1
c1 senddoseAlert_2
d1 receivedoseAlert_2
e1 sendconfirmIntake_3
f1 receiveconfirmIntake_3
g1 sendlogDose_4 (guard a1)
g2 sendescalateMissed_6 (guard a2)
h1 receivelogDose_4 (guard a1)
h2 receiveescalateMissed_6 (guard a2)
i1 sendlogAck_5 (guard a1)
j1 receivelogAck_5 (guard a1)
k1 sendrefillCheck_7
l1 receiverefillCheck_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    └── g2 → h2 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1

model_052 AmbulanceRouting · pass

Guard

a1	useCache = 1
b1	needAck = 1

```
(root)
└── a1  useCache = 1
    └── b1  needAck = 1
```

เส้นทาง

```
a1 -> b1
```

Event

a1 sendapproveEta_1
b1 receiveapproveEta_1
c1 sendackPatient_2
d1 receiveackPatient_2
e1 sendassignPatient_3_opt1 (guard a1)
f1 receiveassignPatient_3_opt1 (guard a1)
g1 sendnotifyRoute_4
h1 receivenotifyRoute_4
i1 sendreserveEta_5
j1 receivereserveEta_5
k1 sendrequestRoute_6_opt2 (guard b1)
l1 receiverequestRoute_6_opt2 (guard b1)
m1 sendcheckPatient_7
n1 receivecheckPatient_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
```

model_058 BloodDonationDrive · fail

Guard

a1	t_obs ≤ 8
b1	withExtra = 1
a2	t_obs > 8

```
(root)
├── a1  t_obs ≤ 8
│   └── b1  withExtra = 1
└── a2  t_obs > 8
```

เส้นทาง

```
a1 -> b1
a2
```

Event

a1 sendackScreening_1 (guard a1 b1)
b1 receiveackScreening_1
c1 sendlogSlot_2 (guard a1 b1)
d1 receivelogSlot_2
e1 sendsyncSlot_3 (guard a1)
e2 sendlogDonation_6 (guard a2)
f1 receivesyncSlot_3 (guard a1)
f2 receivelogDonation_6 (guard a2)
g1 sendnotifyDonation_4_opt1
h1 receivenotifyDonation_4_opt1 (guard a1 b1)
i1 sendvalidateScreening_5 (guard a1)
j1 receivevalidateScreening_5 (guard a1)
k1 sendfetchScreening_7
l1 receivefetchScreening_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1
    └── e2 → f2 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> k1 -> l1
```

model_082 DentalAppointment · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendqueryAppointment_1
b1 receivequeryAppointment_1
c1 sendassignTreatment_2
d1 receiveassignTreatment_2
e1 sendupdateTreatment_3
f1 receiveupdateTreatment_3
g1 sendqueryClaim_4
h1 receivequeryClaim_4
i1 sendconfirmAppointment_5
j1 receiveconfirmAppointment_5
k1 sendreserveClaim_6
l1 receivereserveClaim_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
```

model_101 GlucoseMeterSync · pass

Guard

a1	needAck = 1

```
(root)
└── a1  needAck = 1
```

เส้นทาง
a1

Event

a1 sendconfirmInsulin_1
b1 receiveconfirmInsulin_1
c1 sendsyncReading_2
d1 receivesyncReading_2
e1 sendupdateAlert_3
f1 receiveupdateAlert_3
g1 sendqueryReading_4
h1 receivequeryReading_4
i1 sendnotifyAlert_5
j1 receivenotifyAlert_5
k1 sendreleaseAlert_6
l1 receivereleaseAlert_6
m1 sendqueryAlert_7
n1 receivequeryAlert_7
o1 sendreleaseInsulin_8_opt1 (guard a1)
p1 receivereleaseInsulin_8_opt1 (guard a1)
q1 sendrequestAlert_9_opt1 (guard a1)
r1 receiverequestAlert_9_opt1 (guard a1)
s1 sendreportInsulin_10_opt1 (guard a1)
t1 receivereportInsulin_10_opt1 (guard a1)
u1 sendupdateInsulin_11_opt1 (guard a1)
v1 receiveupdateInsulin_11_opt1 (guard a1)
w1 sendsendAlert_12_opt1 (guard a1)
x1 receivesendAlert_12_opt1 (guard a1)
y1 sendapproveAlert_13_opt1 (guard a1)
z1 receiveapproveAlert_13_opt1 (guard a1)
aa1 sendreserveAlert_14
ab1 receivereserveAlert_14

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1 → u1 → v1 → w1 → x1 →
      y1 → z1 → aa1 → ab1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1 -> y1 -> z1 -> aa1 -> ab1

model_106 HomeOxygenConcentrator · pass

Guard

a1	withExtra = 1
b1	needAck = 1

```
(root)
└── a1  withExtra = 1
    └── b1  needAck = 1
```

เส้นทาง
a1 -> b1

Event

a1 sendreserveFlow_1
b1 receivereserveFlow_1
c1 sendupdateAlert_2
d1 receiveupdateAlert_2
e1 sendreserveAlert_3_opt1 (guard a1)
f1 receivereserveAlert_3_opt1 (guard a1)
g1 sendnotifyAlert_4
h1 receivenotifyAlert_4
i1 sendsyncAlert_5
j1 receivesyncAlert_5
k1 sendrequestFlow_6_opt2 (guard b1)
l1 receiverequestFlow_6_opt2 (guard b1)
m1 sendlogFlow_7_opt2 (guard b1)
n1 receivelogFlow_7_opt2 (guard b1)
o1 sendschedulePurity_8
p1 receiveschedulePurity_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1

model_108 HospitalBedAllocation · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendsendBed_1
b1 receivesendBed_1
c1 sendassignBed_2
d1 receiveassignBed_2
e1 sendsyncTransfer_3
f1 receivesyncTransfer_3
g1 sendscheduleBed_4
h1 receivescheduleBed_4
i1 sendlogTransfer_5
j1 receivelogTransfer_5
k1 sendreleasePatient_6
l1 receivereleasePatient_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1
```

model_109 HospitalLabRequest · fail

Guard

a1	cycleNo < 8
b1	cycleNo mod 3 = 0
c1	attemptNo < 10
b2	cycleNo mod 3 ≠ 0

```
(root)
└── a1  cycleNo < 8
    ├── b1  cycleNo mod 3 = 0
    │   └── c1  attemptNo < 10
    └── b2  cycleNo mod 3 ≠ 0
```

เส้นทาง

```
a1 -> b1 -> c1
a1 -> b2
```

Event

a1 sendcheckOrder_1
b1 receivecheckOrder_1
c1 sendrequestResult_2
d1 receiverequestResult_2
e1 sendconfirmOrder_3 (guard a1)
f1 receiveconfirmOrder_3 (guard a1)
g1 sendscheduleOrder_4 (guard a1 b1)
g2 sendreportSample_9 (guard a1 b2)
h1 receivescheduleOrder_4 (guard a1 b1)
h2 receivereportSample_9 (guard a1 b2)
i1 sendackSample_5 (guard a1 b1 c1)
j1 receiveackSample_5 (guard a1 b1 c1)
k1 sendapproveSample_6 (guard a1 b1 c1)
l1 receiveapproveSample_6 (guard a1 b1 c1)
m1 sendissueResult_7 (guard a1 b1 c1)
n1 receiveissueResult_7 (guard a1 b1 c1)
o1 sendreportResult_8 (guard a1 b1)
o2 checkloop_ackSample_5_loop2 (guard a1 b1 c1)
p1 receivereportResult_8 (guard a1 b1)
q1 sendrequestSample_10 (guard a1)
r1 receiverequestSample_10 (guard a1)
s1 sendreserveSample_11
s2 checkloop_confirmOrder_3_loop1 (guard a1)
t1 receivereserveSample_11

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1
    │   ├── o1 → p1 → q1 → r1
    │   │   ├── s1 → t1
    │   │   └── s2 ↺ e1
    │   └── o2 ↺ i1
    └── g2 → h2 → q1 → r1
        ├── s1 → t1
        └── s2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1 -> o2 ↺ i1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> q1 -> r1 -> s1 -
> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> q1 -> r1 -> s2 ↺
e1
```

model_110 HospitalPharmacyRobot · pass

Guard

a1	t_obs ≤ 13
a2	t_obs > 13

```
(root)
├── a1  t_obs ≤ 13
└── a2  t_obs > 13
```

เส้นทาง

a1
a2

Event

a1 sendnotifyDrawer_1
b1 receivenotifyDrawer_1
c1 sendconfirmOrder_2
d1 receiveconfirmOrder_2
e1 sendreportOrder_3 (guard a1)
e2 sendcheckOrder_5 (guard a2)
f1 receivereportOrder_3 (guard a1)
f2 receivecheckOrder_5 (guard a2)
g1 sendackDrawer_4 (guard a1)
g2 sendassignDrawer_6 (guard a2)
h1 receiveackDrawer_4 (guard a1)
h2 receiveassignDrawer_6 (guard a2)
i1 sendreleaseDose_7
j1 receivereleaseDose_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    └── e2 → f2 → g2 → h2 → i1 → j1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g2 -> h2 -> i1 -> j1

model_125 MedicalImagingPACS · fail

Guard

a1	roundNo < 6
b1	roundNo mod 3 = 0
b2	roundNo mod 3 ≠ 0

```
(root)
└── a1  roundNo < 6
    ├── b1  roundNo mod 3 = 0
    └── b2  roundNo mod 3 ≠ 0
```

เส้นทาง

a1 -> b1
a1 -> b2

Event

a1 sendcheckImage_1
b1 receivecheckImage_1
c1 sendassignImage_2
d1 receiveassignImage_2
e1 sendassignReport_3 (guard a1)
f1 receiveassignReport_3 (guard a1)
g1 sendapproveReport_4 (guard a1 b1)
g2 sendrequestImage_5 (guard a1 b2)
h1 receiveapproveReport_4 (guard a1 b1)
h2 receiverequestImage_5 (guard a1 b2)
i1 sendvalidateImage_6 (guard a1)
j1 receivevalidateImage_6 (guard a1)
k1 sendsyncReport_7
k2 checkloop_assignReport_3_loop1 (guard a1)
l1 receivesyncReport_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1
    │   ├── k1 → l1
    │   └── k2 ↺ e1
    └── g2 → h2 → i1 → j1
        ├── k1 → l1
        └── k2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> i1 -> j1 -> k2 ↺ e1

model_126 MedicalTranscription · pass

Guard

a1	optIn = 1

```
(root)
└── a1  optIn = 1
```

เส้นทาง

a1

Event

a1 sendcheckTranscript_1
b1 receivecheckTranscript_1
c1 sendlogTranscript_2
d1 receivelogTranscript_2
e1 sendsendNote_3_opt1 (guard a1)
f1 receivesendNote_3_opt1 (guard a1)
g1 sendupdateDictation_4_opt1 (guard a1)
h1 receiveupdateDictation_4_opt1 (guard a1)
i1 sendnotifyDictation_5
j1 receivenotifyDictation_5
k1 sendconfirmDictation_6
l1 receiveconfirmDictation_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_139 PatientIntakeForm · fail

Guard

a1	attemptNo < 5
b1	attemptNo mod 2 = 0

```
(root)
└── a1  attemptNo < 5
    └── b1  attemptNo mod 2 = 0
```

เส้นทาง

a1 -> b1

Event

a1 sendreleaseForm_1
b1 receivereleaseForm_1
c1 sendcancelForm_2
d1 receivecancelForm_2
e1 sendconfirmForm_3 (guard a1)
f1 receiveconfirmForm_3 (guard a1)
g1 sendreportForm_4_opt1 (guard a1 b1)
h1 receivereportForm_4_opt1 (guard a1 b1)
i1 sendscheduleConsent_5 (guard a1)
j1 receivescheduleConsent_5 (guard a1)
k1 sendconfirmConsent_6 (guard a1)
l1 receiveconfirmConsent_6 (guard a1)
m1 sendissueForm_7 (guard a1)
n1 receiveissueForm_7 (guard a1)
o1 sendassignRecord_8
o2 checkloop_confirmForm_3_loop1 (guard a1)
p1 receiveassignRecord_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1
    └── o2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o2 ↺ e1

model_141 PharmacyDelivery · fail

Guard

a1	needAck = 1
b1	mode = GOLD
b2	mode = SILVER
b3	mode = BRONZE

```
(root)
└── a1  needAck = 1
    ├── b1  mode = GOLD
    ├── b2  mode = SILVER
    └── b3  mode = BRONZE
```

เส้นทาง

a1 -> b1
a1 -> b2
a1 -> b3

Event

a1 sendsyncSignature_1
b1 receivesyncSignature_1
c1 sendvalidateSignature_2
d1 receivevalidateSignature_2
e1 sendfetchRoute_3_opt1 (guard a1)
f1 receivefetchRoute_3_opt1 (guard a1)
g1 sendqueryRoute_4_opt1 (guard a1)
h1 receivequeryRoute_4_opt1 (guard a1)
i1 sendconfirmRoute_5
j1 receiveconfirmRoute_5
k1 sendissuePackage_6
l1 receiveissuePackage_6
m1 sendcancelSignature_7 (guard b1)
m2 sendqueryPackage_9 (guard b2)
m3 sendscheduleSignature_11 (guard b3)
n1 receivecancelSignature_7 (guard b1)
n2 receivequeryPackage_9 (guard b2)
n3 receivescheduleSignature_11 (guard b3)
o1 sendconfirmPackage_8 (guard b1)
o2 sendackSignature_10 (guard b2)
p1 receiveconfirmPackage_8 (guard b1)
p2 receiveackSignature_10 (guard b2)
q1 sendapproveRoute_12
r1 receiveapproveRoute_12

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1 → o1 → p1 → q1 → r1
    ├── m2 → n2 → o2 → p2 → q1 → r1
    └── m3 → n3 → q1 → r1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 -> n2 -> o2 -> p2 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m3 -> n3 -> q1 -> r1

model_142 PharmacyRefill · fail

Guard

a1	express = 1

```
(root)
└── a1  express = 1
```

เส้นทาง

a1

Event

a1 sendissuePrescription_1
b1 receiveissuePrescription_1
c1 sendcheckClaim_2
d1 receivecheckClaim_2
e1 sendschedulePrescription_3_opt1 (guard a1)
f1 receiveschedulePrescription_3_opt1 (guard a1)
g1 sendqueryRefill_4_opt1 (guard a1)
h1 receivequeryRefill_4_opt1 (guard a1)
i1 sendackRefill_5
j1 receiveackRefill_5
k1 sendreservePrescription_6
l1 receivereservePrescription_6
m1 sendcancelRefill_7
n1 receivecancelRefill_7
o1 sendvalidateClaim_8
p1 receivevalidateClaim_8
q1 sendconfirmPrescription_9
r1 receiveconfirmPrescription_9

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1

model_153 RemotePatientMonitor · fail

Guard

a1	passNo < 12

```
(root)
└── a1  passNo < 12
```

เส้นทาง
a1

Event

a1 sendsyncAlert_1
b1 receivesyncAlert_1
c1 sendcancelReport_2
d1 receivecancelReport_2
e1 sendapproveVital_3 (guard a1)
f1 receiveapproveVital_3 (guard a1)
g1 sendscheduleReport_4 (guard a1)
h1 receivescheduleReport_4 (guard a1)
i1 sendreleaseReport_5
i2 checkloop_approveVital_3_loop1 (guard a1)
j1 receivereleaseReport_5
k1 sendsendAlert_6
l1 receivesendAlert_6
m1 sendreportVital_7
n1 receivereportVital_7
o1 sendnotifyAlert_8
p1 receivenotifyAlert_8
q1 sendfetchVital_9
r1 receivefetchVital_9
s1 sendreportAlert_10
t1 receivereportAlert_10
u1 sendreportReport_11
v1 receivereportReport_11
w1 sendconfirmVital_12
x1 receiveconfirmVital_12
y1 sendassignAlert_13
z1 receiveassignAlert_13

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
    │     q1 → r1 → s1 → t1 → u1 → v1 → w1 → x1 →
    │     y1 → z1
    └── i2 ↺ e1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1 -> y1 -> z1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ e1

model_158 RuralTelehealthKiosk · fail

Guard

a1	passNo < 6
b1	passNo mod 4 = 0
b2	passNo mod 4 ≠ 0

```
(root)
└── a1  passNo < 6
    ├── b1  passNo mod 4 = 0
    └── b2  passNo mod 4 ≠ 0
```

เส้นทาง

a1 -> b1
a1 -> b2

Event

a1 sendreportVital_1
b1 receivereportVital_1
c1 sendvalidateVital_2
d1 receivevalidateVital_2
e1 sendfetchPrescription_3 (guard a1)
f1 receivefetchPrescription_3 (guard a1)
g1 sendschedulePrescription_4 (guard a1 b1)
g2 sendreportPrescription_11 (guard a1 b2)
h1 receiveschedulePrescription_4 (guard a1 b1)
h2 receivereportPrescription_11 (guard a1 b2)
i1 sendcancelPrescription_5 (guard a1 b1)
j1 receivecancelPrescription_5 (guard a1 b1)
k1 sendnotifyConsult_6 (guard a1 b1)
l1 receivenotifyConsult_6 (guard a1 b1)
m1 sendsendVital_7 (guard a1 b1)
n1 receivesendVital_7 (guard a1 b1)
o1 sendfetchVital_8 (guard a1 b1)
p1 receivefetchVital_8 (guard a1 b1)
q1 sendackConsult_9 (guard a1 b1)
r1 receiveackConsult_9 (guard a1 b1)
s1 sendupdateConsult_10 (guard a1 b1)
t1 receiveupdateConsult_10 (guard a1 b1)
u1 sendconfirmVital_12 (guard a1)
v1 receiveconfirmVital_12 (guard a1)
w1 sendreleaseVital_13
w2 checkloop_fetchPrescription_3_loop1 (guard a1)
x1 receivereleaseVital_13

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1 →
    │     o1 → p1 → q1 → r1 → s1 → t1 → u1 → v1
    │   ├── w1 → x1
    │   └── w2 ↺ e1
    └── g2 → h2 → u1 → v1
        ├── w1 → x1
        └── w2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> u1 -> v1 -> w1 -> x1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> u1 -> v1 -> w2 ↺ e1

model_172 SmartMirrorFitness · fail

Guard

a1	attemptNo < 6
b1	attemptNo mod 2 = 0

```
(root)
└── a1  attemptNo < 6
    └── b1  attemptNo mod 2 = 0
```

เส้นทาง

```
a1 -> b1
```

Event

a1 sendreleaseWorkout_1
b1 receivereleaseWorkout_1
c1 sendreportRep_2
d1 receivereportRep_2
e1 sendapprovePlan_3 (guard a1)
f1 receiveapprovePlan_3 (guard a1)
g1 sendconfirmWorkout_4_opt1 (guard a1 b1)
h1 receiveconfirmWorkout_4_opt1 (guard a1 b1)
i1 sendlogWorkout_5_opt1 (guard a1 b1)
j1 receivelogWorkout_5_opt1 (guard a1 b1)
k1 sendissueWorkout_6_opt1 (guard a1 b1)
l1 receiveissueWorkout_6_opt1 (guard a1 b1)
m1 sendcancelRep_7_opt1 (guard a1 b1)
n1 receivecancelRep_7_opt1 (guard a1 b1)
o1 sendackRep_8_opt1 (guard a1 b1)
p1 receiveackRep_8_opt1 (guard a1 b1)
q1 sendreservePlan_9_opt1 (guard a1 b1)
r1 receivereservePlan_9_opt1 (guard a1 b1)
s1 sendsendWorkout_10_opt1 (guard a1 b1)
t1 receivesendWorkout_10_opt1 (guard a1 b1)
u1 sendscheduleWorkout_11 (guard a1)
v1 receivescheduleWorkout_11 (guard a1)
w1 sendsyncPlan_12
w2 checkloop_approvePlan_3_loop1 (guard a1)
x1 receivesyncPlan_12

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1 → u1 → v1
    ├── w1 → x1
    └── w2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w2 ↺ e1
```

model_173 SmartPillBox · pass

Guard

a1	needAck = 1
b1	mode = GOLD
b2	mode = SILVER
b3	mode = BRONZE

```
(root)
└── a1  needAck = 1
    ├── b1  mode = GOLD
    ├── b2  mode = SILVER
    └── b3  mode = BRONZE
```

เส้นทาง

```
a1 -> b1
a1 -> b2
a1 -> b3
```

Event

a1 sendsyncRefill_1
b1 receivesyncRefill_1
c1 sendvalidateRefill_2
d1 receivevalidateRefill_2
e1 sendfetchReminder_3_opt1 (guard a1)
f1 receivefetchReminder_3_opt1 (guard a1)
g1 sendqueryReminder_4_opt1 (guard a1)
h1 receivequeryReminder_4_opt1 (guard a1)
i1 sendconfirmReminder_5
j1 receiveconfirmReminder_5
k1 sendissueDose_6
l1 receiveissueDose_6
m1 sendcancelRefill_7 (guard b1)
m2 sendqueryDose_9 (guard b2)
m3 sendscheduleRefill_11 (guard b3)
n1 receivecancelRefill_7 (guard b1)
n2 receivequeryDose_9 (guard b2)
n3 receivescheduleRefill_11 (guard b3)
o1 sendconfirmDose_8 (guard b1)
o2 sendackRefill_10 (guard b2)
p1 receiveconfirmDose_8 (guard b1)
p2 receiveackRefill_10 (guard b2)
q1 sendapproveReminder_12
r1 receiveapproveReminder_12

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1 → o1 → p1 → q1 → r1
    ├── m2 → n2 → o2 → p2 → q1 → r1
    └── m3 → n3 → q1 → r1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 -> n2 -> o2 -> p2 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m3 -> n3 -> q1 -> r1
```

model_186 TelemedicineVisit · fail

Guard

a1	stepNo < 2
b1	stepNo mod 2 = 0
b2	stepNo mod 2 ≠ 0

```
(root)
└── a1  stepNo < 2
    ├── b1  stepNo mod 2 = 0
    └── b2  stepNo mod 2 ≠ 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

a1 sendissuePrescription_1
b1 receiveissuePrescription_1
c1 sendconfirmPrescription_2
d1 receiveconfirmPrescription_2
e1 sendupdateVisit_3 (guard a1)
f1 receiveupdateVisit_3 (guard a1)
g1 sendschedulePrescription_4 (guard a1 b1)
g2 sendrequestVisit_5 (guard a1 b2)
h1 receiveschedulePrescription_4 (guard a1 b1)
h2 receiverequestVisit_5 (guard a1 b2)
i1 sendsyncPrescription_6 (guard a1)
j1 receivesyncPrescription_6 (guard a1)
k1 sendsendNote_7
k2 checkloop_updateVisit_3_loop1 (guard a1)
l1 receivesendNote_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1
    │   ├── k1 → l1
    │   └── k2 ↺ e1
    └── g2 → h2 → i1 → j1
        ├── k1 → l1
        └── k2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> i1 -> j1 -> k2 ↺ e1
```

model_187 ThermalCameraScreening · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendcancelScan_1
b1 receivecancelScan_1
c1 sendackScan_2
d1 receiveackScan_2
e1 sendassignAlert_3
f1 receiveassignAlert_3
g1 sendconfirmAlert_4
h1 receiveconfirmAlert_4
i1 sendackAlert_5
j1 receiveackAlert_5
k1 sendqueryFever_6
l1 receivequeryFever_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_192 VaccineColdBox · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendreleaseAlarm_1
b1 receivereleaseAlarm_1
c1 sendackVaccine_2
d1 receiveackVaccine_2
e1 sendsyncReading_3
f1 receivesyncReading_3
g1 sendsyncAlarm_4
h1 receivesyncAlarm_4
i1 sendapproveAlarm_5
j1 receiveapproveAlarm_5
k1 sendsendVaccine_6
l1 receivesendVaccine_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_193 VeterinaryClinic · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendreleaseAppointment_1
b1 receivereleaseAppointment_1
c1 sendcancelAppointment_2
d1 receivecancelAppointment_2
e1 sendreportInvoice_3
f1 receivereportInvoice_3
g1 sendsendAppointment_4
h1 receivesendAppointment_4
i1 sendapproveAppointment_5
j1 receiveapproveAppointment_5
k1 sendvalidateInvoice_6
l1 receivevalidateInvoice_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

การเดินทางและยานพาหนะ · Transportation & Mobility (26)

model_002 AirportBaggage · fail

Guard

a1	bagCleared = 1
a2	bagCleared = 0

```
(root)
├── a1  bagCleared = 1
└── a2  bagCleared = 0
```

เส้นทาง

a1
a2

Event

a1 sendtagBag_1
b1 receivetagBag_1
c1 sendrouteBag_2
d1 receiverouteBag_2
e1 sendsorted_3
f1 receivesorted_3
g1 sendloadBag_4 (guard a1)
g2 senddivertBag_6 (guard a2)
h1 receiveloadBag_4 (guard a1)
h2 receivedivertBag_6 (guard a2)
i1 sendloadedAck_5 (guard a1)
j1 receiveloadedAck_5 (guard a1)
k1 sendmanifestUpdate_7
l1 receivemanifestUpdate_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    └── g2 → h2 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1

model_007 CarRentalBooking · pass

Guard

a1	searchCount < 3
b1	insurance = 1
c1	t_pay ≤ 10
c2	t_pay > 10

```
(root)
└── a1  searchCount < 3
    └── b1  insurance = 1
        ├── c1  t_pay ≤ 10
        └── c2  t_pay > 10
```

เส้นทาง

a1 -> b1 -> c1
a1 -> b1 -> c2

Event

a1 sendopenApp_1
b1 receiveopenApp_1
c1 sendsearchCar_2 (guard a1)
d1 receivesearchCar_2 (guard a1)
e1 sendcarList_3 (guard a1)
f1 receivecarList_3 (guard a1)
g1 sendaddInsurance_4_opt1 (guard b1)
g2 checkloop_searchCar_2_loop1 (guard a1)
h1 receiveaddInsurance_4_opt1 (guard b1)
i1 sendrequestPayment_5
j1 receiverequestPayment_5
k1 sendpaymentResult_6
l1 receivepaymentResult_6
m1 sendconfirmBooking_7 (guard c1)
m2 sendcancelBooking_8 (guard c2)
n1 receiveconfirmBooking_7 (guard c1)
n2 receivecancelBooking_8 (guard c2)

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    │   ├── m1 → n1
    │   └── m2 → n2
    └── g2 ↺ c1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 -> n2
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 ↺ c1

model_019 FerryTicketGate · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendtapTicket_1
b1 receivetapTicket_1
c1 sendvalidate_2
d1 receivevalidate_2
e1 sendopenBarrier_3
f1 receiveopenBarrier_3
g1 sendpassThrough_4
h1 receivepassThrough_4

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1

model_040 SmartParking · pass

Guard

a1	slotFree = 1

```
(root)
└── a1  slotFree = 1
```

เส้นทาง
a1

Event

a1 sendrequestEntry_1
b1 receiverequestEntry_1
c1 sendcheckSlot_2
d1 receivecheckSlot_2
e1 sendslotStatus_3
f1 receiveslotStatus_3
g1 sendopenGate_4_opt1 (guard a1)
h1 receiveopenGate_4_opt1 (guard a1)
i1 sendissueTicket_5_opt1 (guard a1)
j1 receiveissueTicket_5_opt1 (guard a1)
k1 sendlogEntry_6
l1 receivelogEntry_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_042 TaxiDispatch · fail

Guard

a1	shareRide = 1

```
(root)
└── a1  shareRide = 1
```

เส้นทาง
a1

Event

a1 sendrequestRide_1
b1 receiverequestRide_1
c1 sendfindDriver_2
d1 receivefindDriver_2
e1 senddriverAccept_3
f1 receivedriverAccept_3
g1 sendrideConfirmed_4
h1 receiverideConfirmed_4
i1 sendaddPassenger_5_opt1 (guard a1)
j1 receiveaddPassenger_5_opt1 (guard a1)
k1 sendtripComplete_6
l1 receivetripComplete_6
m1 sendchargeRider_7
n1 receivechargeRider_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1

model_045 TrafficLightController · fail

Guard

a1	cycleCount < 10

```
(root)
└── a1  cycleCount < 10
```

เส้นทาง
a1

Event

a1 sendsetGreen_1
b1 receivesetGreen_1
c1 sendgreenAck_2
d1 receivegreenAck_2
e1 sendsetYellow_3 (guard a1)
f1 receivesetYellow_3 (guard a1)
g1 sendsetRed_4 (guard a1)
h1 receivesetRed_4 (guard a1)
i1 sendphaseDone_5 (guard a1)
j1 receivephaseDone_5 (guard a1)
k1 sendshutdown_6
k2 checkloop_setYellow_3_loop1 (guard a1)
l1 receiveshutdown_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1
    └── k2 ↺ e1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ e1

model_049 AirportSecurityLane · fail

Guard

a1	usage < 10
a2	usage ≥ 10

```
(root)
├── a1  usage < 10
└── a2  usage ≥ 10
```

เส้นทาง

a1
a2

Event

a1 sendcancelTray_1
b1 receivecancelTray_1
c1 sendvalidateTray_2
d1 receivevalidateTray_2
e1 sendrequestScan_3 (guard a1)
e2 sendreleaseScan_4 (guard a2)
f1 receiverequestScan_3 (guard a1)
f2 receivereleaseScan_4 (guard a2)
g1 sendreserveTray_6
h1 receivereserveTray_6
i1 sendupdateTray_5 (guard a2)
j1 receiveupdateTray_5 (guard a2)

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1
    └── e2 → f2 → i1 → j1 → g1 → h1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> i1 -> j1 -> g1 -> h1

model_050 AirportShuttle · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendsendRoute_1
b1 receivesendRoute_1
c1 sendnotifyPickup_2
d1 receivenotifyPickup_2
e1 sendupdateSeat_3
f1 receiveupdateSeat_3
g1 sendconfirmPickup_4
h1 receiveconfirmPickup_4
i1 sendscheduleRoute_5
j1 receivescheduleRoute_5
k1 sendfetchSeat_6
l1 receivefetchSeat_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_057 BikeSharing · pass

Guard

a1	t_obs ≤ 6
b1	useCache = 1
a2	t_obs > 6

```
(root)
├── a1  t_obs ≤ 6
│   └── b1  useCache = 1
└── a2  t_obs > 6
```

เส้นทาง

a1 -> b1
a2

Event

a1 sendfetchRide_1
b1 receivefetchRide_1
c1 sendconfirmBike_2
d1 receiveconfirmBike_2
e1 sendsyncFee_3 (guard a1)
e2 sendvalidateFee_7 (guard a2)
f1 receivesyncFee_3 (guard a1)
f2 receivevalidateFee_7 (guard a2)
g1 sendfetchBike_4_opt1 (guard a1 b1)
h1 receivefetchBike_4_opt1 (guard a1 b1)
i1 sendreserveBike_5_opt1 (guard a1 b1)
j1 receivereserveBike_5_opt1 (guard a1 b1)
k1 sendapproveBike_6 (guard a1)
l1 receiveapproveBike_6 (guard a1)
m1 sendnotifyFee_8
n1 receivenotifyFee_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1 →
    │     m1 → n1
    └── e2 → f2 → m1 → n1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> m1 -> n1

model_062 BusTicketBooking · fail

Guard

a1	retryNo < 3
b1	retryNo mod 3 = 0

```
(root)
└── a1  retryNo < 3
    └── b1  retryNo mod 3 = 0
```

เส้นทาง

```
a1 -> b1
```

Event

a1 sendvalidateSeat_1
b1 receivevalidateSeat_1
c1 sendlogTicket_2
d1 receivelogTicket_2
e1 sendsendSeat_3 (guard a1)
f1 receivesendSeat_3 (guard a1)
g1 sendreserveSeat_4_opt1 (guard a1 b1)
h1 receivereserveSeat_4_opt1 (guard a1 b1)
i1 sendrequestRoute_5_opt1 (guard a1 b1)
j1 receiverequestRoute_5_opt1 (guard a1 b1)
k1 sendlogSeat_6 (guard a1)
l1 receivelogSeat_6 (guard a1)
m1 sendscheduleTicket_7 (guard a1)
n1 receivescheduleTicket_7 (guard a1)
o1 sendapproveSeat_8
o2 checkloop_sendSeat_3_loop1 (guard a1)
p1 receiveapproveSeat_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1
    └── o2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1 -> o2 ↺ e1
```

model_064 CampusShuttleBus · pass

Guard

a1	optIn = 1
b1	load < 10
b2	load ≥ 10

```
(root)
└── a1  optIn = 1
    ├── b1  load < 10
    └── b2  load ≥ 10
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

a1 sendscheduleArrival_1
b1 receivescheduleArrival_1
c1 sendscheduleSeat_2
d1 receivescheduleSeat_2
e1 sendvalidateArrival_3
f1 receivevalidateArrival_3
g1 sendupdateStop_4
h1 receiveupdateStop_4
i1 sendvalidateStop_5
j1 receivevalidateStop_5
k1 sendreserveSeat_6
l1 receivereserveSeat_6
m1 sendupdateArrival_7_opt1 (guard a1)
n1 receiveupdateArrival_7_opt1 (guard a1)
o1 sendapproveArrival_8_opt1 (guard a1 b1)
o2 sendsendStop_10_opt1 (guard a1 b2)
p1 receiveapproveArrival_8_opt1 (guard a1 b1)
p2 receivesendStop_10_opt1 (guard a1 b2)
q1 sendquerySeat_9_opt1 (guard a1 b1)
r1 receivequerySeat_9_opt1 (guard a1 b1)
s1 sendissueStop_11_opt1 (guard a1)
t1 receiveissueStop_11_opt1 (guard a1)
u1 sendissueArrival_12
v1 receiveissueArrival_12

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1 → q1 → r1 → s1 → t1 → u1 → v1
    └── o2 → p2 → s1 → t1 → u1 → v1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1 -> o2 -> p2 -> s1 -> t1 -> u1 -> v1
```

model_068 CarRentalReturn · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendapproveDamage_1
b1 receiveapproveDamage_1
c1 sendfetchCar_2
d1 receivefetchCar_2
e1 sendvalidateDamage_3
f1 receivevalidateDamage_3
g1 sendcancelCar_4
h1 receivecancelCar_4
i1 sendfetchDamage_5
j1 receivefetchDamage_5
k1 sendqueryInvoice_6
l1 receivequeryInvoice_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1
```

model_087 DroneTrafficControl · pass

Guard

a1	approved = 1
b1	express = 1
a2	approved = 0

```
(root)
├── a1  approved = 1
│   └── b1  express = 1
└── a2  approved = 0
```

เส้นทาง

```
a1 -> b1
a2
```

Event

a1 sendreportCorridor_1
b1 receivereportCorridor_1
c1 sendsyncFlight_2
d1 receivesyncFlight_2
e1 sendackFlight_3 (guard a1)
e2 sendcancelCorridor_7 (guard a2)
f1 receiveackFlight_3 (guard a1)
f2 receivecancelCorridor_7 (guard a2)
g1 sendupdateCorridor_4_opt1 (guard a1 b1)
h1 receiveupdateCorridor_4_opt1 (guard a1 b1)
i1 sendqueryCorridor_5_opt1 (guard a1 b1)
j1 receivequeryCorridor_5_opt1 (guard a1 b1)
k1 sendconfirmFlight_6 (guard a1)
l1 receiveconfirmFlight_6 (guard a1)
m1 sendconfirmCorridor_8
n1 receiveconfirmCorridor_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1 →
    │     m1 → n1
    └── e2 → f2 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> m1 -> n1
```

model_095 FlightCheckIn · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendapproveSeat_1
b1 receiveapproveSeat_1
c1 sendcancelSeat_2
d1 receivecancelSeat_2
e1 sendfetchBoarding_3
f1 receivefetchBoarding_3
g1 sendcancelBag_4
h1 receivecancelBag_4
i1 sendassignSeat_5
j1 receiveassignSeat_5
k1 sendrequestBoarding_6
l1 receiverequestBoarding_6
m1 sendackBag_7
n1 receiveackBag_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1
```

model_096 FlightCrewScheduling · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendcancelDuty_1
b1 receivecancelDuty_1
c1 sendackDuty_2
d1 receiveackDuty_2
e1 sendassignSwap_3
f1 receiveassignSwap_3
g1 sendconfirmSwap_4
h1 receiveconfirmSwap_4
i1 sendackSwap_5
j1 receiveackSwap_5
k1 sendqueryRoster_6
l1 receivequeryRoster_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1
```

model_099 GarageGateControl · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendsendTicket_1
b1 receivesendTicket_1
c1 sendassignTicket_2
d1 receiveassignTicket_2
e1 sendreserveTicket_3
f1 receivereserveTicket_3
g1 sendreserveFee_4
h1 receivereserveFee_4
i1 sendrequestSlot_5
j1 receiverequestSlot_5
k1 sendassignFee_6
l1 receiveassignFee_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1
```

model_137 ParkingPermitPortal · pass

Guard

a1	retryNo < 9

```
(root)
└── a1  retryNo < 9
```

เส้นทาง

```
a1
```

Event

a1 sendfetchPermit_1
b1 receivefetchPermit_1
c1 sendapprovePlate_2
d1 receiveapprovePlate_2
e1 sendqueryFine_3 (guard a1)
f1 receivequeryFine_3 (guard a1)
g1 sendconfirmPlate_4 (guard a1)
h1 receiveconfirmPlate_4 (guard a1)
i1 sendfetchPlate_5 (guard a1)
j1 receivefetchPlate_5 (guard a1)
k1 sendreservePermit_6
k2 checkloop_queryFine_3_loop1 (guard a1)
l1 receivereservePermit_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1
    └── k2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺
e1
```

model_150 RailFareGate · pass

Guard

a1	needAck = 1
b1	withExtra = 1

```
(root)
└── a1  needAck = 1
    └── b1  withExtra = 1
```

เส้นทาง
a1 -> b1

Event

a1 sendqueryBalance_1 (guard b1)
b1 receivequeryBalance_1
c1 sendassignBalance_2
d1 receiveassignBalance_2
e1 sendcheckFare_3_opt1 (guard a1)
f1 receivecheckFare_3_opt1 (guard a1)
g1 sendvalidateCard_4_opt1 (guard a1)
h1 receivevalidateCard_4_opt1 (guard a1)
i1 sendcheckCard_5
j1 receivecheckCard_5
k1 sendqueryFare_6
l1 receivequeryFare_6
m1 sendvalidateFare_7_opt2 (guard b1)
n1 receivevalidateFare_7_opt2
o1 sendrequestCard_8_opt2 (guard b1)
p1 receiverequestCard_8_opt2 (guard b1)
q1 sendreportBalance_9
r1 receivereportBalance_9
s1 sendreleaseFare_10
t1 receivereleaseFare_10
u1 sendissueFare_11
v1 receiveissueFare_11
w1 sendlogBalance_12
x1 receivelogBalance_12
y1 sendvalidateBalance_13
z1 receivevalidateBalance_13
aa1 sendassignFare_14
ab1 receiveassignFare_14
ac1 sendcancelFare_15
ad1 receivecancelFare_15

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1 → u1 → v1 → w1 → x1 →
      y1 → z1 → aa1 → ab1 → ac1 → ad1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1 -> y1 -> z1 -> aa1 -> ab1 -> ac1 -> ad1

model_154 RentalScooter · pass

Guard

a1	express = 1

```
(root)
└── a1  express = 1
```

เส้นทาง
a1

Event

a1 sendfetchBattery_1 (guard a1)
b1 receivefetchBattery_1
c1 sendrequestBattery_2
d1 receiverequestBattery_2
e1 sendupdateUnlock_3_opt1 (guard a1)
f1 receiveupdateUnlock_3_opt1
g1 sendissueTrip_4
h1 receiveissueTrip_4
i1 sendscheduleTrip_5
j1 receivescheduleTrip_5
k1 sendfetchTrip_6
l1 receivefetchTrip_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_156 RideHailingSurge · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendnotifyFare_1
b1 receivenotifyFare_1
c1 sendcancelSurge_2
d1 receivecancelSurge_2
e1 sendreleaseTrip_3
f1 receivereleaseTrip_3
g1 sendrequestTrip_4
h1 receiverequestTrip_4
i1 sendlogSurge_5
j1 receivelogSurge_5
k1 sendfetchFare_6
l1 receivefetchFare_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_160 SchoolBusTracking · pass

Guard

a1	t_obs ≤ 2
a2	t_obs > 2

```
(root)
├── a1  t_obs ≤ 2
└── a2  t_obs > 2
```

เส้นทาง
a1
a2

Event

a1 sendupdateArrival_1
b1 receiveupdateArrival_1
c1 sendconfirmStop_2
d1 receiveconfirmStop_2
e1 sendreportBus_3 (guard a1)
e2 sendsyncBus_5 (guard a2)
f1 receivereportBus_3 (guard a1)
f2 receivesyncBus_5 (guard a2)
g1 sendcancelStop_4 (guard a1)
h1 receivecancelStop_4 (guard a1)
i1 sendcheckBus_6
j1 receivecheckBus_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    └── e2 → f2 → i1 → j1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> i1 -> j1

model_163 ShipLockOperation · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendqueryLock_1
b1 receivequeryLock_1
c1 sendscheduleLock_2
d1 receivescheduleLock_2
e1 sendreleaseLevel_3
f1 receivereleaseLevel_3
g1 sendapproveLevel_4
h1 receiveapproveLevel_4
i1 sendconfirmLock_5
j1 receiveconfirmLock_5
k1 sendassignLevel_6
l1 receiveassignLevel_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_183 StreetParkingMeter · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendreserveFee_1
b1 receivereserveFee_1
c1 sendlogFee_2
d1 receivelogFee_2
e1 sendreportFee_3
f1 receivereportFee_3
g1 sendreserveSession_4
h1 receivereserveSession_4
i1 sendqueryFine_5
j1 receivequeryFine_5
k1 sendissueSession_6
l1 receiveissueSession_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_188 TollPlazaETC · pass

Guard

a1	withExtra = 1

```
(root)
└── a1  withExtra = 1
```

เส้นทาง
a1

Event

a1 sendrequestPlate_1
b1 receiverequestPlate_1
c1 sendschedulePlate_2
d1 receiveschedulePlate_2
e1 sendvalidatePlate_3_opt1 (guard a1)
f1 receivevalidatePlate_3_opt1 (guard a1)
g1 sendvalidateToll_4_opt1 (guard a1)
h1 receivevalidateToll_4_opt1 (guard a1)
i1 sendqueryAccount_5
j1 receivequeryAccount_5
k1 sendreservePlate_6
l1 receivereservePlate_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_190 TrafficSignalPriority · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendqueryPriority_1
b1 receivequeryPriority_1
c1 sendassignGreen_2
d1 receiveassignGreen_2
e1 sendupdateGreen_3
f1 receiveupdateGreen_3
g1 sendqueryLane_4
h1 receivequeryLane_4
i1 sendconfirmPriority_5
j1 receiveconfirmPriority_5
k1 sendreserveLane_6
l1 receivereserveLane_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_191 TrainSeatReservation · fail

Guard

a1	passNo < 11
b1	passNo mod 2 = 0
b2	passNo mod 2 ≠ 0

```
(root)
└── a1  passNo < 11
    ├── b1  passNo mod 2 = 0
    └── b2  passNo mod 2 ≠ 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

a1 sendqueryTicket_1
b1 receivequeryTicket_1
c1 sendassignCoach_2
d1 receiveassignCoach_2
e1 sendscheduleCoach_3 (guard a1)
f1 receivescheduleCoach_3 (guard a1)
g1 sendconfirmSeat_4 (guard a1 b1)
g2 sendapproveTicket_8 (guard a1 b2)
h1 receiveconfirmSeat_4 (guard a1 b1)
h2 receiveapproveTicket_8 (guard a1 b2)
i1 sendreserveTicket_5 (guard a1 b1)
j1 receivereserveTicket_5 (guard a1 b1)
k1 sendissueTicket_6 (guard a1 b1)
l1 receiveissueTicket_6 (guard a1 b1)
m1 sendrequestCoach_7 (guard a1 b1)
n1 receiverequestCoach_7 (guard a1 b1)
o1 sendcheckTicket_9 (guard a1)
p1 receivecheckTicket_9 (guard a1)
q1 sendcancelTicket_10 (guard a1)
r1 receivecancelTicket_10 (guard a1)
s1 sendreportCoach_11 (guard a1)
t1 receivereportCoach_11 (guard a1)
u1 sendissueCoach_12
u2 checkloop_scheduleCoach_3_loop1 (guard a1)
v1 receiveissueCoach_12

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1 →
    │     o1 → p1 → q1 → r1 → s1 → t1
    │   ├── u1 → v1
    │   └── u2 ↺ e1
    └── g2 → h2 → o1 → p1 → q1 → r1 → s1 → t1
        ├── u1 → v1
        └── u2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> o1 -> p1 -> q1 -
> r1 -> s1 -> t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> o1 -> p1 -> q1 -
> r1 -> s1 -> t1 -> u2 ↺ e1
```

โลจิสติกส์และคลังสินค้า · Logistics & Supply Chain (14)

model_011 ColdChainMonitor · fail

Guard

a1	tempBreach = 1
a2	tempBreach = 0

```
(root)
├── a1  tempBreach = 1
└── a2  tempBreach = 0
```

เส้นทาง

```
a1
a2
```

Event

a1 sendlogTemperature_1
b1 receivelogTemperature_1
c1 senduploadLog_2
d1 receiveuploadLog_2
e1 senduploadAck_3
f1 receiveuploadAck_3
g1 sendraiseAlert_4 (guard a1)
g2 senddailySummary_6 (guard a2)
h1 receiveraiseAlert_4 (guard a1)
h2 receivedailySummary_6 (guard a2)
i1 sendalertAck_5 (guard a1)
j1 receivealertAck_5 (guard a1)
k1 sendcloseTrip_7
l1 receivecloseTrip_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    └── g2 → h2 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1
```

model_014 DroneDelivery · pass

Guard

a1	t_deliver ≤ 20
a2	t_deliver > 20

```
(root)
├── a1  t_deliver ≤ 20
└── a2  t_deliver > 20
```

เส้นทาง

a1
a2

Event

a1 sendplaceOrder_1
b1 receiveplaceOrder_1
c1 sendprepareParcel_2
d1 receiveprepareParcel_2
e1 sendassignDrone_3
f1 receiveassignDrone_3
g1 sendpickupParcel_4
h1 receivepickupParcel_4
i1 senddeliverParcel_5
j1 receivedeliverParcel_5
k1 sendconfirmDelivery_6 (guard a1)
k2 sendlateCompensation_7 (guard a2)
l1 receiveconfirmDelivery_6 (guard a1)
l2 receivelateCompensation_7 (guard a2)
m1 sendreturnToBase_8
n1 receivereturnToBase_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1 → m1 → n1
    └── k2 → l2 → m1 → n1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 -> l2 -> m1 -> n1

model_032 PackageTracking · pass

Guard

a1	customsCheck = 1
b1	delivered = YES
b2	delivered = NO

```
(root)
└── a1  customsCheck = 1
    ├── b1  delivered = YES
    └── b2  delivered = NO
```

เส้นทาง

a1 -> b1
a1 -> b2

Event

a1 sendcreateShipment_1
b1 receivecreateShipment_1
c1 sendpickupConfirm_2
d1 receivepickupConfirm_2
e1 sendarriveHub_3
f1 receivearriveHub_3
g1 sendscanIn_4
h1 receivescanIn_4
i1 sendcustomsHold_5_opt1 (guard a1)
j1 receivecustomsHold_5_opt1 (guard a1)
k1 senddispatchOut_6
l1 receivedispatchOut_6
m1 senddeliver_7 (guard b1)
m2 sendleaveNotice_9 (guard b2)
n1 receivedeliver_7 (guard b1)
n2 receiveleaveNotice_9 (guard b2)
o1 sendsignReceipt_8 (guard b1)
p1 receivesignReceipt_8 (guard b1)
q1 sendcloseShipment_10
r1 receivecloseShipment_10
s1 sendrateService_11
t1 receiverateService_11

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1 → o1 → p1 → q1 → r1 → s1 → t1
    └── m2 → n2 → q1 → r1 → s1 → t1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 -> n2 -> q1 -> r1 -> s1 -> t1

model_033 ParcelLocker · fail

Guard

a1	codeValid = 1

```
(root)
└── a1  codeValid = 1
```

เส้นทาง

a1

Event

a1 senddepositParcel_1
b1 receivedepositParcel_1
c1 sendlockerReceipt_2
d1 receivelockerReceipt_2
e1 sendsendPickupCode_3
f1 receivesendPickupCode_3
g1 sendenterCode_4
h1 receiveenterCode_4
i1 sendopenSlot_5_opt1 (guard a1)
j1 receiveopenSlot_5_opt1 (guard a1)
k1 sendslotClosed_6_opt1 (guard a1)
l1 receiveslotClosed_6_opt1 (guard a1)
m1 sendnotifyCourier_7
n1 receivenotifyCourier_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1

model_048 WarehouseRobotFleet · pass

Guard

a1	shift < 8
b1	lowBattery = 1

```
(root)
└── a1  shift < 8
    └── b1  lowBattery = 1
```

เส้นทาง

a1 -> b1

Event

a1 sendstartShift_1
b1 receivestartShift_1
c1 sendassignTaskA_2 (guard a1)
d1 receiveassignTaskA_2 (guard a1)
e1 sendassignTaskB_3 (guard a1)
f1 receiveassignTaskB_3 (guard a1)
g1 sendtaskDone_4 (guard a1)
h1 receivetaskDone_4 (guard a1)
i1 sendreserveDock_5_opt1 (guard a1 b1)
j1 receivereserveDock_5_opt1 (guard a1 b1)
k1 senddockReady_6_opt1 (guard a1 b1)
l1 receivedockReady_6_opt1 (guard a1 b1)
m1 sendendShift_7
m2 checkloop_assignTaskA_2_loop1 (guard a1)
n1 receiveendShift_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1
    └── m2 ↺ c1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 ↺ c1

model_067 CargoShipTracking · fail

Guard

a1	cycleNo < 7
b1	useCache = 1

```
(root)
└── a1  cycleNo < 7
    └── b1  useCache = 1
```

เส้นทาง

```
a1 -> b1
```

Event

a1 sendsyncManifest_1 (guard a1 b1)
b1 receivesyncManifest_1
c1 sendassignContainer_2
d1 receiveassignContainer_2
e1 sendackContainer_3 (guard a1)
f1 receiveackContainer_3 (guard a1)
g1 sendupdateManifest_4_opt1 (guard a1 b1)
h1 receiveupdateManifest_4_opt1
i1 sendupdateBerth_5 (guard a1)
j1 receiveupdateBerth_5 (guard a1)
k1 sendreleaseContainer_6
k2 checkloop_ackContainer_3_loop1 (guard a1)
l1 receivereleaseContainer_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1
    └── k2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ e1
```

model_081 CustomsClearance · pass

Guard

a1	useCache = 1

```
(root)
└── a1  useCache = 1
```

เส้นทาง

```
a1
```

Event

a1 sendsendRelease_1 (guard a1)
b1 receivesendRelease_1
c1 sendrequestDuty_2
d1 receiverequestDuty_2
e1 sendcancelDuty_3_opt1 (guard a1)
f1 receivecancelDuty_3_opt1
g1 sendapproveDeclaration_4
h1 receiveapproveDeclaration_4
i1 sendsyncDeclaration_5
j1 receivesyncDeclaration_5
k1 sendfetchRelease_6
l1 receivefetchRelease_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
```

model_114 InventoryReorder · fail

Guard

a1	retryNo < 12
b1	retryNo mod 3 = 0
c1	express = 1

```
(root)
└── a1  retryNo < 12
    └── b1  retryNo mod 3 = 0
        └── c1  express = 1
```

เส้นทาง

a1 -> b1 -> c1

Event

a1 sendcheckInvoice_1
b1 receivecheckInvoice_1
c1 sendreportInvoice_2
d1 receivereportInvoice_2
e1 sendsendInvoice_3 (guard a1)
f1 receivesendInvoice_3 (guard a1)
g1 sendackInvoice_4_opt1 (guard a1 b1)
h1 receiveackInvoice_4_opt1 (guard a1 b1)
i1 sendlogStock_5_opt1 (guard a1 b1)
j1 receivelogStock_5_opt1 (guard a1 b1)
k1 sendrequestInvoice_6 (guard a1)
l1 receiverequestInvoice_6 (guard a1)
m1 sendscheduleStock_7_opt2 (guard a1 c1)
n1 receivescheduleStock_7_opt2 (guard a1 c1)
o1 sendscheduleOrder_8_opt2 (guard a1 c1)
p1 receivescheduleOrder_8_opt2 (guard a1 c1)
q1 sendupdateInvoice_9 (guard a1)
r1 receiveupdateInvoice_9 (guard a1)
s1 sendissueOrder_10 (guard a1)
t1 receiveissueOrder_10 (guard a1)
u1 sendfetchOrder_11
u2 checkloop_sendInvoice_3_loop1 (guard a1)
v1 receivefetchOrder_11

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1
    ├── u1 → v1
    └── u2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u2 ↺ e1

model_122 MailroomSorting · pass

Guard

a1	withExtra = 1
b1	pollNo < 5

```
(root)
└── a1  withExtra = 1
    └── b1  pollNo < 5
```

เส้นทาง

a1 -> b1

Event

a1 sendassignSlot_1
b1 receiveassignSlot_1
c1 sendackNotice_2
d1 receiveackNotice_2
e1 sendcancelNotice_3
f1 receivecancelNotice_3
g1 sendissueNotice_4
h1 receiveissueNotice_4
i1 sendfetchSlot_5
j1 receivefetchSlot_5
k1 sendcheckParcel_6
l1 receivecheckParcel_6
m1 sendapproveNotice_7_opt1 (guard a1)
n1 receiveapproveNotice_7_opt1 (guard a1)
o1 sendlogParcel_8_opt1 (guard a1 b1)
p1 receivelogParcel_8_opt1 (guard a1 b1)
q1 sendcancelParcel_9_opt1 (guard a1 b1)
r1 receivecancelParcel_9_opt1 (guard a1 b1)
s1 sendreserveParcel_10_opt1 (guard a1 b1)
t1 receivereserveParcel_10_opt1 (guard a1 b1)
u1 sendreportParcel_11_opt1 (guard a1)
u2 checkloop_logParcel_8_loop1 (guard a1 b1)
v1 receivereportParcel_11_opt1 (guard a1)
w1 sendvalidateSlot_12
x1 receivevalidateSlot_12

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1
    ├── u1 → v1 → w1 → x1
    └── u2 ↺ o1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u2 ↺ o1

model_135 ParcelReturnDropoff · fail

Guard

a1	useCache = 1

```
(root)
└── a1  useCache = 1
```

เส้นทาง
a1

Event

a1 sendsyncReturn_1
b1 receivesyncReturn_1
c1 sendsendRefund_2
d1 receivesendRefund_2
e1 sendnotifyReturn_3_opt1 (guard a1)
f1 receivenotifyReturn_3_opt1 (guard a1)
g1 sendscheduleReturn_4
h1 receivescheduleReturn_4
i1 sendackReturn_5
j1 receiveackReturn_5
k1 sendreportReturn_6
l1 receivereportReturn_6
m1 sendlogRefund_7
n1 receivelogRefund_7
o1 sendapproveRefund_8
p1 receiveapproveRefund_8
q1 sendassignLabel_9
r1 receiveassignLabel_9
s1 sendvalidateRefund_10
t1 receivevalidateRefund_10

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1

model_146 PortCraneControl · pass

Guard

a1	express = 1

```
(root)
└── a1  express = 1
```

เส้นทาง
a1

Event

a1 sendreportContainer_1
b1 receivereportContainer_1
c1 sendcheckSlot_2
d1 receivecheckSlot_2
e1 sendrequestLift_3_opt1 (guard a1)
f1 receiverequestLift_3_opt1 (guard a1)
g1 sendackContainer_4_opt1 (guard a1)
h1 receiveackContainer_4_opt1 (guard a1)
i1 sendassignLift_5
j1 receiveassignLift_5
k1 sendupdateSlot_6
l1 receiveupdateSlot_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_161 SeaportPilotBooking · pass

Guard

a1	needAck = 1

```
(root)
└── a1  needAck = 1
```

เส้นทาง
a1

Event

a1 sendnotifyArrival_1
b1 receivenotifyArrival_1
c1 sendreserveArrival_2
d1 receivereserveArrival_2
e1 sendsendBerth_3_opt1 (guard a1)
f1 receivesendBerth_3_opt1 (guard a1)
g1 sendlogPilot_4
h1 receivelogPilot_4
i1 sendrequestBerth_5
j1 receiverequestBerth_5
k1 sendcheckArrival_6
l1 receivecheckArrival_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_195 WarehouseForklift · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendscheduleAisle_1
b1 receivescheduleAisle_1
c1 sendcancelAisle_2
d1 receivecancelAisle_2
e1 sendvalidatePallet_3
f1 receivevalidatePallet_3
g1 sendapprovePallet_4
h1 receiveapprovePallet_4
i1 sendfetchAisle_5
j1 receivefetchAisle_5
k1 sendapproveAisle_6
l1 receiveapproveAisle_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_196 WarehousePicking · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendreportTote_1
b1 receivereportTote_1
c1 sendsendOrder_2
d1 receivesendOrder_2
e1 sendlogTote_3
f1 receivelogTote_3
g1 sendlogBin_4
h1 receivelogBin_4
i1 sendassignBin_5
j1 receiveassignBin_5
k1 sendreserveTote_6
l1 receivereserveTote_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

บ้านและอาคารอัจฉริยะ · Smart Home & Building (21)

model_015 ElevatorControl · pass

Guard

a1	doorState = CLEAR
a2	doorState = BLOCKED
b1	retryDoor < 3

```
(root)
├── a1  doorState = CLEAR
└── a2  doorState = BLOCKED
    └── b1  retryDoor < 3
```

เส้นทาง

a1
a2 -> b1

Event

a1 sendcallElevator_1
b1 receivecallElevator_1
c1 sendmoveCar_2
d1 receivemoveCar_2
e1 sendarrived_3
f1 receivearrived_3
g1 sendopenDoor_4 (guard a1)
g2 sendcloseDoor_6 (guard a2 b1)
h1 receiveopenDoor_4 (guard a1)
h2 receivecloseDoor_6 (guard a2 b1)
i1 senddoorOpened_5 (guard a1)
i2 senddoorSensor_7 (guard a2 b1)
j1 receivedoorOpened_5 (guard a1)
j2 receivedoorSensor_7 (guard a2 b1)
k1 sendnotifyPassenger_8
k2 checkloop_closeDoor_6_loop1 (guard a2 b1)
l1 receivenotifyPassenger_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    └── g2 → h2 → i2 → j2
        ├── k1 → l1
        └── k2 ↺ g2
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> i2 -> j2 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> i2 -> j2 -> k2 ↺ g2

model_037 SmartDoorLock · pass

Guard

a1	authResult = PASS
b1	autoLock = 1
a2	authResult = FAIL

```
(root)
├── a1  authResult = PASS
│   └── b1  autoLock = 1
└── a2  authResult = FAIL
```

เส้นทาง

```
a1 -> b1
a2
```

Event

a1 sendunlockRequest_1
b1 receiveunlockRequest_1
c1 sendchallenge_2
d1 receivechallenge_2
e1 sendrespond_3
f1 receiverespond_3
g1 senddoorUnlocked_4 (guard a1)
g2 sendaccessDenied_6 (guard a2)
h1 receivedoorUnlocked_4 (guard a1)
h2 receiveaccessDenied_6 (guard a2)
i1 sendscheduleRelock_5_opt1 (guard a1 b1)
j1 receivescheduleRelock_5_opt1 (guard a1 b1)
k1 sendstatusUpdate_7
l1 receivestatusUpdate_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    └── g2 → h2 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1
```

model_041 SmartPlugControl · fail

Guard

a1	relayOn = 1
a2	relayOn = 0

```
(root)
├── a1  relayOn = 1
└── a2  relayOn = 0
```

เส้นทาง

```
a1
a2
```

Event

a1 sendqueryState_1
b1 receivequeryState_1
c1 sendplugState_2
d1 receiveplugState_2
e1 sendturnOff_3 (guard a1)
e2 sendturnOn_5 (guard a2)
f1 receiveturnOff_3 (guard a1)
f2 receiveturnOn_5 (guard a2)
g1 sendoffAck_4 (guard a1)
h1 receiveoffAck_4 (guard a1)
i1 sendenergyReport_6
j1 receiveenergyReport_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    └── e2 → f2 → i1 → j1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> i1 -> j1
```

model_043 ThermostatControl · pass

Guard

a1	cycle < 5
b1	overheat = 1
b2	overheat = 0

```
(root)
└── a1  cycle < 5
    ├── b1  overheat = 1
    └── b2  overheat = 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

a1 sendstartHeating_1
b1 receivestartHeating_1
c1 sendheatingAck_2
d1 receiveheatingAck_2
e1 sendqueryTemp_3 (guard a1)
f1 receivequeryTemp_3 (guard a1)
g1 sendtempReport_4 (guard a1)
h1 receivetempReport_4 (guard a1)
i1 sendreducePower_5 (guard a1 b1)
i2 sendkeepPower_6 (guard a1 b2)
j1 receivereducePower_5 (guard a1 b1)
j2 receivekeepPower_6 (guard a1 b2)
k1 sendstopHeating_7
k2 checkloop_queryTemp_3_loop1 (guard a1)
l1 receivestopHeating_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1
    │   ├── k1 → l1
    │   └── k2 ↺ e1
    └── i2 → j2
        ├── k1 → l1
        └── k2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 -> j2 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 -> j2 -> k2 ↺ e1
```

model_047 VoiceAssistant · pass

Guard

a1	intent = MUSIC
a2	intent = WEATHER
a3	intent = UNKNOWN

```
(root)
├── a1  intent = MUSIC
├── a2  intent = WEATHER
└── a3  intent = UNKNOWN
```

เส้นทาง

```
a1
a2
a3
```

Event

a1 sendwakeWord_1
b1 receivewakeWord_1
c1 sendstreamAudio_2
d1 receivestreamAudio_2
e1 sendintentResult_3
f1 receiveintentResult_3
g1 sendplaySong_4 (guard a1)
g2 sendfetchWeather_5 (guard a2)
g3 sendaskRepeat_8 (guard a3)
h1 receiveplaySong_4 (guard a1)
h2 receivefetchWeather_5 (guard a2)
h3 receiveaskRepeat_8 (guard a3)
i1 sendsleepMode_9
j1 receivesleepMode_9
k1 sendweatherInfo_6 (guard a2)
l1 receiveweatherInfo_6 (guard a2)
m1 sendspeakWeather_7 (guard a2)
n1 receivespeakWeather_7 (guard a2)

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1
    ├── g2 → h2 → k1 → l1 → m1 → n1 → i1 → j1
    └── g3 → h3 → i1 → j1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m1 -> n1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g3 -> h3 -> i1 -> j1
```

model_053 ApartmentIntercom · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendsyncCall_1
b1 receivesyncCall_1
c1 sendreserveCall_2
d1 receivereserveCall_2
e1 sendreserveDoor_3
f1 receivereserveDoor_3
g1 sendcheckDoor_4
h1 receivecheckDoor_4
i1 sendqueryDoor_5
j1 receivequeryDoor_5
k1 sendcancelCall_6
l1 receivecancelCall_6
m1 sendlogCode_7
n1 receivelogCode_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1

model_060 BuildingVisitorPass · pass

Guard

a1	t_obs ≤ 4
a2	t_obs > 4

```
(root)
├── a1  t_obs ≤ 4
└── a2  t_obs > 4
```

เส้นทาง

a1
a2

Event

a1 sendsendBadge_1
b1 receivesendBadge_1
c1 sendcheckBadge_2
d1 receivecheckBadge_2
e1 sendqueryVisit_3 (guard a1)
e2 sendnotifyPass_5 (guard a2)
f1 receivequeryVisit_3 (guard a1)
f2 receivenotifyPass_5 (guard a2)
g1 sendreleasePass_4 (guard a1)
g2 sendvalidateVisit_6 (guard a2)
h1 receivereleasePass_4 (guard a1)
h2 receivevalidateVisit_6 (guard a2)
i1 sendsendPass_7
j1 receivesendPass_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    └── e2 → f2 → g2 → h2 → i1 → j1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g2 -> h2 -> i1 -> j1

model_086 DormLaundry · pass

Guard

a1	cycleNo < 3

```
(root)
└── a1  cycleNo < 3
```

เส้นทาง

a1

Event

a1 sendackCredit_1
b1 receiveackCredit_1
c1 sendconfirmCredit_2
d1 receiveconfirmCredit_2
e1 sendreserveMachine_3 (guard a1)
f1 receivereserveMachine_3 (guard a1)
g1 sendnotifyCredit_4 (guard a1)
h1 receivenotifyCredit_4 (guard a1)
i1 sendreleaseCycle_5
i2 checkloop_reserveMachine_3_loop1 (guard a1)
j1 receivereleaseCycle_5
k1 sendfetchCredit_6
l1 receivefetchCredit_6
m1 sendsendCredit_7
n1 receivesendCredit_7
o1 sendfetchCycle_8
p1 receivefetchCycle_8
q1 sendcancelCycle_9
r1 receivecancelCycle_9
s1 sendqueryCredit_10
t1 receivequeryCredit_10

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
    │     q1 → r1 → s1 → t1
    └── i2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ e1

model_093 FireSprinklerSystem · pass

Guard

a1	optIn = 1

```
(root)
└── a1  optIn = 1
```

เส้นทาง
a1

Event

a1 sendcheckValve_1
b1 receivecheckValve_1
c1 sendlogValve_2
d1 receivelogValve_2
e1 sendsendAlarm_3_opt1 (guard a1)
f1 receivesendAlarm_3_opt1 (guard a1)
g1 sendupdateFire_4_opt1 (guard a1)
h1 receiveupdateFire_4_opt1 (guard a1)
i1 sendnotifyFire_5
j1 receivenotifyFire_5
k1 sendconfirmFire_6
l1 receiveconfirmFire_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_107 HomeSecurityAlarm · fail

Guard

a1	needAck = 1

```
(root)
└── a1  needAck = 1
```

เส้นทาง
a1

Event

a1 sendlogCode_1
b1 receivelogCode_1
c1 sendreportAlert_2
d1 receivereportAlert_2
e1 sendcheckAlert_3_opt1 (guard a1)
f1 receivecheckAlert_3_opt1 (guard a1)
g1 sendnotifyZone_4_opt1 (guard a1)
h1 receivenotifyZone_4_opt1 (guard a1)
i1 sendsyncZone_5
j1 receivesyncZone_5
k1 sendqueryCode_6
l1 receivequeryCode_6
m1 sendassignAlert_7
n1 receiveassignAlert_7
o1 sendqueryZone_8
p1 receivequeryZone_8
q1 sendscheduleZone_9
r1 receivescheduleZone_9

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1

model_115 IoTDoorbell · pass

Guard

a1	useCache = 1
b1	needAck = 1

```
(root)
└── a1  useCache = 1
    └── b1  needAck = 1
```

เส้นทาง
a1 -> b1

Event

a1 sendapproveMessage_1
b1 receiveapproveMessage_1
c1 sendackVideo_2
d1 receiveackVideo_2
e1 sendassignVideo_3_opt1 (guard a1)
f1 receiveassignVideo_3_opt1 (guard a1)
g1 sendnotifyRing_4
h1 receivenotifyRing_4
i1 sendreserveMessage_5
j1 receivereserveMessage_5
k1 sendrequestRing_6_opt2 (guard b1)
l1 receiverequestRing_6_opt2 (guard b1)
m1 sendcheckVideo_7
n1 receivecheckVideo_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1

model_119 LiftEmergencyCall · fail

Guard

a1	needAck = 1
b1	t_obs ≤ 23
b2	t_obs > 23

```
(root)
└── a1  needAck = 1
    ├── b1  t_obs ≤ 23
    └── b2  t_obs > 23
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

a1 sendreleaseCall_1
b1 receivereleaseCall_1
c1 sendqueryCall_2
d1 receivequeryCall_2
e1 sendsyncFault_3_opt1 (guard a1)
f1 receivesyncFault_3_opt1 (guard a1)
g1 sendlogCall_4
h1 receivelogCall_4
i1 sendnotifyFault_5
j1 receivenotifyFault_5
k1 sendreportFault_6 (guard b1)
k2 sendcancelFault_7 (guard b2)
l1 receivereportFault_6 (guard b1)
l2 receivecancelFault_7 (guard b2)
m1 sendsendFault_8
n1 receivesendFault_8
o1 sendsyncCall_9
p1 receivesyncCall_9
q1 sendackFault_10
r1 receiveackFault_10
s1 sendnotifyDispatch_11
t1 receivenotifyDispatch_11
u1 sendconfirmDispatch_12
v1 receiveconfirmDispatch_12
w1 sendvalidateFault_13
x1 receivevalidateFault_13

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1 → m1 → n1 → o1 → p1 → q1 → r1 →
    │     s1 → t1 → u1 → v1 → w1 → x1
    └── k2 → l2 → m1 → n1 → o1 → p1 → q1 → r1 →
          s1 → t1 → u1 → v1 → w1 → x1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 -> l2 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1
```

model_120 LiftMaintenance · fail

Guard

a1	attemptNo < 5

```
(root)
└── a1  attemptNo < 5
```

เส้นทาง

```
a1
```

Event

a1 sendlogJob_1
b1 receivelogJob_1
c1 sendvalidateJob_2
d1 receivevalidateJob_2
e1 sendrequestJob_3 (guard a1)
f1 receiverequestJob_3 (guard a1)
g1 sendcheckJob_4 (guard a1)
h1 receivecheckJob_4 (guard a1)
i1 sendsyncLock_5 (guard a1)
j1 receivesyncLock_5 (guard a1)
k1 sendcancelFault_6
k2 checkloop_requestJob_3_loop1 (guard a1)
l1 receivecancelFault_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1
    └── k2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ e1
```

model_130 OfficeLightingControl · fail

Guard

a1	needAck = 1
b1	express = 1

```
(root)
└── a1  needAck = 1
    └── b1  express = 1
```

เส้นทาง

a1 -> b1

Event

a1 sendlogDim_1
b1 receivelogDim_1
c1 sendsyncSchedule_2
d1 receivesyncSchedule_2
e1 sendreleaseOccupancy_3_opt1 (guard a1)
f1 receivereleaseOccupancy_3_opt1 (guard a1)
g1 sendapproveOccupancy_4
h1 receiveapproveOccupancy_4
i1 sendlogSchedule_5
j1 receivelogSchedule_5
k1 sendqueryDim_6_opt2 (guard b1)
l1 receivequeryDim_6_opt2 (guard b1)
m1 sendfetchSchedule_7
n1 receivefetchSchedule_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1

model_140 PetDoorController · pass

Guard

a1	t_obs ≤ 58
a2	t_obs > 58

```
(root)
├── a1  t_obs ≤ 58
└── a2  t_obs > 58
```

เส้นทาง

a1
a2

Event

a1 sendrequestLog_1
b1 receiverequestLog_1
c1 sendscheduleEntry_2
d1 receivescheduleEntry_2
e1 sendcheckLog_3
f1 receivecheckLog_3
g1 sendcancelCurfew_4
h1 receivecancelCurfew_4
i1 sendreportCurfew_5
j1 receivereportCurfew_5
k1 sendqueryEntry_6
l1 receivequeryEntry_6
m1 sendrequestCurfew_7
n1 receiverequestCurfew_7
o1 sendapproveLog_8 (guard a1)
o2 sendupdateEntry_10 (guard a2)
p1 receiveapproveLog_8 (guard a1)
p2 receiveupdateEntry_10 (guard a2)
q1 sendsendCurfew_9 (guard a1)
q2 sendsyncCurfew_11 (guard a2)
r1 receivesendCurfew_9 (guard a1)
r2 receivesyncCurfew_11 (guard a2)
s1 sendsendEntry_12
t1 receivesendEntry_12

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1 → q1 → r1 → s1 → t1
    └── o2 → p2 → q2 → r2 → s1 → t1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o2 -> p2 -> q2 -> r2 -> s1 -> t1

model_157 RoboticLawnMower · fail

Guard

a1	useCache = 1

```
(root)
└── a1  useCache = 1
```

เส้นทาง

a1

Event

a1 sendsendDock_1
b1 receivesendDock_1
c1 sendrequestBlade_2
d1 receiverequestBlade_2
e1 sendcancelBlade_3_opt1 (guard a1)
f1 receivecancelBlade_3_opt1 (guard a1)
g1 sendapproveZone_4
h1 receiveapproveZone_4
i1 sendsyncZone_5
j1 receivesyncZone_5
k1 sendfetchDock_6
l1 receivefetchDock_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_164 SmartAquarium · fail

Guard

a1	cycleNo < 4
b1	cycleNo mod 2 = 0
b2	cycleNo mod 2 ≠ 0

```
(root)
└── a1  cycleNo < 4
    ├── b1  cycleNo mod 2 = 0
    └── b2  cycleNo mod 2 ≠ 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

a1 sendvalidatePh_1
b1 receivevalidatePh_1
c1 sendupdateFeed_2
d1 receiveupdateFeed_2
e1 sendreportFeed_3 (guard a1)
f1 receivereportFeed_3 (guard a1)
g1 sendapproveLight_4 (guard a1 b1)
g2 sendnotifyFeed_6 (guard a1 b2)
h1 receiveapproveLight_4 (guard a1 b1)
h2 receivenotifyFeed_6 (guard a1 b2)
i1 sendqueryFeed_5 (guard a1 b1)
j1 receivequeryFeed_5 (guard a1 b1)
k1 sendsendLight_7 (guard a1)
l1 receivesendLight_7 (guard a1)
m1 sendsyncLight_8
m2 checkloop_reportFeed_3_loop1 (guard a1)
n1 receivesyncLight_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    │   ├── m1 → n1
    │   └── m2 ↺ e1
    └── g2 → h2 → k1 → l1
        ├── m1 → n1
        └── m2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m2 ↺ e1
```

model_165 SmartBabyMonitor · pass

Guard

a1	useCache = 1
b1	score < 50
b2	score ≥ 50

```
(root)
└── a1  useCache = 1
    ├── b1  score < 50
    └── b2  score ≥ 50
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

a1 sendqueryClimate_1
b1 receivequeryClimate_1
c1 sendconfirmCry_2
d1 receiveconfirmCry_2
e1 sendreserveCry_3_opt1 (guard a1)
f1 receivereserveCry_3_opt1 (guard a1)
g1 sendcheckAlert_4_opt1 (guard a1 b1)
g2 sendscheduleAlert_9_opt1 (guard a1 b2)
h1 receivecheckAlert_4_opt1 (guard a1 b1)
h2 receivescheduleAlert_9_opt1 (guard a1 b2)
i1 sendapproveCry_5_opt1 (guard a1 b1)
j1 receiveapproveCry_5_opt1 (guard a1 b1)
k1 sendreserveAlert_6_opt1 (guard a1 b1)
l1 receivereserveAlert_6_opt1 (guard a1 b1)
m1 sendvalidateAlert_7_opt1 (guard a1 b1)
n1 receivevalidateAlert_7_opt1 (guard a1 b1)
o1 sendupdateClimate_8_opt1 (guard a1 b1)
p1 receiveupdateClimate_8_opt1 (guard a1 b1)
q1 sendsyncCry_10_opt1 (guard a1)
r1 receivesyncCry_10_opt1 (guard a1)
s1 sendcancelCry_11
t1 receivecancelCry_11

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1 →
    │     o1 → p1 → q1 → r1 → s1 → t1
    └── g2 → h2 → q1 → r1 → s1 → t1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> q1 -> r1 -> s1 -> t1
```

model_167 SmartCurtainControl · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendapprovePosition_1
b1 receiveapprovePosition_1
c1 sendupdateScene_2
d1 receiveupdateScene_2
e1 sendconfirmLux_3
f1 receiveconfirmLux_3
g1 sendfetchScene_4
h1 receivefetchScene_4
i1 sendreleaseLux_5
j1 receivereleaseLux_5
k1 sendcheckScene_6
l1 receivecheckScene_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_168 SmartFridgeInventory · fail

Guard

a1	attemptNo < 2
b1	tier = GREEN
b2	tier = RED
c1	stepNo < 11

```
(root)
└── a1  attemptNo < 2
    ├── b1  tier = GREEN
    │   └── c1  stepNo < 11
    └── b2  tier = RED
        └── c1  stepNo < 11
```

เส้นทาง

a1 -> b1 -> c1
a1 -> b2 -> c1

Event

a1 sendapproveOrder_1
b1 receiveapproveOrder_1
c1 sendassignExpiry_2
d1 receiveassignExpiry_2
e1 sendsendOrder_3 (guard a1)
f1 receivesendOrder_3 (guard a1)
g1 sendassignItem_4 (guard a1 b1)
g2 sendlogOrder_6 (guard a1 b2)
h1 receiveassignItem_4 (guard a1 b1)
h2 receivelogOrder_6 (guard a1 b2)
i1 sendscheduleOrder_5 (guard a1 b1)
j1 receivescheduleOrder_5 (guard a1 b1)
k1 sendapproveExpiry_7 (guard a1)
l1 receiveapproveExpiry_7 (guard a1)
m1 sendqueryItem_8
m2 checkloop_sendOrder_3_loop1 (guard a1)
n1 receivequeryItem_8
o1 sendcheckOrder_9
p1 receivecheckOrder_9
q1 sendissueExpiry_10 (guard c1)
r1 receiveissueExpiry_10 (guard c1)
s1 sendsyncOrder_11
s2 checkloop_issueExpiry_10_loop2 (guard c1)
t1 receivesyncOrder_11

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    │   ├── m1 → n1 → o1 → p1 → q1 → r1
    │   │   ├── s1 → t1
    │   │   └── s2 ↺ q1
    │   └── m2 ↺ e1
    └── g2 → h2 → k1 → l1
        ├── m1 → n1 → o1 → p1 → q1 → r1
        │   ├── s1 → t1
        │   └── s2 ↺ q1
        └── m2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s2 ↺ q1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s2 ↺ q1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m2 ↺ e1

model_197 WaterLeakDetection · fail

Guard

a1	cycleNo < 10
b1	cycleNo mod 2 = 0

```
(root)
└── a1  cycleNo < 10
    └── b1  cycleNo mod 2 = 0
```

เส้นทาง

```
a1 -> b1
```

Event

a1 sendfetchTicket_1
b1 receivefetchTicket_1
c1 sendupdateLeak_2
d1 receiveupdateLeak_2
e1 sendreserveValve_3 (guard a1)
f1 receivereserveValve_3 (guard a1)
g1 sendreleaseLeak_4_opt1 (guard a1 b1)
h1 receivereleaseLeak_4_opt1 (guard a1 b1)
i1 sendreleaseTicket_5 (guard a1)
j1 receivereleaseTicket_5 (guard a1)
k1 sendreportValve_6 (guard a1)
l1 receivereportValve_6 (guard a1)
m1 sendsyncTicket_7 (guard a1)
n1 receivesyncTicket_7 (guard a1)
o1 sendissueTicket_8 (guard a1)
p1 receiveissueTicket_8 (guard a1)
q1 sendcancelLeak_9 (guard a1)
r1 receivecancelLeak_9 (guard a1)
s1 sendackLeak_10 (guard a1)
t1 receiveackLeak_10 (guard a1)
u1 sendreportLeak_11
u2 checkloop_reserveValve_3_loop1 (guard a1)
v1 receivereportLeak_11

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1
    ├── u1 → v1
    └── u2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u2 ↺ e1
```

อุตสาหกรรม พลังงาน และโครงสร้างพื้นฐาน · Industrial, Energy & Infrastructure (13)

model_005 BatteryChargeCycle · fail

Guard

a1	pulseNo < 8

```
(root)
└── a1  pulseNo < 8
```

เส้นทาง

```
a1
```

Event

a1 sendstartCharge_1
b1 receivestartCharge_1
c1 sendchargeAck_2
d1 receivechargeAck_2
e1 sendchargePulse_3 (guard a1)
f1 receivechargePulse_3 (guard a1)
g1 sendcellVoltage_4 (guard a1)
h1 receivecellVoltage_4 (guard a1)
i1 sendstopCharge_5
i2 checkloop_chargePulse_3_loop1 (guard a1)
j1 receivestopCharge_5
k1 sendfullReport_6
l1 receivefullReport_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1
    └── i2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ e1
```

model_018 FactoryRobotArm · pass

Guard

a1	pickCount < 10
b1	defect = 1

```
(root)
└── a1  pickCount < 10
    └── b1  defect = 1
```

เส้นทาง

```
a1 -> b1
```

Event

a1 sendstartBatch_1
b1 receivestartBatch_1
c1 sendinitArm_2
d1 receiveinitArm_2
e1 sendcalibrateCamera_3
f1 receivecalibrateCamera_3
g1 sendlogStart_4
h1 receivelogStart_4
i1 sendpickItem_5 (guard a1)
j1 receivepickItem_5 (guard a1)
k1 sendinspectItem_6 (guard a1)
l1 receiveinspectItem_6 (guard a1)
m1 sendinspectionResult_7 (guard a1)
n1 receiveinspectionResult_7 (guard a1)
o1 senddiscardItem_8_opt1 (guard a1 b1)
p1 receivediscardItem_8_opt1 (guard a1 b1)
q1 sendreportSummary_9
q2 checkloop_pickItem_5_loop1 (guard a1)
r1 receivereportSummary_9

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
    ├── q1 → r1
    └── q2 ↺ i1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q2 ↺ i1
```

model_039 SmartMeterBilling · fail

Guard

a1	autoPay = 1

```
(root)
└── a1  autoPay = 1
```

เส้นทาง

```
a1
```

Event

a1 sendsendUsage_1
b1 receivesendUsage_1
c1 sendusageAck_2
d1 receiveusageAck_2
e1 sendgenerateBill_3
f1 receivegenerateBill_3
g1 sendpayBill_4
h1 receivepayBill_4
i1 sendsetAutoPay_5_opt1 (guard a1)
j1 receivesetAutoPay_5_opt1 (guard a1)
k1 sendautoPayOK_6_opt1 (guard a1)
l1 receiveautoPayOK_6_opt1 (guard a1)
m1 sendreceiptIssued_7
n1 receivereceiptIssued_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
```

model_079 CraneLoadMonitor · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendcancelLimit_1
b1 receivecancelLimit_1
c1 sendvalidateAlarm_2
d1 receivevalidateAlarm_2
e1 sendackLimit_3
f1 receiveackLimit_3
g1 sendissueLimit_4
h1 receiveissueLimit_4
i1 sendcancelLoad_5
j1 receivecancelLoad_5
k1 sendscheduleLoad_6
l1 receivescheduleLoad_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
```

model_088 EVBatterySwap · pass

Guard

a1	useCache = 1
b1	optIn = 1
c1	retryNo < 11

```
(root)
└── a1  useCache = 1
    └── b1  optIn = 1
        └── c1  retryNo < 11
```

เส้นทาง

a1 -> b1 -> c1

Event

a1 sendreportCredit_1 (guard b1)
b1 receivereportCredit_1
c1 sendissueCredit_2
d1 receiveissueCredit_2
e1 sendreserveSwap_3_opt1 (guard a1)
f1 receivereserveSwap_3_opt1 (guard a1)
g1 sendnotifyCredit_4
h1 receivenotifyCredit_4
i1 sendreportBattery_5
j1 receivereportBattery_5
k1 sendissueSwap_6_opt2 (guard b1)
l1 receiveissueSwap_6_opt2
m1 sendcheckCredit_7
n1 receivecheckCredit_7
o1 sendscheduleSwap_8
p1 receivescheduleSwap_8
q1 sendlogBattery_9 (guard c1)
r1 receivelogBattery_9 (guard c1)
s1 sendfetchSwap_10 (guard c1)
t1 receivefetchSwap_10 (guard c1)
u1 sendconfirmBattery_11
u2 checkloop_logBattery_9_loop1 (guard c1)
v1 receiveconfirmBattery_11

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1
    ├── u1 → v1
    └── u2 ↺ q1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u2 ↺ q1

model_089 EVChargingSession · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendcancelTariff_1
b1 receivecancelTariff_1
c1 sendnotifySession_2
d1 receivenotifySession_2
e1 sendupdateTariff_3
f1 receiveupdateTariff_3
g1 sendrequestEnergy_4
h1 receiverequestEnergy_4
i1 sendvalidateSession_5
j1 receivevalidateSession_5
k1 sendackSession_6
l1 receiveackSession_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_091 FactoryQualityGate · pass

Guard

a1	roundNo < 7

```
(root)
└── a1  roundNo < 7
```

เส้นทาง

a1

Event

a1 sendcancelPart_1 (guard a1)
b1 receivecancelPart_1
c1 sendfetchDefect_2
d1 receivefetchDefect_2
e1 sendlogDefect_3 (guard a1)
f1 receivelogDefect_3 (guard a1)
g1 sendnotifyPart_4 (guard a1)
h1 receivenotifyPart_4 (guard a1)
i1 sendreportPart_5
i2 checkloop_logDefect_3_loop1 (guard a1)
j1 receivereportPart_5
k1 sendreleaseBatch_6
l1 receivereleaseBatch_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1
    └── i2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ e1

model_105 HomeEnergyBattery · pass

Guard

a1	express = 1
b1	attemptNo < 4

```
(root)
└── a1  express = 1
    └── b1  attemptNo < 4
```

เส้นทาง

```
a1 -> b1
```

Event

a1 sendnotifyExport_1
b1 receivenotifyExport_1
c1 sendvalidateExport_2
d1 receivevalidateExport_2
e1 sendlogCharge_3_opt1 (guard a1)
f1 receivelogCharge_3_opt1 (guard a1)
g1 sendreserveTariff_4
h1 receivereserveTariff_4
i1 sendassignCharge_5
j1 receiveassignCharge_5
k1 sendreleaseTariff_6 (guard b1)
l1 receivereleaseTariff_6 (guard b1)
m1 sendfetchCharge_7 (guard b1)
n1 receivefetchCharge_7 (guard b1)
o1 sendrequestCharge_8
o2 checkloop_releaseTariff_6_loop1 (guard b1)
p1 receiverequestCharge_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1
    └── o2 ↺ k1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1 -> o2 ↺ k1
```

model_144 PipelineLeakSCADA · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendscheduleWork_1
b1 receivescheduleWork_1
c1 sendcancelWork_2
d1 receivecancelWork_2
e1 sendvalidatePressure_3
f1 receivevalidatePressure_3
g1 sendapprovePressure_4
h1 receiveapprovePressure_4
i1 sendfetchWork_5
j1 receivefetchWork_5
k1 sendapproveWork_6
l1 receiveapproveWork_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1
```

model_170 SmartHelmetMining · pass

Guard

a1	temp < 50
a2	temp ≥ 50

```
(root)
├── a1  temp < 50
└── a2  temp ≥ 50
```

เส้นทาง

```
a1
a2
```

Event

a1 sendsyncAlarm_1
b1 receivesyncAlarm_1
c1 sendackGas_2
d1 receiveackGas_2
e1 sendscheduleGas_3 (guard a1)
e2 sendcheckAlarm_4 (guard a2)
f1 receivescheduleGas_3 (guard a1)
f2 receivecheckAlarm_4 (guard a2)
g1 sendupdateAlarm_6
h1 receiveupdateAlarm_6
i1 sendreserveLocation_5 (guard a2)
j1 receivereserveLocation_5 (guard a2)

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1
    └── e2 → f2 → i1 → j1 → g1 → h1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> i1 -> j1 -> g1 -> h1
```

model_176 SmartStreetLight · fail

Guard

a1	urgent = 1
a2	urgent = 0

```
(root)
├── a1  urgent = 1
└── a2  urgent = 0
```

เส้นทาง

a1
a2

Event

a1 sendcancelDim_1
b1 receivecancelDim_1
c1 sendqueryDim_2
d1 receivequeryDim_2
e1 sendrequestFault_3
f1 receiverequestFault_3
g1 sendreleaseLamp_4
h1 receivereleaseLamp_4
i1 sendsendDim_5
j1 receivesendDim_5
k1 sendlogLamp_6
l1 receivelogLamp_6
m1 sendconfirmDim_7
n1 receiveconfirmDim_7
o1 sendconfirmFault_8 (guard a1)
o2 sendapproveDim_10 (guard a2)
p1 receiveconfirmFault_8 (guard a1)
p2 receiveapproveDim_10 (guard a2)
q1 sendnotifyDim_9 (guard a1)
r1 receivenotifyDim_9 (guard a1)
s1 sendackFault_11
t1 receiveackFault_11

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1 → q1 → r1 → s1 → t1
    └── o2 → p2 → s1 → t1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o2 -> p2 -> s1 -> t1

model_181 SolarPanelMonitor · fail

Guard

a1	attemptNo < 8
b1	attemptNo mod 2 = 0
b2	attemptNo mod 2 ≠ 0

```
(root)
└── a1  attemptNo < 8
    ├── b1  attemptNo mod 2 = 0
    └── b2  attemptNo mod 2 ≠ 0
```

เส้นทาง

a1 -> b1
a1 -> b2

Event

a1 sendackPower_1
b1 receiveackPower_1
c1 sendapproveExport_2
d1 receiveapproveExport_2
e1 sendissuePower_3 (guard a1)
f1 receiveissuePower_3 (guard a1)
g1 sendconfirmFault_4 (guard a1 b1)
g2 sendcheckPower_6 (guard a1 b2)
h1 receiveconfirmFault_4 (guard a1 b1)
h2 receivecheckPower_6 (guard a1 b2)
i1 sendreportFault_5 (guard a1 b1)
j1 receivereportFault_5 (guard a1 b1)
k1 sendreportExport_7 (guard a1)
l1 receivereportExport_7 (guard a1)
m1 sendreserveExport_8
m2 checkloop_issuePower_3_loop1 (guard a1)
n1 receivereserveExport_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    │   ├── m1 → n1
    │   └── m2 ↺ e1
    └── g2 → h2 → k1 → l1
        ├── m1 → n1
        └── m2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m2 ↺ e1

model_198 WindTurbineMonitor · pass

Guard

a1	withExtra = 1
b1	needAck = 1

```
(root)
└── a1  withExtra = 1
    └── b1  needAck = 1
```

เส้นทาง
a1 -> b1

Event

a1 sendreserveRpm_1
b1 receivereserveRpm_1
c1 sendupdatePower_2
d1 receiveupdatePower_2
e1 sendreservePower_3_opt1 (guard a1)
f1 receivereservePower_3_opt1 (guard a1)
g1 sendnotifyPower_4
h1 receivenotifyPower_4
i1 sendsyncPower_5
j1 receivesyncPower_5
k1 sendrequestRpm_6_opt2 (guard b1)
l1 receiverequestRpm_6_opt2 (guard b1)
m1 sendlogRpm_7_opt2 (guard b1)
n1 receivelogRpm_7_opt2 (guard b1)
o1 sendscheduleFault_8
p1 receivescheduleFault_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1

ค้าปลีก อาหาร และ e-commerce · Retail, Food & E-commerce (14)

model_010 CoffeeOrderKiosk · pass

Guard

a1	addShot = 1

```
(root)
└── a1  addShot = 1
```

เส้นทาง
a1

Event

a1 sendchooseDrink_1
b1 receivechooseDrink_1
c1 sendshowPrice_2
d1 receiveshowPrice_2
e1 sendaddExtraShot_3_opt1 (guard a1)
f1 receiveaddExtraShot_3_opt1 (guard a1)
g1 sendpayCash_4
h1 receivepayCash_4
i1 sendserveDrink_5
j1 receiveserveDrink_5

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1

model_021 FoodOrderApp · pass

Guard

a1	hasCoupon = 1

```
(root)
└── a1  hasCoupon = 1
```

เส้นทาง
a1

Event

a1 sendbrowseMenu_1
b1 receivebrowseMenu_1
c1 sendmenuList_2
d1 receivemenuList_2
e1 sendapplyCoupon_3_opt1 (guard a1)
f1 receiveapplyCoupon_3_opt1 (guard a1)
g1 senddiscountApplied_4_opt1 (guard a1)
h1 receivediscountApplied_4_opt1 (guard a1)
i1 sendsubmitOrder_5
j1 receivesubmitOrder_5
k1 sendforwardOrder_6
l1 receiveforwardOrder_6
m1 sendorderReady_7
n1 receiveorderReady_7
o1 sendnotifyCustomer_8
p1 receivenotifyCustomer_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1

model_030 OnlineAuction · fail

Guard

a1	roundNo < 5

```
(root)
└── a1  roundNo < 5
```

เส้นทาง
a1

Event

a1 sendplaceBid_1
b1 receiveplaceBid_1
c1 sendbidAccepted_2
d1 receivebidAccepted_2
e1 sendoutbidNotice_3 (guard a1)
f1 receiveoutbidNotice_3 (guard a1)
g1 sendraiseBid_4 (guard a1)
h1 receiveraiseBid_4 (guard a1)
i1 sendcloseAuction_5
i2 checkloop_outbidNotice_3_loop1 (guard a1)
j1 receivecloseAuction_5
k1 sendsaleConfirm_6
l1 receivesaleConfirm_6
m1 sendwinnerNotice_7
n1 receivewinnerNotice_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1 → m1 → n1
    └── i2 ↺ e1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ e1

model_044 ToolRentalLocker · fail

Guard

a1	lateReturn = 1

```
(root)
└── a1  lateReturn = 1
```

เส้นทาง
a1

Event

a1 sendreserveTool_1
b1 receivereserveTool_1
c1 sendreservationCode_2
d1 receivereservationCode_2
e1 sendunlockLocker_3
f1 receiveunlockLocker_3
g1 sendverifyCode_4
h1 receiveverifyCode_4
i1 sendcodeValid_5
j1 receivecodeValid_5
k1 sendreleaseTool_6
l1 receivereleaseTool_6
m1 sendchargeLateFee_7_opt1 (guard a1)
n1 receivechargeLateFee_7_opt1 (guard a1)

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1

model_046 VendingMachine · pass

Guard

a1	coinStatus = ENOUGH
a2	coinStatus = SHORT
a3	coinStatus = INVALID

```
(root)
├── a1  coinStatus = ENOUGH
├── a2  coinStatus = SHORT
└── a3  coinStatus = INVALID
```

เส้นทาง
a1
a2
a3

Event

a1 sendinsertCoin_1
b1 receiveinsertCoin_1
c1 sendselectItem_2
d1 receiveselectItem_2
e1 senddispenseItem_3 (guard a1)
e2 sendrequestMoreCoins_5 (guard a2)
e3 sendrejectCoin_6 (guard a3)
f1 receivedispenseItem_3 (guard a1)
f2 receiverequestMoreCoins_5 (guard a2)
f3 receiverejectCoin_6 (guard a3)
g1 sendreturnChange_4 (guard a1)
h1 receivereturnChange_4 (guard a1)
i1 sendshowIdle_7
j1 receiveshowIdle_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    ├── e2 → f2 → i1 → j1
    └── e3 → f3 → i1 → j1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e3 -> f3 -> i1 -> j1

model_063 CafeLoyaltyPoints · fail

Guard

a1	roundNo < 12
b1	roundNo mod 2 = 0
c1	needAck = 1

```
(root)
└── a1  roundNo < 12
    └── b1  roundNo mod 2 = 0
        └── c1  needAck = 1
```

เส้นทาง

a1 -> b1 -> c1

Event

a1 sendapproveVisit_1
b1 receiveapproveVisit_1
c1 sendfetchPoint_2
d1 receivefetchPoint_2
e1 sendcheckPoint_3 (guard a1)
f1 receivecheckPoint_3 (guard a1)
g1 sendissueVisit_4_opt1 (guard a1 b1)
h1 receiveissueVisit_4_opt1 (guard a1 b1)
i1 sendupdateVisit_5 (guard a1)
j1 receiveupdateVisit_5 (guard a1)
k1 sendreleaseVisit_6_opt2 (guard a1 c1)
l1 receivereleaseVisit_6_opt2 (guard a1 c1)
m1 sendsendPoint_7 (guard a1)
n1 receivesendPoint_7 (guard a1)
o1 sendsyncReward_8 (guard a1)
p1 receivesyncReward_8 (guard a1)
q1 sendfetchReward_9
q2 checkloop_checkPoint_3_loop1 (guard a1)
r1 receivefetchReward_9

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
    ├── q1 → r1
    └── q2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q2 ↺ e1

model_075 ConcessionStandOrder · pass

Guard

a1	retryNo < 8
b1	roundNo < 6
c1	pollNo < 11

```
(root)
└── a1  retryNo < 8
    └── b1  roundNo < 6
        └── c1  pollNo < 11
```

เส้นทาง

a1 -> b1 -> c1

Event

a1 sendsyncPickup_1
b1 receivesyncPickup_1
c1 sendreportPickup_2
d1 receivereportPickup_2
e1 sendconfirmOrder_3 (guard a1)
f1 receiveconfirmOrder_3 (guard a1)
g1 sendapproveSnack_4
g2 checkloop_confirmOrder_3_loop1 (guard a1)
h1 receiveapproveSnack_4
i1 sendassignOrder_5
j1 receiveassignOrder_5
k1 sendscheduleSnack_6 (guard b1)
l1 receivescheduleSnack_6 (guard b1)
m1 sendnotifySnack_7 (guard b1 c1)
n1 receivenotifySnack_7 (guard b1 c1)
o1 sendnotifyPickup_8 (guard b1 c1)
p1 receivenotifyPickup_8 (guard b1 c1)
q1 sendschedulePickup_9 (guard b1)
q2 checkloop_notifySnack_7_loop3 (guard b1 c1)
r1 receiveschedulePickup_9 (guard b1)
s1 sendupdatePickup_10
s2 checkloop_scheduleSnack_6_loop2 (guard b1)
t1 receiveupdatePickup_10

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1 →
    │     o1 → p1
    │   ├── q1 → r1
    │   │   ├── s1 → t1
    │   │   └── s2 ↺ k1
    │   └── q2 ↺ m1
    └── g2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s2 ↺ k1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q2 ↺ m1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 ↺ e1

model_097 FoodTruckOrdering · pass

Guard

a1	pollNo < 5
b1	needAck = 1
c1	express = 1
d1	tier = GREEN
d2	tier = RED

```
(root)
└── a1  pollNo < 5
    └── b1  needAck = 1
        └── c1  express = 1
            ├── d1  tier = GREEN
            └── d2  tier = RED
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1
a1 -> b1 -> c1 -> d2
```

Event

a1 sendvalidateOrder_1
b1 receivevalidateOrder_1
c1 sendcheckPickup_2
d1 receivecheckPickup_2
e1 sendreleasePickup_3 (guard a1)
f1 receivereleasePickup_3 (guard a1)
g1 sendissueDish_4_opt1 (guard a1 b1)
h1 receiveissueDish_4_opt1 (guard a1 b1)
i1 sendcheckDish_5 (guard a1)
j1 receivecheckDish_5 (guard a1)
k1 sendcancelDish_6_opt2 (guard a1 c1)
l1 receivecancelDish_6_opt2 (guard a1 c1)
m1 sendcancelPickup_7_opt2 (guard a1 c1 d1)
m2 sendassignPickup_8_opt2 (guard a1 c1 d2)
n1 receivecancelPickup_7_opt2 (guard a1 c1 d1)
n2 receiveassignPickup_8_opt2 (guard a1 c1 d2)
o1 sendapprovePickup_9_opt2 (guard a1 c1)
p1 receiveapprovePickup_9_opt2 (guard a1 c1)
q1 sendackOrder_10 (guard a1)
r1 receiveackOrder_10 (guard a1)
s1 sendcheckOrder_11
s2 checkloop_releasePickup_3_loop1 (guard a1)
t1 receivecheckOrder_11

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1 → o1 → p1 → q1 → r1
    │   ├── s1 → t1
    │   └── s2 ↺ e1
    └── m2 → n2 → o1 → p1 → q1 → r1
        ├── s1 → t1
        └── s2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 -> n2 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 -> n2 -> o1 -> p1 -> q1 -> r1 -> s2 ↺ e1
```

model_104 GroceryDeliverySlot · fail

Guard

a1	retryNo < 8
b1	retryNo mod 3 = 0
b2	retryNo mod 3 ≠ 0

```
(root)
└── a1  retryNo < 8
    ├── b1  retryNo mod 3 = 0
    └── b2  retryNo mod 3 ≠ 0
```

เส้นทาง

a1 -> b1
a1 -> b2

Event

a1 sendapproveOrder_1
b1 receiveapproveOrder_1
c1 sendfetchSlot_2
d1 receivefetchSlot_2
e1 sendreleaseOrder_3 (guard a1)
f1 receivereleaseOrder_3 (guard a1)
g1 sendqueryOrder_4 (guard a1 b1)
g2 sendreportSlot_5 (guard a1 b2)
h1 receivequeryOrder_4 (guard a1 b1)
h2 receivereportSlot_5 (guard a1 b2)
i1 sendnotifySlot_12 (guard a1)
j1 receivenotifySlot_12 (guard a1)
k1 sendcancelSlot_13
k2 checkloop_releaseOrder_3_loop1 (guard a1)
l1 receivecancelSlot_13
m1 sendsyncSlot_6 (guard a1 b2)
n1 receivesyncSlot_6 (guard a1 b2)
o1 sendnotifyOrder_7 (guard a1 b2)
p1 receivenotifyOrder_7 (guard a1 b2)
q1 sendreportDelivery_8 (guard a1 b2)
r1 receivereportDelivery_8 (guard a1 b2)
s1 sendreserveOrder_9 (guard a1 b2)
t1 receivereserveOrder_9 (guard a1 b2)
u1 sendassignDelivery_10 (guard a1 b2)
v1 receiveassignDelivery_10 (guard a1 b2)
w1 sendfetchDelivery_11 (guard a1 b2)
x1 receivefetchDelivery_11 (guard a1 b2)

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1
    │   ├── k1 → l1
    │   └── k2 ↺ e1
    └── g2 → h2 → m1 → n1 → o1 → p1 → q1 → r1 →
          s1 → t1 → u1 → v1 → w1 → x1 → i1 → j1
        ├── k1 → l1
        └── k2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1 -> i1 -> j1 -> k2 ↺ e1

model_133 OnlineGroceryPicking · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendlogOrder_1
b1 receivelogOrder_1
c1 sendassignOrder_2
d1 receiveassignOrder_2
e1 sendcancelOrder_3
f1 receivecancelOrder_3
g1 sendfetchOrder_4
h1 receivefetchOrder_4
i1 sendfetchItem_5
j1 receivefetchItem_5
k1 sendcancelItem_6
l1 receivecancelItem_6
m1 sendconfirmBag_7
n1 receiveconfirmBag_7
o1 sendapproveOrder_8
p1 receiveapproveOrder_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1

model_143 PhotoPrintKiosk · pass

Guard

a1	t_obs ≤ 13
a2	t_obs > 13

```
(root)
├── a1  t_obs ≤ 13
└── a2  t_obs > 13
```

เส้นทาง

a1
a2

Event

a1 sendcheckPrint_1
b1 receivecheckPrint_1
c1 sendassignPrint_2
d1 receiveassignPrint_2
e1 sendassignOrder_3 (guard a1)
e2 sendupdatePhoto_5 (guard a2)
f1 receiveassignOrder_3 (guard a1)
f2 receiveupdatePhoto_5 (guard a2)
g1 sendreservePhoto_4 (guard a1)
h1 receivereservePhoto_4 (guard a1)
i1 sendschedulePrint_6
j1 receiveschedulePrint_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    └── e2 → f2 → i1 → j1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> i1 -> j1

model_162 SelfCheckoutLane · pass

Guard

a1	t_obs ≤ 10
a2	t_obs > 10
b1	stepNo < 3

```
(root)
├── a1  t_obs ≤ 10
│   └── b1  stepNo < 3
└── a2  t_obs > 10
    └── b1  stepNo < 3
```

เส้นทาง

a1 -> b1
a2 -> b1

Event

a1 sendreportItem_1
b1 receivereportItem_1
c1 sendassignItem_2
d1 receiveassignItem_2
e1 sendcancelItem_3 (guard a1)
e2 sendscheduleItem_4 (guard a2)
f1 receivecancelItem_3 (guard a1)
f2 receivescheduleItem_4 (guard a2)
g1 sendnotifyReceipt_5
h1 receivenotifyReceipt_5
i1 sendissueItem_6
j1 receiveissueItem_6
k1 sendissueReceipt_7 (guard b1)
l1 receiveissueReceipt_7 (guard b1)
m1 sendassignReceipt_8 (guard b1)
n1 receiveassignReceipt_8 (guard b1)
o1 sendreportReceipt_9
o2 checkloop_issueReceipt_7_loop1 (guard b1)
p1 receivereportReceipt_9

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1 →
    │     m1 → n1
    │   ├── o1 → p1
    │   └── o2 ↺ k1
    └── e2 → f2 → g1 → h1 → i1 → j1 → k1 → l1 →
          m1 → n1
        ├── o1 → p1
        └── o2 ↺ k1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o2 ↺ k1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o2 ↺ k1

model_175 SmartShoppingCart · pass

Guard

a1	useCache = 1

```
(root)
└── a1  useCache = 1
```

เส้นทาง
a1

Event

a1 sendsyncTotal_1 (guard a1)
b1 receivesyncTotal_1
c1 sendcancelCoupon_2
d1 receivecancelCoupon_2
e1 sendapproveCoupon_3_opt1 (guard a1)
f1 receiveapproveCoupon_3_opt1
g1 sendqueryItem_4
h1 receivequeryItem_4
i1 sendupdateCoupon_5
j1 receiveupdateCoupon_5
k1 sendissueCoupon_6
l1 receiveissueCoupon_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_178 SmartVendingFridge · fail

Guard

a1	attemptNo < 9
b1	attemptNo mod 4 = 0

```
(root)
└── a1  attemptNo < 9
    └── b1  attemptNo mod 4 = 0
```

เส้นทาง
a1 -> b1

Event

a1 sendackItem_1 a1 b1)
b1 receiveackItem_1
c1 sendreportCharge_2
d1 receivereportCharge_2
e1 sendassignItem_3 (guard a1)
f1 receiveassignItem_3 (guard a1)
g1 sendfetchItem_4_opt1 (guard a1 b1)
h1 receivefetchItem_4_opt1 (guard a1 b1)
i1 sendsendRestock_5_opt1 (guard
j1 receivesendRestock_5_opt1 (guard a1 b1)
k1 sendreserveCharge_6 (guard a1)
l1 receivereserveCharge_6 (guard a1)
m1 sendlogRestock_7
m2 checkloop_assignItem_3_loop1 (guard a1)
n1 receivelogRestock_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1
    └── m2 ↺ e1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 ↺ e1

การจอง ที่พัก และกิจกรรม · Booking, Hospitality & Events (19)

model_009 CinemaTicketing · pass

Guard

a1	seatTry < 3
b1	wantSnack = 1

```
(root)
└── a1  seatTry < 3
    └── b1  wantSnack = 1
```

เส้นทาง

```
a1 -> b1
```

Event

a1 sendselectMovie_1
b1 receiveselectMovie_1
c1 sendshowtimes_2
d1 receiveshowtimes_2
e1 sendpickSeat_3 (guard a1)
f1 receivepickSeat_3 (guard a1)
g1 sendseatStatus_4 (guard a1)
h1 receiveseatStatus_4 (guard a1)
i1 sendreserveSeat_5
i2 checkloop_pickSeat_3_loop1 (guard a1)
j1 receivereserveSeat_5
k1 sendaddSnack_6_opt1 (guard b1)
l1 receiveaddSnack_6_opt1 (guard b1)
m1 sendissueTicket_7
n1 receiveissueTicket_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1 → m1 → n1
    └── i2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ e1
```

model_012 ConferenceRoomBooking · fail

Guard

a1	needDisplay = 1

```
(root)
└── a1  needDisplay = 1
```

เส้นทาง

```
a1
```

Event

a1 sendsearchRoom_1
b1 receivesearchRoom_1
c1 sendavailability_2
d1 receiveavailability_2
e1 sendbookRoom_3
f1 receivebookRoom_3
g1 sendcreateEvent_4
h1 receivecreateEvent_4
i1 sendeventCreated_5
j1 receiveeventCreated_5
k1 sendupdatePanel_6_opt1 (guard a1)
l1 receiveupdatePanel_6_opt1 (guard a1)
m1 sendbookingDone_7
n1 receivebookingDone_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
```

model_022 GymCheckIn · fail

Guard

a1	membership = ACTIVE
a2	membership = EXPIRED

```
(root)
├── a1  membership = ACTIVE
└── a2  membership = EXPIRED
```

เส้นทาง

```
a1
a2
```

Event

a1 sendscanQR_1
b1 receivescanQR_1
c1 sendaccessResult_2
d1 receiveaccessResult_2
e1 sendopenGate_3 (guard a1)
e2 sendrenewPrompt_4 (guard a2)
f1 receiveopenGate_3 (guard a1)
f2 receiverenewPrompt_4 (guard a2)
g1 sendexitScan_5
h1 receiveexitScan_5

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1
    └── e2 → f2 → g1 → h1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1
```

model_024 HotelCheckIn · pass

Guard

a1	roomReady = 1
a2	roomReady = 0

```
(root)
├── a1  roomReady = 1
└── a2  roomReady = 0
```

เส้นทาง

a1
a2

Event

a1 sendarrive_1
b1 receivearrive_1
c1 sendlookupBooking_2
d1 receivelookupBooking_2
e1 sendbookingInfo_3
f1 receivebookingInfo_3
g1 sendassignRoom_4 (guard a1)
g2 sendofferLounge_6 (guard a2)
h1 receiveassignRoom_4 (guard a1)
h2 receiveofferLounge_6 (guard a2)
i1 sendissueKeycard_5 (guard a1)
j1 receiveissueKeycard_5 (guard a1)
k1 sendupdateOccupancy_7
l1 receiveupdateOccupancy_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    └── g2 → h2 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1

model_055 ArtGalleryTicketing · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendvalidateEntry_1
b1 receivevalidateEntry_1
c1 sendapproveEntry_2
d1 receiveapproveEntry_2
e1 sendlogEntry_3
f1 receivelogEntry_3
g1 sendrequestEntry_4
h1 receiverequestEntry_4
i1 sendfetchEntry_5
j1 receivefetchEntry_5
k1 sendnotifyEntry_6
l1 receivenotifyEntry_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_069 CarWashBooking · pass

Guard

a1	roundNo < 12
b1	withExtra = 1

```
(root)
└── a1  roundNo < 12
    └── b1  withExtra = 1
```

เส้นทาง

a1 -> b1

Event

a1 sendcancelSlot_1
b1 receivecancelSlot_1
c1 sendfetchWash_2
d1 receivefetchWash_2
e1 sendcancelPayment_3 (guard a1)
f1 receivecancelPayment_3 (guard a1)
g1 sendapproveWash_4 (guard a1)
h1 receiveapproveWash_4 (guard a1)
i1 sendissueWash_5_opt1 (guard a1 b1)
j1 receiveissueWash_5_opt1 (guard a1 b1)
k1 sendconfirmPayment_6 (guard a1)
l1 receiveconfirmPayment_6 (guard a1)
m1 sendconfirmWash_7
m2 checkloop_cancelPayment_3_loop1 (guard a1)
n1 receiveconfirmWash_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1
    └── m2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 ↺ e1

model_074 ConcertTicketQueue · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendsyncSeat_1
b1 receivesyncSeat_1
c1 sendassignSeat_2
d1 receiveassignSeat_2
e1 sendfetchTicket_3
f1 receivefetchTicket_3
g1 sendvalidateQueue_4
h1 receivevalidateQueue_4
i1 sendquerySeat_5
j1 receivequerySeat_5
k1 sendreserveQueue_6
l1 receivereserveQueue_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_076 ConferenceBadgeScan · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendreleaseBadge_1
b1 receivereleaseBadge_1
c1 sendscheduleSession_2
d1 receivescheduleSession_2
e1 sendissueBadge_3
f1 receiveissueBadge_3
g1 sendquerySession_4
h1 receivequerySession_4
i1 sendcancelSession_5
j1 receivecancelSession_5
k1 sendvalidateSession_6
l1 receivevalidateSession_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_078 CoworkingDeskBooking · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendsendDesk_1
b1 receivesendDesk_1
c1 sendreserveBooking_2
d1 receivereserveBooking_2
e1 sendvalidateBooking_3
f1 receivevalidateBooking_3
g1 sendissuePass_4
h1 receiveissuePass_4
i1 sendconfirmPass_5
j1 receiveconfirmPass_5
k1 sendqueryPass_6
l1 receivequeryPass_6
m1 sendsyncBooking_7
n1 receivesyncBooking_7
o1 sendqueryDesk_8
p1 receivequeryDesk_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1

model_090 EventCheckInQR · fail

Guard

a1	passNo < 8
b1	passNo mod 4 = 0
b2	passNo mod 4 ≠ 0
c1	withExtra = 1

```
(root)
└── a1  passNo < 8
    ├── b1  passNo mod 4 = 0
    │   └── c1  withExtra = 1
    └── b2  passNo mod 4 ≠ 0
        └── c1  withExtra = 1
```

เส้นทาง

```
a1 -> b1 -> c1
a1 -> b2 -> c1
```

Event

a1 sendreleaseBadge_1
b1 receivereleaseBadge_1
c1 sendsyncBadge_2
d1 receivesyncBadge_2
e1 sendnotifyEntry_3 (guard a1)
f1 receivenotifyEntry_3 (guard a1)
g1 sendsendBadge_4 (guard a1 b1)
g2 sendupdateBadge_6 (guard a1 b2)
h1 receivesendBadge_4 (guard a1 b1)
h2 receiveupdateBadge_6 (guard a1 b2)
i1 sendreportEntry_5 (guard a1 b1)
j1 receivereportEntry_5 (guard a1 b1)
k1 sendsyncPass_7 (guard a1)
l1 receivesyncPass_7 (guard a1)
m1 sendfetchBadge_8
m2 checkloop_notifyEntry_3_loop1 (guard a1)
n1 receivefetchBadge_8
o1 sendreportBadge_9
p1 receivereportBadge_9
q1 sendassignEntry_10_opt1 (guard c1)
r1 receiveassignEntry_10_opt1 (guard c1)
s1 sendlogEntry_11_opt1 (guard c1)
t1 receivelogEntry_11_opt1 (guard c1)
u1 sendissuePass_12
v1 receiveissuePass_12

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    │   ├── m1 → n1 → o1 → p1 → q1 → r1 → s1 → t1 →
    │   │     u1 → v1
    │   └── m2 ↺ e1
    └── g2 → h2 → k1 → l1
        ├── m1 → n1 → o1 → p1 → q1 → r1 → s1 → t1 →
        │     u1 → v1
        └── m2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m2 ↺ e1
```

model_102 GolfCartFleet · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendfetchFee_1
b1 receivefetchFee_1
c1 sendscheduleRound_2
d1 receivescheduleRound_2
e1 sendfetchCart_3
f1 receivefetchCart_3
g1 sendconfirmCart_4
h1 receiveconfirmCart_4
i1 sendapproveCart_5
j1 receiveapproveCart_5
k1 sendissueCart_6
l1 receiveissueCart_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
```

model_111 HotelKeylessEntry · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendfetchAccess_1
b1 receivefetchAccess_1
c1 sendscheduleRoom_2
d1 receivescheduleRoom_2
e1 sendfetchKey_3
f1 receivefetchKey_3
g1 sendconfirmKey_4
h1 receiveconfirmKey_4
i1 sendapproveKey_5
j1 receiveapproveKey_5
k1 sendissueKey_6
l1 receiveissueKey_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1
```

model_112 HotelRoomService · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendconfirmMeal_1
b1 receiveconfirmMeal_1
c1 sendackBill_2
d1 receiveackBill_2
e1 sendvalidateMeal_3
f1 receivevalidateMeal_3
g1 sendapproveBill_4
h1 receiveapproveBill_4
i1 sendrequestBill_5
j1 receiverequestBill_5
k1 sendupdateBill_6
l1 receiveupdateBill_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1
```

model_123 MarathonTiming · fail

Guard

a1	roundNo < 5
b1	roundNo mod 2 = 0
c1	load < 10
d1	retryNo < 11
c2	load ≥ 10

```
(root)
└── a1  roundNo < 5
    └── b1  roundNo mod 2 = 0
        ├── c1  load < 10
        │   └── d1  retryNo < 11
        └── c2  load ≥ 10
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1
a1 -> b1 -> c2
```

Event

a1 sendreleaseSplit_1
b1 receivereleaseSplit_1
c1 sendquerySplit_2
d1 receivequerySplit_2
e1 sendcancelBib_3 (guard a1)
f1 receivecancelBib_3 (guard a1)
g1 sendrequestBib_4_opt1 (guard a1 b1)
h1 receiverequestBib_4_opt1 (guard a1 b1)
i1 sendrequestFinish_5 (guard a1)
j1 receiverequestFinish_5 (guard a1)
k1 sendassignSplit_6 (guard a1)
l1 receiveassignSplit_6 (guard a1)
m1 sendcancelFinish_7
m2 checkloop_cancelBib_3_loop1 (guard a1)
n1 receivecancelFinish_7
o1 sendsendFinish_8
p1 receivesendFinish_8
q1 sendfetchFinish_9 (guard c1)
q2 sendreleaseFinish_13 (guard c2)
r1 receivefetchFinish_9 (guard c1)
r2 receivereleaseFinish_13 (guard c2)
s1 sendreserveSplit_10 (guard c1 d1)
t1 receivereserveSplit_10 (guard c1 d1)
u1 sendreserveFinish_11 (guard c1 d1)
v1 receivereserveFinish_11 (guard c1 d1)
w1 sendvalidateFinish_12 (guard c1)
w2 checkloop_reserveSplit_10_loop2 (guard c1 d1)
x1 receivevalidateFinish_12 (guard c1)
y1 sendsendBib_14
z1 receivesendBib_14

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1 → o1 → p1
    │   ├── q1 → r1 → s1 → t1 → u1 → v1
    │   │   ├── w1 → x1 → y1 → z1
    │   │   └── w2 ↺ s1
    │   └── q2 → r2 → y1 → z1
    └── m2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1 -> y1 -> z1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w2 ↺ s1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q2 -> r2 -> y1 -> z1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 ↺ e1
```

model_127 MeetingRoomDisplay · pass

Guard

a1	isVip = 1
a2	isVip = 0

```
(root)
├── a1  isVip = 1
└── a2  isVip = 0
```

เส้นทาง

```
a1
a2
```

Event

a1 sendcancelMeeting_1
b1 receivecancelMeeting_1
c1 sendconfirmCheckin_2
d1 receiveconfirmCheckin_2
e1 sendconfirmMeeting_3 (guard a1)
e2 sendrequestMeeting_5 (guard a2)
f1 receiveconfirmMeeting_3 (guard a1)
f2 receiverequestMeeting_5 (guard a2)
g1 sendupdateRoom_4 (guard a1)
h1 receiveupdateRoom_4 (guard a1)
i1 sendconfirmRoom_6
j1 receiveconfirmRoom_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    └── e2 → f2 → i1 → j1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> i1 -> j1
```

model_129 MuseumAudioGuide · pass

Guard

a1	useCache = 1
b1	stepNo < 11

```
(root)
└── a1  useCache = 1
    └── b1  stepNo < 11
```

เส้นทาง

```
a1 -> b1
```

Event

a1 sendconfirmTour_1
b1 receiveconfirmTour_1
c1 sendnotifyTour_2
d1 receivenotifyTour_2
e1 sendreserveTour_3_opt1 (guard a1)
f1 receivereserveTour_3_opt1 (guard a1)
g1 sendcancelTour_4
h1 receivecancelTour_4
i1 sendvalidateTour_5
j1 receivevalidateTour_5
k1 sendcheckTour_6 (guard b1)
l1 receivecheckTour_6 (guard b1)
m1 sendvalidateAudio_7
m2 checkloop_checkTour_6_loop1 (guard b1)
n1 receivevalidateAudio_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1
    └── m2 ↺ k1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m2 ↺ k1
```

model_155 RestaurantReservation · pass

Guard

a1	express = 1
b1	roundNo < 8
c1	needAck = 1

```
(root)
└── a1  express = 1
    └── b1  roundNo < 8
        └── c1  needAck = 1
```

เส้นทาง

```
a1 -> b1 -> c1
```

Event

a1 sendfetchMenu_1
b1 receivefetchMenu_1
c1 sendreserveTable_2
d1 receivereserveTable_2
e1 sendqueryBooking_3_opt1 (guard a1)
f1 receivequeryBooking_3_opt1 (guard a1)
g1 sendsyncMenu_4
h1 receivesyncMenu_4
i1 sendlogMenu_5
j1 receivelogMenu_5
k1 sendapproveMenu_6 (guard b1)
l1 receiveapproveMenu_6 (guard b1)
m1 sendackBooking_7_opt2 (guard b1 c1)
n1 receiveackBooking_7_opt2 (guard b1 c1)
o1 sendreportMenu_8 (guard b1)
p1 receivereportMenu_8 (guard b1)
q1 sendassignBooking_9
q2 checkloop_approveMenu_6_loop1 (guard b1)
r1 receiveassignBooking_9

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
    ├── q1 → r1
    └── q2 ↺ k1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1 -> o1 -> p1 -> q2 ↺ k1
```

model_169 SmartGymLocker · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendsendLocker_1
b1 receivesendLocker_1
c1 sendreservePin_2
d1 receivereservePin_2
e1 sendvalidatePin_3
f1 receivevalidatePin_3
g1 sendissueSession_4
h1 receiveissueSession_4
i1 sendconfirmSession_5
j1 receiveconfirmSession_5
k1 sendquerySession_6
l1 receivequerySession_6
m1 sendsyncPin_7
n1 receivesyncPin_7
o1 sendqueryLocker_8
p1 receivequeryLocker_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1

model_182 StadiumEntryGate · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendreserveSeat_1
b1 receivereserveSeat_1
c1 sendlogSeat_2
d1 receivelogSeat_2
e1 sendconfirmEntry_3
f1 receiveconfirmEntry_3
g1 sendlogEntry_4
h1 receivelogEntry_4
i1 sendsyncTicket_5
j1 receivesyncTicket_5
k1 sendscheduleEntry_6
l1 receivescheduleEntry_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

การศึกษาและบริการภาครัฐ · Education & Public Services (17)

model_013 CourseEnrollment · fail

Guard

a1	needsApproval = 1

```
(root)
└── a1  needsApproval = 1
```

เส้นทาง

a1

Event

a1 sendbrowseCourses_1
b1 receivebrowseCourses_1
c1 sendcourseList_2
d1 receivecourseList_2
e1 sendenrollRequest_3
f1 receiveenrollRequest_3
g1 sendcheckPrereq_4
h1 receivecheckPrereq_4
i1 sendprereqOK_5
j1 receiveprereqOK_5
k1 sendrequestApproval_6_opt1 (guard a1)
l1 receiverequestApproval_6_opt1 (guard a1)
m1 sendapproved_7_opt1 (guard a1)
n1 receiveapproved_7_opt1 (guard a1)
o1 sendenrollConfirm_8
p1 receiveenrollConfirm_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1

model_017 EmergencyDispatch · pass

Guard

a1	priority = HIGH
b1	updateCount < 4
c1	trafficJam = 1
a2	priority = LOW

```
(root)
├── a1  priority = HIGH
│   └── b1  updateCount < 4
│       └── c1  trafficJam = 1
└── a2  priority = LOW
```

เส้นทาง

```
a1 -> b1 -> c1
a2
```

Event

a1 sendreportEmergency_1
b1 receivereportEmergency_1
c1 sendaskDetails_2
d1 receiveaskDetails_2
e1 sendprovideDetails_3
f1 receiveprovideDetails_3
g1 senddispatchAmbulance_4 (guard a1)
g2 sendscheduleCallback_8 (guard a2)
h1 receivedispatchAmbulance_4 (guard a1)
h2 receivescheduleCallback_8 (guard a2)
i1 sendsendLocation_5 (guard a1 b1)
j1 receivesendLocation_5 (guard a1 b1)
k1 sendrerouteRequest_6_opt1 (guard a1 b1 c1)
l1 receivererouteRequest_6_opt1 (guard a1 b1 c1)
m1 sendnotifyHospital_7 (guard a1)
m2 checkloop_sendLocation_5_loop1 (guard a1 b1)
n1 receivenotifyHospital_7 (guard a1)
o1 sendcloseCase_9
p1 receivecloseCase_9

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    │   ├── m1 → n1 → o1 → p1
    │   └── m2 ↺ i1
    └── g2 → h2 → o1 → p1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 ↺ i1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> o1 -> p1
```

model_027 LibraryLoan · pass

Guard

a1	fine ≤ 100
b1	wantReminder = 1
a2	fine > 100

```
(root)
├── a1  fine ≤ 100
│   └── b1  wantReminder = 1
└── a2  fine > 100
```

เส้นทาง

```
a1 -> b1
a2
```

Event

a1 sendscanCard_1
b1 receivescanCard_1
c1 sendcheckMember_2
d1 receivecheckMember_2
e1 sendmemberInfo_3
f1 receivememberInfo_3
g1 sendborrowBook_4 (guard a1)
g2 sendrejectLoan_7 (guard a2)
h1 receiveborrowBook_4 (guard a1)
h2 receiverejectLoan_7 (guard a2)
i1 sendrecordLoan_5 (guard a1)
j1 receiverecordLoan_5 (guard a1)
k1 sendsetReminder_6_opt1 (guard a1 b1)
l1 receivesetReminder_6_opt1 (guard a1 b1)
m1 sendlogout_8
n1 receivelogout_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1
    └── g2 → h2 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> m1 -> n1
```

model_031 OnlineExam · pass

Guard

a1	questionNo < 20
b1	autosaveCount < 3

```
(root)
└── a1  questionNo < 20
    └── b1  autosaveCount < 3
```

เส้นทาง

a1 -> b1

Event

a1 sendlogin_1
b1 receivelogin_1
c1 sendfetchQuestion_2 (guard a1)
d1 receivefetchQuestion_2 (guard a1)
e1 sendquestionData_3 (guard a1)
f1 receivequestionData_3 (guard a1)
g1 sendautosave_4 (guard a1 b1)
h1 receiveautosave_4 (guard a1 b1)
i1 sendsaveAck_5 (guard a1 b1)
j1 receivesaveAck_5 (guard a1 b1)
k1 sendsubmitAnswer_6 (guard a1)
k2 checkloop_autosave_4_loop2 (guard a1 b1)
l1 receivesubmitAnswer_6 (guard a1)
m1 sendflagBehavior_7
m2 checkloop_fetchQuestion_2_loop1 (guard a1)
n1 receiveflagBehavior_7
o1 sendexamResult_8
p1 receiveexamResult_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1
    │   ├── m1 → n1 → o1 → p1
    │   └── m2 ↺ c1
    └── k2 ↺ g1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 ↺ c1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ g1

model_059 BoardingSchoolGate · fail

Guard

a1	passNo < 5
b1	urgent = 1
b2	urgent = 0

```
(root)
└── a1  passNo < 5
    ├── b1  urgent = 1
    └── b2  urgent = 0
```

เส้นทาง

a1 -> b1
a1 -> b2

Event

a1 sendsendPass_1
b1 receivesendPass_1
c1 sendscheduleNotice_2
d1 receivescheduleNotice_2
e1 sendrequestExit_3 (guard a1)
f1 receiverequestExit_3 (guard a1)
g1 sendapproveExit_4
g2 checkloop_requestExit_3_loop1 (guard a1)
h1 receiveapproveExit_4
i1 sendvalidateExit_5
j1 receivevalidateExit_5
k1 sendapproveNotice_6 (guard b1)
k2 sendcancelExit_8 (guard b2)
l1 receiveapproveNotice_6 (guard b1)
l2 receivecancelExit_8 (guard b2)
m1 sendnotifyNotice_7 (guard b1)
m2 sendsyncExit_9 (guard b2)
n1 receivenotifyNotice_7 (guard b1)
n2 receivesyncExit_9 (guard b2)
o1 sendupdateExit_10
p1 receiveupdateExit_10

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1
    │   ├── k1 → l1 → m1 → n1 → o1 → p1
    │   └── k2 → l2 → m2 → n2 → o1 → p1
    └── g2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 -> l2 -> m2 -> n2 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 ↺ e1

model_071 ChildcareCheckin · fail

Guard

a1	approved = 1
a2	approved = 0
b1	retryNo < 4

```
(root)
├── a1  approved = 1
│   └── b1  retryNo < 4
└── a2  approved = 0
    └── b1  retryNo < 4
```

เส้นทาง

```
a1 -> b1
a2 -> b1
```

Event

a1 sendissuePickup_1
b1 receiveissuePickup_1
c1 sendschedulePickup_2
d1 receiveschedulePickup_2
e1 sendqueryNote_3 (guard a1)
e2 sendreserveNote_5 (guard a2)
f1 receivequeryNote_3 (guard a1)
f2 receivereserveNote_5 (guard a2)
g1 sendackPickup_4 (guard a1)
h1 receiveackPickup_4 (guard a1)
i1 sendscheduleChild_6
j1 receivescheduleChild_6
k1 sendvalidateNote_7
l1 receivevalidateNote_7
m1 sendcheckChild_8 (guard b1)
n1 receivecheckChild_8 (guard b1)
o1 sendcancelChild_9
o2 checkloop_checkChild_8_loop1 (guard b1)
p1 receivecancelChild_9

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1 →
    │     m1 → n1
    │   ├── o1 → p1
    │   └── o2 ↺ m1
    └── e2 → f2 → i1 → j1 → k1 → l1 → m1 → n1
        ├── o1 → p1
        └── o2 ↺ m1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1 -> m1 -> n1 -> o2 ↺ m1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> i1 -> j1 -> k1 -> l1 -> m1 -
> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> i1 -> j1 -> k1 -> l1 -> m1 -
> n1 -> o2 ↺ m1
```

model_077 CourseAutoGrading · pass

Guard

a1	mode = MANUAL
a2	mode = AUTO
b1	t_obs ≤ 27
b2	t_obs > 27

```
(root)
├── a1  mode = MANUAL
│   ├── b1  t_obs ≤ 27
│   └── b2  t_obs > 27
└── a2  mode = AUTO
    ├── b1  t_obs ≤ 27
    └── b2  t_obs > 27
```

เส้นทาง

```
a1 -> b1
a1 -> b2
a2 -> b1
a2 -> b2
```

Event

a1 sendissueFeedback_1
b1 receiveissueFeedback_1
c1 sendassignFeedback_2
d1 receiveassignFeedback_2
e1 sendreserveGrade_3 (guard a1)
e2 sendlogGrade_4 (guard a2)
f1 receivereserveGrade_3 (guard a1)
f2 receivelogGrade_4 (guard a2)
g1 sendackGrade_6
h1 receiveackGrade_6
i1 sendnotifyGrade_7
j1 receivenotifyGrade_7
k1 sendupdateGrade_8 (guard b1)
k2 sendvalidateSubmission_9 (guard b2)
l1 receiveupdateGrade_8 (guard b1)
l2 receivevalidateSubmission_9 (guard b2)
m1 sendreleaseSubmission_11
n1 receivereleaseSubmission_11
o1 sendackSubmission_10 (guard b2)
p1 receiveackSubmission_10 (guard b2)
q1 sendcheckSubmission_5 (guard a2)
r1 receivecheckSubmission_5 (guard a2)

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    │   ├── k1 → l1 → m1 → n1
    │   └── k2 → l2 → o1 → p1 → m1 → n1
    └── e2 → f2 → q1 → r1 → g1 → h1 → i1 → j1
        ├── k1 → l1 → m1 → n1
        └── k2 → l2 → o1 → p1 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 -> l2 -> o1 -> p1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> q1 -> r1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> q1 -> r1 -> g1 -> h1 -> i1 -> j1 -> k2 -> l2 -> o1 -> p1 -> m1 -> n1
```

model_083 DigitalLibraryAccess · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendsyncLoan_1
b1 receivesyncLoan_1
c1 sendfetchLoan_2
d1 receivefetchLoan_2
e1 sendsendLicense_3
f1 receivesendLicense_3
g1 sendreportLoan_4
h1 receivereportLoan_4
i1 sendreleaseLoan_5
j1 receivereleaseLoan_5
k1 sendcheckEbook_6
l1 receivecheckEbook_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
```

model_084 DigitalVotingBooth · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendcancelBallot_1
b1 receivecancelBallot_1
c1 sendscheduleVote_2
d1 receivescheduleVote_2
e1 sendlogVote_3
f1 receivelogVote_3
g1 sendupdateReceipt_4
h1 receiveupdateReceipt_4
i1 sendvalidateReceipt_5
j1 receivevalidateReceipt_5
k1 sendsendVote_6
l1 receivesendVote_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1
```

model_085 DisasterAlertBroadcast · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendassignQuake_1
b1 receiveassignQuake_1
c1 sendfetchQuake_2
d1 receivefetchQuake_2
e1 sendreportShelter_3
f1 receivereportShelter_3
g1 sendfetchShelter_4
h1 receivefetchShelter_4
i1 sendqueryAlert_5
j1 receivequeryAlert_5
k1 sendissueAlert_6
l1 receiveissueAlert_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1
```

model_118 LabEquipmentBooking · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendreserveSample_1
b1 receivereserveSample_1
c1 sendreleaseSample_2
d1 receivereleaseSample_2
e1 sendsendSample_3
f1 receivesendSample_3
g1 sendcheckSample_4
h1 receivecheckSample_4
i1 sendsendSlot_5
j1 receivesendSlot_5
k1 sendissueSlot_6
l1 receiveissueSlot_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -
> l1
```

model_138 PassportRenewalKiosk · pass

Guard

a1	t_obs ≤ 4
a2	t_obs > 4

```
(root)
├── a1  t_obs ≤ 4
└── a2  t_obs > 4
```

เส้นทาง

a1
a2

Event

a1 sendackPhoto_1
b1 receiveackPhoto_1
c1 sendreportForm_2
d1 receivereportForm_2
e1 sendsendPassport_3 (guard a1)
e2 sendcancelPassport_4 (guard a2)
f1 receivesendPassport_3 (guard a1)
f2 receivecancelPassport_4 (guard a2)
g1 sendreserveForm_5
h1 receivereserveForm_5
i1 sendlogPassport_6
j1 receivelogPassport_6
k1 sendfetchPassport_7
l1 receivefetchPassport_7
m1 sendvalidatePhoto_8
n1 receivevalidatePhoto_8
o1 sendvalidateForm_9
p1 receivevalidateForm_9
q1 sendcheckPhoto_10
r1 receivecheckPhoto_10
s1 sendsendForm_11
t1 receivesendForm_11
u1 sendsyncPhoto_12
v1 receivesyncPhoto_12

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1 →
    │     m1 → n1 → o1 → p1 → q1 → r1 → s1 → t1 →
    │     u1 → v1
    └── e2 → f2 → g1 → h1 → i1 → j1 → k1 → l1 →
          m1 → n1 → o1 → p1 → q1 → r1 → s1 → t1 →
          u1 → v1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1

model_148 PrisonVisitBooking · fail

Guard

a1	passNo < 9
b1	passNo mod 2 = 0
c1	optIn = 1

```
(root)
└── a1  passNo < 9
    └── b1  passNo mod 2 = 0
        └── c1  optIn = 1
```

เส้นทาง

a1 -> b1 -> c1

Event

a1 sendackVisit_1
b1 receiveackVisit_1
c1 sendsendSlot_2
d1 receivesendSlot_2
e1 sendqueryClearance_3 (guard a1)
f1 receivequeryClearance_3 (guard a1)
g1 sendcancelSlot_4_opt1 (guard a1 b1)
h1 receivecancelSlot_4_opt1 (guard a1 b1)
i1 sendissueVisit_5 (guard a1)
j1 receiveissueVisit_5 (guard a1)
k1 sendissueClearance_6_opt2 (guard a1 c1)
l1 receiveissueClearance_6_opt2 (guard a1 c1)
m1 sendrequestClearance_7 (guard a1)
n1 receiverequestClearance_7 (guard a1)
o1 sendsyncVisit_8 (guard a1)
p1 receivesyncVisit_8 (guard a1)
q1 sendapproveClearance_9
q2 checkloop_queryClearance_3_loop1 (guard a1)
r1 receiveapproveClearance_9

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
    ├── q1 → r1
    └── q2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q2 ↺ e1

model_152 RemoteExamProctor · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendackExam_1
b1 receiveackExam_1
c1 sendapproveExam_2
d1 receiveapproveExam_2
e1 sendscheduleSession_3
f1 receivescheduleSession_3
g1 sendrequestExam_4
h1 receiverequestExam_4
i1 sendconfirmFlag_5
j1 receiveconfirmFlag_5
k1 sendnotifyFlag_6
l1 receivenotifyFlag_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_171 SmartLibraryShelf · fail

Guard

a1	roundNo < 12
b1	roundNo mod 4 = 0
c1	grade = ACTIVE
c2	grade = IDLE

```
(root)
└── a1  roundNo < 12
    └── b1  roundNo mod 4 = 0
        ├── c1  grade = ACTIVE
        └── c2  grade = IDLE
```

เส้นทาง

a1 -> b1 -> c1
a1 -> b1 -> c2

Event

a1 sendsyncShelf_1
b1 receivesyncShelf_1
c1 sendreserveShelf_2
d1 receivereserveShelf_2
e1 sendnotifyHold_3 (guard a1)
f1 receivenotifyHold_3 (guard a1)
g1 sendrequestBook_4_opt1 (guard a1 b1)
h1 receiverequestBook_4_opt1 (guard a1 b1)
i1 sendupdateHold_5_opt1 (guard a1 b1)
j1 receiveupdateHold_5_opt1 (guard a1 b1)
k1 sendapproveBook_6 (guard a1)
l1 receiveapproveBook_6 (guard a1)
m1 sendreportBook_7 (guard a1)
n1 receivereportBook_7 (guard a1)
o1 sendsyncBook_8
o2 checkloop_notifyHold_3_loop1 (guard a1)
p1 receivesyncBook_8
q1 sendrequestHold_9
r1 receiverequestHold_9
s1 sendcancelShelf_10 (guard c1)
s2 sendcancelHold_12 (guard c2)
t1 receivecancelShelf_10 (guard c1)
t2 receivecancelHold_12 (guard c2)
u1 sendlogBook_11 (guard c1)
v1 receivelogBook_11 (guard c1)
w1 sendreserveBook_13
x1 receivereserveBook_13

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1 → q1 → r1
    │   ├── s1 → t1 → u1 → v1 → w1 → x1
    │   └── s2 → t2 → w1 → x1
    └── o2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s2 -> t2 -> w1 -> x1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o2 ↺ e1

model_184 StudentAttendance · pass

Guard

a1	useCache = 1

```
(root)
└── a1  useCache = 1
```

เส้นทาง
a1

Event

a1 sendsyncReport_1
b1 receivesyncReport_1
c1 sendcancelAbsence_2
d1 receivecancelAbsence_2
e1 sendapproveAbsence_3_opt1 (guard a1)
f1 receiveapproveAbsence_3_opt1 (guard a1)
g1 sendqueryCheckin_4
h1 receivequeryCheckin_4
i1 sendupdateAbsence_5
j1 receiveupdateAbsence_5
k1 sendissueAbsence_6
l1 receiveissueAbsence_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_185 TaxFilingPortal · pass

Guard
ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendreportReturn_1
b1 receivereportReturn_1
c1 sendreleaseRefund_2
d1 receivereleaseRefund_2
e1 sendassignRefund_3
f1 receiveassignRefund_3
g1 sendschedulePayment_4
h1 receiveschedulePayment_4
i1 sendupdatePayment_5
j1 receiveupdatePayment_5
k1 sendissueReturn_6
l1 receiveissueReturn_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

ไอทีและบริการดิจิทัล · IT & Digital Services (19)

model_008 ChatMessaging · fail

Guard

a1	typing = 1

```
(root)
└── a1  typing = 1
```

เส้นทาง
a1

Event

a1 sendsendText_1
b1 receivesendText_1
c1 senddeliverText_2
d1 receivedeliverText_2
e1 sendreadReceipt_3
f1 receivereadReceipt_3
g1 sendforwardReceipt_4
h1 receiveforwardReceipt_4
i1 sendtypingIndicator_5_opt1 (guard a1)
j1 receivetypingIndicator_5_opt1 (guard a1)
k1 sendrelayTyping_6_opt1 (guard a1)
l1 receiverelayTyping_6_opt1 (guard a1)
m1 sendreplyText_7
n1 receivereplyText_7
o1 senddeliverReply_8
p1 receivedeliverReply_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1

model_016 EmailVerification · pass

Guard

a1	resendCount < 3
b1	verifyAttempt < 5

```
(root)
└── a1  resendCount < 3
    └── b1  verifyAttempt < 5
```

เส้นทาง

```
a1 -> b1
```

Event

a1 sendregister_1
b1 receiveregister_1
c1 sendsendCode_2
d1 receivesendCode_2
e1 sendresendRequest_3 (guard a1)
f1 receiveresendRequest_3 (guard a1)
g1 sendresendCode_4 (guard a1)
h1 receiveresendCode_4 (guard a1)
i1 sendsubmitCode_5 (guard b1)
i2 checkloop_resendRequest_3_loop1 (guard a1)
j1 receivesubmitCode_5 (guard b1)
k1 sendverifyResult_6 (guard b1)
l1 receiveverifyResult_6 (guard b1)
m1 sendaccountActivated_7
m2 checkloop_submitCode_5_loop2 (guard b1)
n1 receiveaccountActivated_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1
    │   ├── m1 → n1
    │   └── m2 ↺ i1
    └── i2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 ↺ i1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ e1
```

model_028 LoginRateLimiter · fail

Guard

a1	failCount < 5
b1	failCount mod 2 = 0

```
(root)
└── a1  failCount < 5
    └── b1  failCount mod 2 = 0
```

เส้นทาง

```
a1 -> b1
```

Event

a1 sendloginForm_1
b1 receiveloginForm_1
c1 sendsubmitLogin_2 (guard a1)
d1 receivesubmitLogin_2 (guard a1)
e1 sendverify_3 (guard a1)
f1 receiveverify_3 (guard a1)
g1 sendverifyResult_4 (guard a1)
h1 receiveverifyResult_4 (guard a1)
i1 sendloginPage_5 (guard a1)
j1 receiveloginPage_5 (guard a1)
k1 sendlockAccount_6_opt1 (guard b1)
k2 checkloop_submitLogin_2_loop1 (guard a1)
l1 receivelockAccount_6_opt1 (guard b1)
m1 sendsessionToken_7
n1 receivesessionToken_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1 → m1 → n1
    └── k2 ↺ c1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ c1
```

model_034 PrinterQueue · fail

Guard

a1	jobCount < 5
b1	retryPrint < 3

```
(root)
└── a1  jobCount < 5
    └── b1  retryPrint < 3
```

เส้นทาง

```
a1 -> b1
```

Event

a1 sendsubmitJob_1
b1 receivesubmitJob_1
c1 sendjobQueued_2
d1 receivejobQueued_2
e1 sendpagePrinted_3 (guard a1)
f1 receivepagePrinted_3 (guard a1)
g1 sendnextPage_4 (guard a1)
h1 receivenextPage_4 (guard a1)
i1 sendreprint_5 (guard b1)
i2 checkloop_pagePrinted_3_loop1 (guard a1)
j1 receivereprint_5 (guard b1)
k1 sendreprintDone_6 (guard b1)
l1 receivereprintDone_6 (guard b1)
m1 sendjobComplete_7
m2 checkloop_reprint_5_loop2 (guard b1)
n1 receivejobComplete_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1
    │   ├── m1 → n1
    │   └── m2 ↺ i1
    └── i2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 ↺ i1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ e1
```

model_035 SensorDataPipeline · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendpublishReading_1
b1 receivepublishReading_1
c1 sendackReading_2
d1 receiveackReading_2
e1 sendaggregateBatch_3
f1 receiveaggregateBatch_3
g1 sendstoreConfirm_4
h1 receivestoreConfirm_4
i1 sendsyncClock_5
j1 receivesyncClock_5
k1 sendclockSynced_6
l1 receiveclockSynced_6
m1 sendheartbeat_7
n1 receiveheartbeat_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
```

model_036 ServerAutoScaling · pass

Guard

a1	checkNo < 12
b1	cpu > 80
b2	cpu ≤ 80

```
(root)
└── a1  checkNo < 12
    ├── b1  cpu > 80
    └── b2  cpu ≤ 80
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

a1 sendstartMonitoring_1
b1 receivestartMonitoring_1
c1 sendreportLoad_2 (guard a1)
d1 receivereportLoad_2 (guard a1)
e1 sendaddVM_3 (guard a1 b1)
e2 sendkeepCapacity_6 (guard a1 b2)
f1 receiveaddVM_3 (guard a1 b1)
f2 receivekeepCapacity_6 (guard a1 b2)
g1 sendvmReady_4 (guard a1 b1)
h1 receivevmReady_4 (guard a1 b1)
i1 sendregisterVM_5 (guard a1 b1)
j1 receiveregisterVM_5 (guard a1 b1)
k1 sendstopMonitoring_7
k2 checkloop_reportLoad_2_loop1 (guard a1)
l1 receivestopMonitoring_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    │   ├── k1 → l1
    │   └── k2 ↺ c1
    └── e2 → f2
        ├── k1 → l1
        └── k2 ↺ c1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ c1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> k2 ↺ c1
```

model_054 ArchiveDocumentScan · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendreserveBatch_1
b1 receivereserveBatch_1
c1 sendreleaseBatch_2
d1 receivereleaseBatch_2
e1 sendsendBatch_3
f1 receivesendBatch_3
g1 sendcheckBatch_4
h1 receivecheckBatch_4
i1 sendsendDocument_5
j1 receivesendDocument_5
k1 sendissueDocument_6
l1 receiveissueDocument_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
```

model_061 BulkSMSGateway · fail

Guard

a1	retryNo < 11
b1	retryNo mod 3 = 0

```
(root)
└── a1  retryNo < 11
    └── b1  retryNo mod 3 = 0
```

เส้นทาง

```
a1 -> b1
```

Event

a1 sendconfirmDelivery_1
b1 receiveconfirmDelivery_1
c1 sendapproveCampaign_2
d1 receiveapproveCampaign_2
e1 sendlogCampaign_3 (guard a1)
f1 receivelogCampaign_3 (guard a1)
g1 sendlogDelivery_4_opt1 (guard a1 b1)
h1 receivelogDelivery_4_opt1 (guard a1 b1)
i1 sendnotifyCampaign_5 (guard a1)
j1 receivenotifyCampaign_5 (guard a1)
k1 sendlogMessage_6 (guard a1)
l1 receivelogMessage_6 (guard a1)
m1 sendconfirmCampaign_7
m2 checkloop_logCampaign_3_loop1 (guard a1)
n1 receiveconfirmCampaign_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1
    └── m2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 ↺ e1
```

model_065 CampusWifiAccess · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendcancelSession_1
b1 receivecancelSession_1
c1 sendvalidateToken_2
d1 receivevalidateToken_2
e1 sendsendToken_3
f1 receivesendToken_3
g1 sendissueSession_4
h1 receiveissueSession_4
i1 sendreserveToken_5
j1 receivereserveToken_5
k1 sendsendSession_6
l1 receivesendSession_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
```

model_072 CloudBackupJob · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendupdateSnapshot_1
b1 receiveupdateSnapshot_1
c1 sendconfirmSnapshot_2
d1 receiveconfirmSnapshot_2
e1 sendcheckJob_3
f1 receivecheckJob_3
g1 sendrequestChunk_4
h1 receiverequestChunk_4
i1 sendnotifyChunk_5
j1 receivenotifyChunk_5
k1 sendsyncJob_6
l1 receivesyncJob_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
```

model_073 CloudCIPipeline · pass

Guard

a1	useCache = 1
b1	optIn = 1
c1	retryNo < 11

```
(root)
└── a1  useCache = 1
    └── b1  optIn = 1
        └── c1  retryNo < 11
```

เส้นทาง

a1 -> b1 -> c1

Event

a1 sendreportImage_1
b1 receivereportImage_1
c1 sendissueImage_2
d1 receiveissueImage_2
e1 sendreserveBuild_3_opt1 (guard a1)
f1 receivereserveBuild_3_opt1 (guard a1)
g1 sendnotifyImage_4
h1 receivenotifyImage_4
i1 sendreportCommit_5
j1 receivereportCommit_5
k1 sendissueBuild_6_opt2 (guard b1)
l1 receiveissueBuild_6_opt2 (guard b1)
m1 sendcheckImage_7
n1 receivecheckImage_7
o1 sendscheduleBuild_8
p1 receivescheduleBuild_8
q1 sendlogCommit_9 (guard c1)
r1 receivelogCommit_9 (guard c1)
s1 sendfetchBuild_10 (guard c1)
t1 receivefetchBuild_10 (guard c1)
u1 sendconfirmCommit_11
u2 checkloop_logCommit_9_loop1 (guard c1)
v1 receiveconfirmCommit_11

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1
    ├── u1 → v1
    └── u2 ↺ q1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u2 ↺ q1

model_117 JobApplicationPortal · pass

Guard

a1	stage = ACTIVE
a2	stage = IDLE

```
(root)
├── a1  stage = ACTIVE
└── a2  stage = IDLE
```

เส้นทาง

a1
a2

Event

a1 sendassignInterview_1
b1 receiveassignInterview_1
c1 sendcheckResume_2
d1 receivecheckResume_2
e1 sendvalidateInterview_3
f1 receivevalidateInterview_3
g1 sendreleaseInterview_4
h1 receivereleaseInterview_4
i1 sendackOffer_5
j1 receiveackOffer_5
k1 sendrequestOffer_6
l1 receiverequestOffer_6
m1 sendrequestInterview_7
n1 receiverequestInterview_7
o1 sendupdateOffer_8 (guard a1)
o2 sendcancelResume_9 (guard a2)
p1 receiveupdateOffer_8 (guard a1)
p2 receivecancelResume_9 (guard a2)
q1 sendassignResume_10
r1 receiveassignResume_10

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1 → q1 → r1
    └── o2 → p2 → q1 → r1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o2 -> p2 -> q1 -> r1

model_134 OnlineQuizGame · pass

Guard

a1	attemptNo < 10
b1	roundNo < 6

```
(root)
└── a1  attemptNo < 10
    └── b1  roundNo < 6
```

เส้นทาง

```
a1 -> b1
```

Event

a1 sendcheckRound_1
b1 receivecheckRound_1
c1 sendscheduleAnswer_2
d1 receivescheduleAnswer_2
e1 sendackRound_3 (guard a1)
f1 receiveackRound_3 (guard a1)
g1 sendfetchAnswer_4
g2 checkloop_ackRound_3_loop1 (guard a1)
h1 receivefetchAnswer_4
i1 sendqueryScore_5
j1 receivequeryScore_5
k1 sendissueAnswer_6 (guard b1)
l1 receiveissueAnswer_6 (guard b1)
m1 sendapproveAnswer_7 (guard b1)
n1 receiveapproveAnswer_7 (guard b1)
o1 sendassignScore_8 (guard b1)
p1 receiveassignScore_8 (guard b1)
q1 sendcancelScore_9 (guard b1)
r1 receivecancelScore_9 (guard b1)
s1 sendrequestScore_10 (guard b1)
t1 receiverequestScore_10 (guard b1)
u1 sendreportScore_11 (guard b1)
v1 receivereportScore_11 (guard b1)
w1 sendreserveRound_12
w2 checkloop_issueAnswer_6_loop2 (guard b1)
x1 receivereserveRound_12

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1 →
    │     o1 → p1 → q1 → r1 → s1 → t1 → u1 → v1
    │   ├── w1 → x1
    │   └── w2 ↺ k1
    └── g2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w2 ↺ k1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 ↺ e1
```

model_136 ParentalControlRouter · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendreleaseRule_1
b1 receivereleaseRule_1
c1 sendreserveReport_2
d1 receivereserveReport_2
e1 sendcancelRule_3
f1 receivecancelRule_3
g1 sendupdateRule_4
h1 receiveupdateRule_4
i1 sendnotifyRule_5
j1 receivenotifyRule_5
k1 sendconfirmBlock_6
l1 receiveconfirmBlock_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
```

model_145 PodcastPublishing · fail

Guard

a1	retryNo < 3
b1	retryNo mod 3 = 0
b2	retryNo mod 3 ≠ 0

```
(root)
└── a1  retryNo < 3
    ├── b1  retryNo mod 3 = 0
    └── b2  retryNo mod 3 ≠ 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

a1 sendvalidateFeed_1
b1 receivevalidateFeed_1
c1 sendassignAudio_2
d1 receiveassignAudio_2
e1 sendsendEpisode_3 (guard a1)
f1 receivesendEpisode_3 (guard a1)
g1 sendreserveEpisode_4 (guard a1 b1)
g2 sendcheckFeed_6 (guard a1 b2)
h1 receivereserveEpisode_4 (guard a1 b1)
h2 receivecheckFeed_6 (guard a1 b2)
i1 sendissueAudio_5 (guard a1 b1)
j1 receiveissueAudio_5 (guard a1 b1)
k1 sendsendAudio_7 (guard a1)
l1 receivesendAudio_7 (guard a1)
m1 sendlogAudio_8
m2 checkloop_sendEpisode_3_loop1 (guard a1)
n1 receivelogAudio_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    │   ├── m1 → n1
    │   └── m2 ↺ e1
    └── g2 → h2 → k1 → l1
        ├── m1 → n1
        └── m2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m2 ↺ e1
```

model_147 PrintFleetMonitor · fail

Guard

a1	isVip = 1
a2	isVip = 0

```
(root)
├── a1  isVip = 1
└── a2  isVip = 0
```

เส้นทาง

```
a1
a2
```

Event

a1 sendvalidateJob_1
b1 receivevalidateJob_1
c1 sendreserveAlert_2
d1 receivereserveAlert_2
e1 sendqueryToner_3 (guard a1)
e2 sendconfirmJob_4 (guard a2)
f1 receivequeryToner_3 (guard a1)
f2 receiveconfirmJob_4 (guard a2)
g1 sendfetchToner_5
h1 receivefetchToner_5
i1 sendscheduleToner_6
j1 receivescheduleToner_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    └── e2 → f2 → g1 → h1 → i1 → j1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1
```

model_149 PublicWifiHotspot · fail

Guard

a1	optIn = 1
b1	load < 10
b2	load ≥ 10

```
(root)
└── a1  optIn = 1
    ├── b1  load < 10
    └── b2  load ≥ 10
```

เส้นทาง

a1 -> b1
a1 -> b2

Event

a1 sendscheduleQuota_1
b1 receivescheduleQuota_1
c1 sendscheduleVoucher_2
d1 receivescheduleVoucher_2
e1 sendvalidateQuota_3
f1 receivevalidateQuota_3
g1 sendupdateSession_4
h1 receiveupdateSession_4
i1 sendvalidateSession_5
j1 receivevalidateSession_5
k1 sendreserveVoucher_6
l1 receivereserveVoucher_6
m1 sendupdateQuota_7_opt1 (guard a1)
n1 receiveupdateQuota_7_opt1 (guard a1)
o1 sendapproveQuota_8_opt1 (guard a1 b1)
o2 sendsendSession_10_opt1 (guard a1 b2)
p1 receiveapproveQuota_8_opt1 (guard a1 b1)
p2 receivesendSession_10_opt1 (guard a1 b2)
q1 sendqueryVoucher_9_opt1 (guard a1 b1)
r1 receivequeryVoucher_9_opt1 (guard a1 b1)
s1 sendissueSession_11_opt1 (guard a1)
t1 receiveissueSession_11_opt1 (guard a1)
u1 sendissueQuota_12
v1 receiveissueQuota_12

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1 → q1 → r1 → s1 → t1 → u1 → v1
    └── o2 → p2 → s1 → t1 → u1 → v1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o2 -> p2 -> s1 -> t1 -> u1 -> v1

model_159 SatelliteGroundStation · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendupdateTelemetry_1
b1 receiveupdateTelemetry_1
c1 sendlogCommand_2
d1 receivelogCommand_2
e1 sendrequestCommand_3
f1 receiverequestCommand_3
g1 sendassignTelemetry_4
h1 receiveassignTelemetry_4
i1 sendcheckCommand_5
j1 receivecheckCommand_5
k1 sendreleasePass_6
l1 receivereleasePass_6
m1 sendreportPass_7
n1 receivereportPass_7
o1 sendupdateCommand_8
p1 receiveupdateCommand_8
q1 sendschedulePass_9
r1 receiveschedulePass_9
s1 sendsendTelemetry_10
t1 receivesendTelemetry_10
u1 sendscheduleTelemetry_11
v1 receivescheduleTelemetry_11
w1 sendvalidatePass_12
x1 receivevalidatePass_12
y1 sendconfirmCommand_13
z1 receiveconfirmCommand_13

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1 → u1 → v1 → w1 → x1 →
      y1 → z1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1 -> y1 -> z1

model_194 VideoStreaming · fail

Guard

a1	stepNo < 7
b1	stepNo mod 2 = 0
b2	stepNo mod 2 ≠ 0

```
(root)
└── a1  stepNo < 7
    ├── b1  stepNo mod 2 = 0
    └── b2  stepNo mod 2 ≠ 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

a1 sendreserveLicense_1
b1 receivereserveLicense_1
c1 sendreportChunk_2
d1 receivereportChunk_2
e1 sendreportStream_3 (guard a1)
f1 receivereportStream_3 (guard a1)
g1 sendvalidateStream_4 (guard a1 b1)
g2 sendsendChunk_5 (guard a1 b2)
h1 receivevalidateStream_4 (guard a1 b1)
h2 receivesendChunk_5 (guard a1 b2)
i1 sendackChunk_7 (guard a1)
j1 receiveackChunk_7 (guard a1)
k1 sendscheduleLicense_8
k2 checkloop_reportStream_3_loop1 (guard a1)
l1 receivescheduleLicense_8
m1 sendfetchStream_9
n1 receivefetchStream_9
o1 sendscheduleStream_10
p1 receivescheduleStream_10
q1 sendupdateStream_11
r1 receiveupdateStream_11
s1 sendreleaseChunk_12
t1 receivereleaseChunk_12
u1 sendrequestStream_13
v1 receiverequestStream_13
w1 sendapproveStream_6 (guard a1 b2)
x1 receiveapproveStream_6 (guard a1 b2)

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1
    │   ├── k1 → l1 → m1 → n1 → o1 → p1 → q1 → r1 →
    │   │     s1 → t1 → u1 → v1
    │   └── k2 ↺ e1
    └── g2 → h2 → w1 → x1 → i1 → j1
        ├── k1 → l1 → m1 → n1 → o1 → p1 → q1 → r1 →
        │     s1 → t1 → u1 → v1
        └── k2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> w1 -> x1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> w1 -> x1 -> i1 -> j1 -> k2 ↺ e1
```

เกษตรและสิ่งแวดล้อม · Agriculture & Environment (15)

model_038 SmartGreenhouse · pass

Guard

a1	cycleNo < 24
b1	humidity < 40
b2	humidity ≥ 40

```
(root)
└── a1  cycleNo < 24
    ├── b1  humidity < 40
    └── b2  humidity ≥ 40
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

a1 sendstartCycle_1
b1 receivestartCycle_1
c1 sendreadClimate_2 (guard a1)
d1 receivereadClimate_2 (guard a1)
e1 sendclimateData_3 (guard a1)
f1 receiveclimateData_3 (guard a1)
g1 sendstartSprinkler_4 (guard a1 b1)
g2 sendstartFan_6 (guard a1 b2)
h1 receivestartSprinkler_4 (guard a1 b1)
h2 receivestartFan_6 (guard a1 b2)
i1 sendsprinklerOn_5 (guard a1 b1)
j1 receivesprinklerOn_5 (guard a1 b1)
k1 sendendCycle_7
k2 checkloop_readClimate_2_loop1 (guard a1)
l1 receiveendCycle_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1
    │   ├── k1 → l1
    │   └── k2 ↺ c1
    └── g2 → h2
        ├── k1 → l1
        └── k2 ↺ c1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ c1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k2 ↺ c1
```

model_051 AirQualityMonitor · pass

Guard

a1	useCache = 1
b1	online = 1
b2	online = 0

```
(root)
└── a1  useCache = 1
    ├── b1  online = 1
    └── b2  online = 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

a1 sendcancelFan_1
b1 receivecancelFan_1
c1 sendreleaseAlert_2
d1 receivereleaseAlert_2
e1 sendsendAlert_3_opt1 (guard a1)
f1 receivesendAlert_3_opt1 (guard a1)
g1 sendfetchFan_4
h1 receivefetchFan_4
i1 sendlogFan_5
j1 receivelogFan_5
k1 sendqueryAlert_6 (guard b1)
k2 sendcancelAlert_13 (guard b2)
l1 receivequeryAlert_6 (guard b1)
l2 receivecancelAlert_13 (guard b2)
m1 sendnotifyAlert_7 (guard b1)
n1 receivenotifyAlert_7 (guard b1)
o1 sendreservePm_8 (guard b1)
p1 receivereservePm_8 (guard b1)
q1 sendapproveFan_9 (guard b1)
r1 receiveapproveFan_9 (guard b1)
s1 sendreportFan_10 (guard b1)
t1 receivereportFan_10 (guard b1)
u1 sendrequestPm_11 (guard b1)
v1 receiverequestPm_11 (guard b1)
w1 sendfetchPm_12 (guard b1)
x1 receivefetchPm_12 (guard b1)
y1 sendvalidateFan_14
z1 receivevalidateFan_14

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1 → m1 → n1 → o1 → p1 → q1 → r1 →
    │     s1 → t1 → u1 → v1 → w1 → x1 → y1 → z1
    └── k2 → l2 → y1 → z1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1 -> y1 -> z1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 -> l2 -> y1 -> z1
```

model_070 CenterPivotIrrigation · fail

Guard

a1	cycleNo < 9
b1	cycleNo mod 3 = 0

```
(root)
└── a1  cycleNo < 9
    └── b1  cycleNo mod 3 = 0
```

เส้นทาง

a1 -> b1

Event

a1 sendcancelWater_1
b1 receivecancelWater_1
c1 sendvalidateWater_2
d1 receivevalidateWater_2
e1 sendreserveWater_3 (guard a1)
f1 receivereserveWater_3 (guard a1)
g1 sendreleaseSector_4_opt1 (guard a1 b1)
h1 receivereleaseSector_4_opt1 (guard a1 b1)
i1 sendupdateWater_5_opt1 (guard a1 b1)
j1 receiveupdateWater_5_opt1 (guard a1 b1)
k1 sendcancelSector_6 (guard a1)
l1 receivecancelSector_6 (guard a1)
m1 sendscheduleWater_7 (guard a1)
n1 receivescheduleWater_7 (guard a1)
o1 sendqueryRain_8
o2 checkloop_reserveWater_3_loop1 (guard a1)
p1 receivequeryRain_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1
    └── o2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o2 ↺ e1

model_080 CropDroneSpraying · fail

Guard

a1	t_obs ≤ 11
a2	t_obs > 11

```
(root)
├── a1  t_obs ≤ 11
└── a2  t_obs > 11
```

เส้นทาง

a1
a2

Event

a1 sendapproveArea_1
b1 receiveapproveArea_1
c1 sendcheckMission_2
d1 receivecheckMission_2
e1 sendsyncSpray_3 (guard a1)
e2 sendassignSpray_4 (guard a2)
f1 receivesyncSpray_3 (guard a1)
f2 receiveassignSpray_4 (guard a2)
g1 sendscheduleSpray_5
h1 receivescheduleSpray_5
i1 sendassignMission_6
j1 receiveassignMission_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    └── e2 → f2 → g1 → h1 → i1 → j1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1

model_092 FarmMilkingRobot · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendreserveYield_1
b1 receivereserveYield_1
c1 sendlogYield_2
d1 receivelogYield_2
e1 sendreportYield_3
f1 receivereportYield_3
g1 sendreserveMilk_4
h1 receivereserveMilk_4
i1 sendqueryHealth_5
j1 receivequeryHealth_5
k1 sendissueMilk_6
l1 receiveissueMilk_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_098 ForestFireSensor · fail

Guard

a1	grade = MANUAL
a2	grade = AUTO
b1	t_obs ≤ 3
c1	retryNo < 11
b2	t_obs > 3

```
(root)
├── a1  grade = MANUAL
│   ├── b1  t_obs ≤ 3
│   │   └── c1  retryNo < 11
│   └── b2  t_obs > 3
└── a2  grade = AUTO
    ├── b1  t_obs ≤ 3
    │   └── c1  retryNo < 11
    └── b2  t_obs > 3
```

เส้นทาง

```
a1 -> b1 -> c1
a1 -> b2
a2 -> b1 -> c1
a2 -> b2
```

Event

a1 sendreleaseAlert_1
b1 receivereleaseAlert_1
c1 sendschedulePatrol_2
d1 receiveschedulePatrol_2
e1 sendsyncPatrol_3 (guard a1)
e2 sendcancelAlert_4 (guard a2)
f1 receivesyncPatrol_3 (guard a1)
f2 receivecancelAlert_4 (guard a2)
g1 sendissueSmoke_5
h1 receiveissueSmoke_5
i1 sendnotifyAlert_6
j1 receivenotifyAlert_6
k1 sendreserveAlert_7 (guard b1)
k2 sendscheduleSmoke_11 (guard b2)
l1 receivereserveAlert_7 (guard b1)
l2 receivescheduleSmoke_11 (guard b2)
m1 sendvalidateSmoke_8 (guard b1 c1)
n1 receivevalidateSmoke_8 (guard b1 c1)
o1 sendcheckAlert_9 (guard b1 c1)
p1 receivecheckAlert_9 (guard b1 c1)
q1 sendassignAlert_10 (guard b1)
q2 checkloop_validateSmoke_8_loop1 (guard b1 c1)
r1 receiveassignAlert_10 (guard b1)
s1 sendqueryPatrol_12
t1 receivequeryPatrol_12

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    │   ├── k1 → l1 → m1 → n1 → o1 → p1
    │   │   ├── q1 → r1 → s1 → t1
    │   │   └── q2 ↺ m1
    │   └── k2 → l2 → s1 → t1
    └── e2 → f2 → g1 → h1 → i1 → j1
        ├── k1 → l1 → m1 → n1 → o1 → p1
        │   ├── q1 → r1 → s1 → t1
        │   └── q2 ↺ m1
        └── k2 → l2 → s1 → t1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q2 ↺ m1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 -> l2 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q2 ↺ m1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1 -> k2 -> l2 -> s1 -> t1
```

model_103 GreenhouseLighting · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendapproveEnergy_1
b1 receiveapproveEnergy_1
c1 sendackLight_2
d1 receiveackLight_2
e1 sendcheckLux_3
f1 receivecheckLux_3
g1 sendreserveLux_4
h1 receivereserveLux_4
i1 sendreleaseLight_5
j1 receivereleaseLight_5
k1 sendnotifyEnergy_6
l1 receivenotifyEnergy_6
m1 sendqueryEnergy_7
n1 receivequeryEnergy_7
o1 sendrequestLight_8
p1 receiverequestLight_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1
```

model_116 IrrigationZoneControl · pass

Guard

a1	attemptNo < 12

```
(root)
└── a1  attemptNo < 12
```

เส้นทาง

a1

Event

a1 sendqueryZone_1
b1 receivequeryZone_1
c1 sendackSchedule_2
d1 receiveackSchedule_2
e1 sendrequestZone_3 (guard a1)
f1 receiverequestZone_3 (guard a1)
g1 sendcheckMoisture_4 (guard a1)
h1 receivecheckMoisture_4 (guard a1)
i1 sendqueryMoisture_5 (guard a1)
j1 receivequeryMoisture_5 (guard a1)
k1 sendcancelZone_6
k2 checkloop_requestZone_3_loop1 (guard a1)
l1 receivecancelZone_6
m1 sendrequestSchedule_7
n1 receiverequestSchedule_7
o1 sendreleaseMoisture_8
p1 receivereleaseMoisture_8
q1 sendreserveMoisture_9
r1 receivereserveMoisture_9
s1 sendscheduleSchedule_10
t1 receivescheduleSchedule_10
u1 sendsendSchedule_11
v1 receivesendSchedule_11

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1 → m1 → n1 → o1 → p1 → q1 → r1 →
    │     s1 → t1 → u1 → v1
    └── k2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ e1

model_121 LivestockFeeder · fail

Guard

a1	approved = 1
b1	express = 1
a2	approved = 0

```
(root)
├── a1  approved = 1
│   └── b1  express = 1
└── a2  approved = 0
```

เส้นทาง

a1 -> b1
a2

Event

a1 sendreportRation_1
b1 receivereportRation_1
c1 sendsyncFeed_2
d1 receivesyncFeed_2
e1 sendackFeed_3 (guard a1)
e2 sendcancelRation_7 (guard a2)
f1 receiveackFeed_3 (guard a1)
f2 receivecancelRation_7 (guard a2)
g1 sendupdateRation_4_opt1 (guard a1 b1)
h1 receiveupdateRation_4_opt1 (guard a1 b1)
i1 sendqueryRation_5_opt1 (guard a1 b1)
j1 receivequeryRation_5_opt1 (guard a1 b1)
k1 sendconfirmFeed_6 (guard a1)
l1 receiveconfirmFeed_6 (guard a1)
m1 sendconfirmRation_8
n1 receiveconfirmRation_8

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1 →
    │     m1 → n1
    └── e2 → f2 → m1 → n1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> m1 -> n1

model_151 RecyclingKiosk · fail

Guard

a1	express = 1
b1	roundNo < 6
c1	urgent = 1
c2	urgent = 0

```
(root)
└── a1  express = 1
    └── b1  roundNo < 6
        ├── c1  urgent = 1
        └── c2  urgent = 0
```

เส้นทาง

```
a1 -> b1 -> c1
a1 -> b1 -> c2
```

Event

a1 sendconfirmCredit_1 (guard a1 b1)
b1 receiveconfirmCredit_1
c1 sendfetchVoucher_2
d1 receivefetchVoucher_2
e1 sendsyncCredit_3_opt1 (guard a1)
f1 receivesyncCredit_3_opt1 (guard a1)
g1 sendnotifyCredit_4_opt1 (guard a1 b1)
h1 receivenotifyCredit_4_opt1 (guard a1 b1)
i1 sendconfirmBottle_5_opt1 (guard a1)
i2 checkloop_notifyCredit_4_loop1
j1 receiveconfirmBottle_5_opt1 (guard a1)
k1 sendconfirmVoucher_6
l1 receiveconfirmVoucher_6
m1 sendackVoucher_7
n1 receiveackVoucher_7
o1 sendassignBottle_8 (guard c1)
o2 sendscheduleVoucher_9 (guard c2)
p1 receiveassignBottle_8 (guard c1)
p2 receivescheduleVoucher_9 (guard c2)
q1 sendqueryCredit_10
r1 receivequeryCredit_10

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1 → m1 → n1
    │   ├── o1 → p1 → q1 → r1
    │   └── o2 → p2 → q1 → r1
    └── i2 ↺ g1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o2 -> p2 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ g1
```

model_166 SmartBeehive · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendupdateHoney_1
b1 receiveupdateHoney_1
c1 sendreleaseWeight_2
d1 receivereleaseWeight_2
e1 sendlogWeight_3
f1 receivelogWeight_3
g1 sendissueWeight_4
h1 receiveissueWeight_4
i1 sendcheckWeight_5
j1 receivecheckWeight_5
k1 sendscheduleSwarm_6
l1 receivescheduleSwarm_6
m1 sendrequestHoney_7
n1 receiverequestHoney_7

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
```

model_174 SmartRainGauge · pass

Guard

a1	useCache = 1
b1	score < 50
b2	score ≥ 50

```
(root)
└── a1  useCache = 1
    ├── b1  score < 50
    └── b2  score ≥ 50
```

เส้นทาง

a1 -> b1
a1 -> b2

Event

a1 sendqueryWarning_1 (guard a1 b1)
b1 receivequeryWarning_1
c1 sendconfirmRainfall_2 (guard a1 b1)
d1 receiveconfirmRainfall_2
e1 sendreserveRainfall_3_opt1 (guard a1)
f1 receivereserveRainfall_3_opt1 (guard a1)
g1 sendcheckLevel_4_opt1 (guard a1 b1)
g2 sendscheduleLevel_9_opt1 (guard a1 b2)
h1 receivecheckLevel_4_opt1 (guard a1 b1)
h2 receivescheduleLevel_9_opt1 (guard a1 b2)
i1 sendapproveRainfall_5_opt1 (guard a1 b1)
j1 receiveapproveRainfall_5_opt1 (guard a1 b1)
k1 sendreserveLevel_6_opt1 (guard a1 b1)
l1 receivereserveLevel_6_opt1 (guard a1 b1)
m1 sendvalidateLevel_7_opt1 (guard a1 b1)
n1 receivevalidateLevel_7_opt1 (guard a1 b1)
o1 sendupdateWarning_8_opt1 (guard a1 b1)
p1 receiveupdateWarning_8_opt1 (guard a1 b1)
q1 sendsyncRainfall_10_opt1 (guard a1)
r1 receivesyncRainfall_10_opt1 (guard a1)
s1 sendcancelRainfall_11
t1 receivecancelRainfall_11

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1 →
    │     o1 → p1 → q1 → r1 → s1 → t1
    └── g2 → h2 → q1 → r1 → s1 → t1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> q1 -> r1 -> s1 -> t1

model_177 SmartTrashCompactor · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ (ลำดับข้อความอย่างเดียว)

Event

a1 sendcheckRoute_1
b1 receivecheckRoute_1
c1 sendassignLevel_2
d1 receiveassignLevel_2
e1 sendscheduleRoute_3
f1 receivescheduleRoute_3
g1 sendscheduleLevel_4
h1 receivescheduleLevel_4
i1 sendnotifyLevel_5
j1 receivenotifyLevel_5
k1 sendackService_6
l1 receiveackService_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_179 SmartWasteBin · pass

Guard

a1	hasKey = 1
a2	hasKey = 0
b1	phase = GOLD
b2	phase = SILVER
b3	phase = BRONZE

```
(root)
├── a1  hasKey = 1
│   ├── b1  phase = GOLD
│   ├── b2  phase = SILVER
│   └── b3  phase = BRONZE
└── a2  hasKey = 0
    ├── b1  phase = GOLD
    ├── b2  phase = SILVER
    └── b3  phase = BRONZE
```

เส้นทาง

```
a1 -> b1
a1 -> b2
a1 -> b3
a2 -> b1
a2 -> b2
a2 -> b3
```

Event

a1 sendreserveRoute_1
b1 receivereserveRoute_1
c1 sendsendRoute_2
d1 receivesendRoute_2
e1 sendassignRoute_3 (guard a1)
e2 sendreservePickup_4 (guard a2)
f1 receiveassignRoute_3 (guard a1)
f2 receivereservePickup_4 (guard a2)
g1 sendscheduleFill_5
h1 receivescheduleFill_5
i1 sendupdatePickup_6
j1 receiveupdatePickup_6
k1 sendlogPickup_7 (guard b1)
k2 sendissueRoute_8 (guard b2)
k3 sendsyncPickup_9 (guard b3)
l1 receivelogPickup_7 (guard b1)
l2 receiveissueRoute_8 (guard b2)
l3 receivesyncPickup_9 (guard b3)
m1 sendreleaseRoute_10
n1 receivereleaseRoute_10

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    │   ├── k1 → l1 → m1 → n1
    │   ├── k2 → l2 → m1 → n1
    │   └── k3 → l3 → m1 → n1
    └── e2 → f2 → g1 → h1 → i1 → j1
        ├── k1 → l1 → m1 → n1
        ├── k2 → l2 → m1 → n1
        └── k3 → l3 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 -> l2 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k3 -> l3 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1 -> k2 -> l2 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1 -> k3 -> l3 -> m1 -> n1
```

model_180 SoilTestingLab · pass

Guard

a1	roundNo < 12

```
(root)
└── a1  roundNo < 12
```

เส้นทาง

```
a1
```

Event

a1 sendassignReport_1
b1 receiveassignReport_1
c1 sendsendNutrient_2
d1 receivesendNutrient_2
e1 sendreleaseReport_3 (guard a1)
f1 receivereleaseReport_3 (guard a1)
g1 sendcancelNutrient_4
g2 checkloop_releaseReport_3_loop1 (guard a1)
h1 receivecancelNutrient_4
i1 sendquerySample_5
j1 receivequerySample_5
k1 sendnotifyReport_6
l1 receivenotifyReport_6

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    └── g2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 ↺ e1
```